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Capitalism and the dollar: the AI race, tokenization, and the dollar's future

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A massive race for capital

The US is sprinting in its reach for capital. Private markets, debt, and public equities have been setting records, and the US is enjoying unprecedented amounts of foreign equity inflows. In this paper, we discuss how the US is leveraging the power of capitalism – supercharged by blockchain – to try and win the AI race. But the US reach for capital is not only about AI spending; it also reflects growing fiscal deficits and the need for foreign private funding. The US is increasing its reliance on private capital to offset public sector vulnerabilities. Meanwhile, China is running a very different race – given large trade surpluses, domestic excess savings, and a "low-cost" AI approach, China is raising less capital, with much less from abroad.

America embraces asset tokenization

The US is using blockchain technology to support US dollar and capital market dominance. We wrote last year about the crucial role of stablecoins. We are now at the next stage: asset tokenization – converting ownership rights to financial assets into digital tokens recorded on blockchain. Asset tokenization is not futuristic – it is happening now at the heart of US markets. Securities depositories and exchanges are building the infrastructure to tokenize US assets, with regulatory oversight.

TINA meets Tokenization: the future of the dollar

TINA already plays a key role attracting capital to the US. Tokenization could add to this by making US assets more attractive and accessible. The collateral velocity of tokenized assets could be a superpower. And if US assets are trading 24/7 with instant settlement, this could impact global behaviour, particularly for retail. But lowering the hurdles for inbound capital also exposes the US to capital going out. The dollar is becoming riskier, more equity sensitive, and leveraged to the AI race. If AI proves uneconomical, or the US starts to fall behind, the dollar is very exposed.

Open markets meet Open technology: the future of capitalism

The US is taking an “open markets-closed tech” approach to the AI race, while China is taking an “open tech-closed markets” one. The US is widening the doors for capital but keeping technology closed to preserve corporate pricing power. Meanwhile, China is pushing out low-cost models with open-weights. This is not just a contest between AI models, but economic models. If China's approach undercuts US profits, this could threaten the US model of shareholder capitalism itself. China's challenge is not only about technology, but the central maxim of US power: that free and open capital markets always deliver the winning innovation.

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Part 1: A massive reach for capital in the US

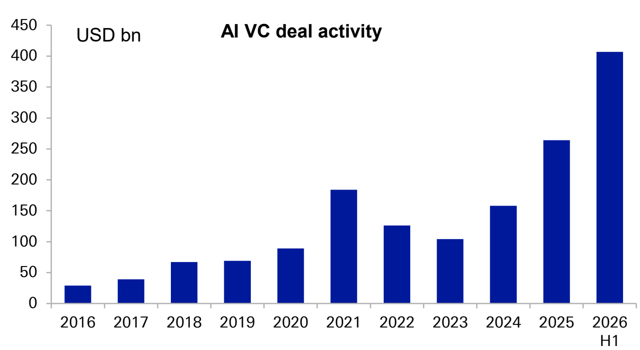
The US is sprinting at record speed in its reach for capital. Huge sums of money are being raised to fund the buildout of AI infrastructure and development of frontier AI models, with US companies expected to spend around $800bn in AI capex this year. **We argue in this paper that the US is leveraging the full power of capitalism to fund capital investments seen as crucial to winning the AI race**

The US is raising enormous amounts of private capital

We simplify the key trends in the US' capital-raising landscape into a few observations. First, AI labs are turning to private markets to raise large sums of equity capital. Second, hyperscalers are turning to investment grade debt markets to issue meaningful amounts of debt for the first time. And third, public equity markets are back in focus for capital funding with some of the world's biggest companies returning to issue stock, and major IPOs in the pipeline.

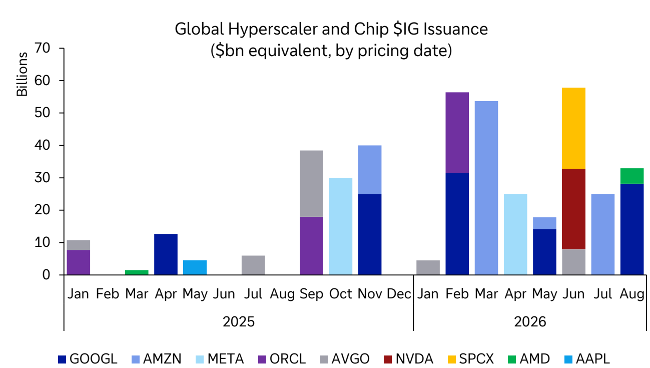
- The AI venture capital industry has raised over $400bn this year alone, with over 90% of the biggest deal volumes in the US (Pitchbook). The two largest AI labs in the US have raised a combined $217bn this year – valuing each company near $1tn. We note that xAI had raised $20bn earlier in the year, before being acquired by SpaceX in February. On an annualized basis, AI VC capital is running at three times last year's pace.
- In debt capital markets, the shift has also been staggering. While free cash flows remain an important source of funding for hyperscalers, the capital intensity of the infrastructure buildout is outrunning cash flows, forcing companies to turn to debt. Hyperscalers like Google, Meta, Oracle and Amazon raised an average of just USD25bn per year between them over 2020-24. In 2026 YTD, they have raised around ten times that amount through IG credit markets. Microsoft has yet to turn to debt markets.
- In public equity markets, Google also had its first primary equity raise since its 2004 IPO in June 2026, raising $85bn. Returning to issue is a big reversal from years of buying back stock. This is also setting up to be a historic year for IPOs. SpaceX achieved the largest total market valuation of a newly listed company in history near $2tn in its June 2026 IPO. The largest US AI labs are expected to come to the IPO market soon which could change the AI boom as we have written about here.

Figure 1: US AI companies are attracting record amounts of private capital



Source : Deutsche Bank Research, Pitchbook

Figure 2: US hyperscalers have turned to public credit markets to issue huge amounts of debt



Source : Deutsche Bank Research, Bloomberg Finance LP

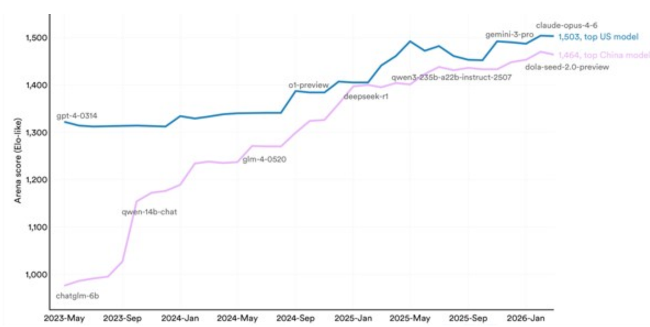
China is running a different race

On many metrics, Chinese models are closing the gap to the US in terms of performance and capabilities (Figure 3). While the US remains ahead in frontier development, Chinese models account for more than half of US company token usage on Open Router (Yahoo Finance).

But from a capital perspective, China appears to be running a different race - raising much less capital, less from private sources, and less from abroad. While US companies are expected to spend around $800bn in AI capex this year, that number is closer to $100bn in China. While the biggest US AI labs have raised nearly $250bn in private funding rounds this year, Chinese labs have raised far less. The largest capital raise came from DeepSeek which raised $7.4bn earlier in the year, and may be returning for a further $8bn. China's National AI fund has participated in a number of capital raisings. Stanford University's recent AI report made a similar observation: noting that in 2025, US private AI investment was 23 times the $12.4bn invested in China, although Chinese numbers are likely understated given government guided funds.

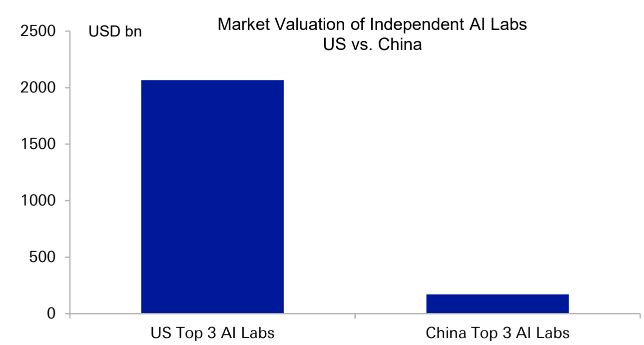
The stark difference in the approach to capital raising is also reflected in markets. With US AI labs raising far more capital for ownership stakes, the market valuations of US AI companies compared with their Chinese peers are many orders of magnitude apart (Figure 4). Furthermore, while Chinese hyperscalers like Tencent, Baidu and Alibaba have been tapping debt markets, they are doing so in smaller size, with a no major uptick compared with earlier years. They are, however, increasingly funding in dim sum bonds, suggestive of a desire to de-dollarize funding. See more on the Rise of the RMB Financing Ecosystem here.

Figure 3: The gap in performance between US and Chinese models has narrowed a lot



Source : Deutsche Bank, Stanford University HAI

Figure 4: US AI labs have much larger implied market valuations than their Chinese peers



Source : Deutsche Bank, Bloomberg Finance LP, Various news sources; Note: Valuations are based on market cap for publicly listed companies and announced or intended funding rounds for private companies.

The US will likely want all the foreign capital it can get

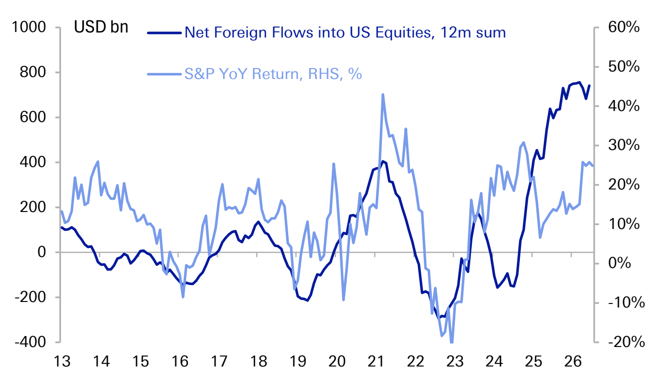
The dramatic reach for capital in the US is not only reflective of an American economic model that prioritizes open capital markets, it also reflects the US' growing need for foreign financing given the US' large twin deficit position. The US is running a 6%+ fiscal deficit combined with a near 4% of GDP current account deficit. In the absence of fiscal savings, incremental private sector capex will require funding from abroad. At a time when private sector financing requirements for AI capex are rising, it is not a surprise the US is widening its reach for capital from abroad.

The US is in fact enjoying an unprecedented amount of equity capital inflow. In Q2 2026 alone, the US received over USD400bn of foreign equity capital, well above any previous quarter. The amount of capital coming into US equities is also well above what US equity performance alone would imply (Figure 5), and now well above debt flows (Figure 6) – the historical mainstay of US capital account financing.

We have written in the past about the big rotation underway in US financing. Geopolitical ruptures in long-standing alliances in defence and trade are hurting long-term appetite for USD government debt, particularly by the foreign official sector. But at the same time, technology is pulling enormous equity capital in from other types of foreign investors like retail. The gap between net equity flows to the US (increasing) and debt flows (falling) has never been wider, and the shift from official to private funding sources is becoming more noticeable. With the US on the cusp of a potentially significant IPO cycle, we find it notable that US companies are picking up on the funding shift. SpaceX reportedly initially earmarked 30% of its IPO for retail investors, more than three times as much as a traditional IPO offering.

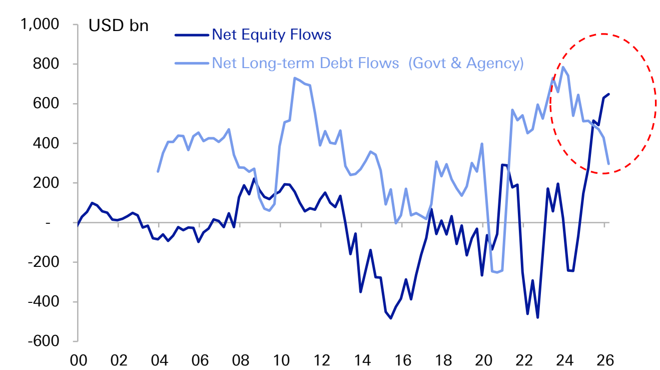
As we show in this paper, the US will likely remain very focused on attracting foreign private capital to fund both its AI ambitions and its deficits.

Figure 5: The US is attracting a record amount of foreign equity capital - more than market returns alone would suggest



Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 6: Equity flows have replaced debt flows as the main source of foreign capital financing for the US



Source : Deutsche Bank, Haver Analytics

Where blockchain technology comes in

The US is taking a very capital-intensive approach to the AI race, raising record amounts of capital, and attracting huge amounts of it from overseas. Foreign capital not only helps support the dollar by providing deficit funding, it also improves the cost of financing for US companies. It is natural for the US to leverage technology that supports these capital flow trends.

We have written at length about how the US is using the power of blockchain technology and a new form of private money – stablecoins – to democratize access to the dollar. **We are now at the second stage of evolution, where the power of blockchain technology is being used to democratize access to US *capital markets.* This is called real world asset (RWA) tokenization.** While stablecoins aim put the US dollar "on chain" as a digital money token, asset market tokenization could look to put the entirety of US asset markets and their underlying financial architecture on chain.

The two forces of tokenized money and tokenized assets are themselves highly symbiotic. The existence of tokenized assets should widen the appeal of holding tokenized money like stablecoins, that can then be more easily deployed for returns and yields "on chain", without needing to be moved off-chain back into fiat currency to invest in traditional assets. Both stablecoins and tokenized bank deposits could become important means of payment for tokenized assets, with a meaningful impact on the settlement process. We will explore the nuances of this in a forthcoming FX Special Report on Programmable Money.

From a macro perspective, we argue that US tokenization ambitions are deeply linked to supporting capital inflows into the US. Our earlier research on stablecoins discussed how they could support the diversification of dollar debt to private sector holders. Not only did the GENIUS Act introduce 1:1 reserve backing requirements for dollar stablecoin issuance, that helps power demand for front-end US T-bills; more importantly, if stablecoins help keep the world paying in dollars, they help retain the role of the dollar as a private sector savings instrument, which could help demand for assets like longer-dated Treasuries. If stablecoins are now combined with a wider universe of tokenized assets, this could encourage demand for many more US assets. This should help what we have termed a "geopolitics-to-technology funding swap". The US is moving away from attracting official, long-term, geopolitically motivated capital flows, towards private sector, shorter-term, technology motivated capital.

America's full-hearted embrace of asset tokenization – compared with China's turn away from it – can be understood in the context of both the AI race and the relative external funding position of both states. If the US is taking a capital-focused approach to the AI race, with a starting point of twin deficits, the US will naturally benefit from lowering the hurdles for foreign and private capital as much as possible. That is indeed what it is doing.

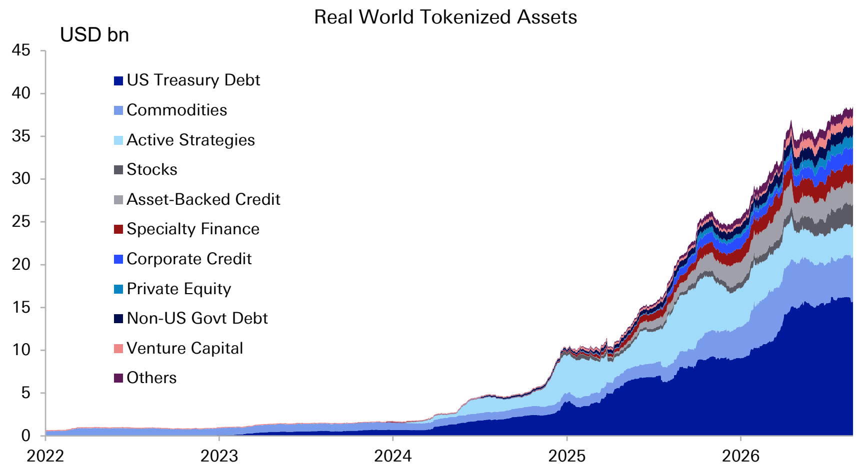
Part 2: America embraces Asset Tokenization

We begin with a definition for asset tokenization, borrowing from the DTCC, the custodian for close to $115 trillion in US assets and an organization that finds itself at forefront of this development. **Asset tokenization is the "process of converting ownership rights to a financial asset – such as stocks, bonds or real estate – into a digital token recorded on a blockchain. These tokens represent the same legal rights as the original asset but can be transferred, tracked and managed digitally.**" They note that tokenization introduces new capabilities, from fractional ownership, faster cross-border transfers, transparent records and smart contracts. In a tokenized world, instead of waiting days for a U.S. Treasury bond to settle through multiple intermediaries, a tokenized version could settle in minutes. Also see "Asset Tokenization 101" from the DB Research Institute here.

Only around $40bn of real-world assets have been tokenized to date, but the size of the tokenized asset universe is expected to grow much further and faster from here given adoption by traditional finance that is just getting underway.

In Part 1 of this paper we explored the *macro* motivations for America's embrace of tokenization: to reduce the hurdles for foreign capital to flow into the US and fund its AI ambitions in an increasingly competitive and existential AI race. In this section, we look at some of the more micro motivations for tokenization, and how they come together.

Figure 7: Real world tokenized assets have been growing, but are still a fraction of the over USD100tn US asset universe - leaving big headroom



Source : Deutsche Bank, RWA.xyz

Some further motivating drivers for asset tokenization

The underlying technology that enables tokenization is distributed ledger technology (DLT) which allows the ownership rights to anything from money, to a stock, to a painting, to be represented as a digital token on a decentralized network like blockchain. But new technology alone cannot drive real world adoption and change, without the financial and economic incentives to use it.

So, why is DLT being used to tokenize real-world assets today and how have incentives aligned across economic agents – from retail investors, asset managers, corporates, financial firms, securities depositories and exchanges – to motivate this adoption? Beyond the macro, state-level motivation for attracting capital to fund the AI race and US deficits, we argue that private agents in the financial system are also likely to be deeply invested for their own reasons.

First, the revolution in overseas retail trading demonstrates the case for widening access to markets. Retail trading has been spurred on in recent years by a number of developments: from Covid lockdowns, the rise of zero-commission platforms, social media, fintech brokerages, improved access to information and charting, and even tax incentives to increase financialization of household savings (i.e., Japan's NISA program). Korea is arguably the poster child of the retail revolution, and offers a potential template for what might become possible at a global scale if technology is used to lower hurdles for entry. Korea represents 2% of global nominal GDP, but accounted for 10% of the $740bn of foreign equity inflows to the US last year, a lot of which were powered by retail investors, or the "seohakaemi" (overseas stock ants). Access arguably played a big role. Korea began allowing for the fractional trading of foreign stocks by certain Korean brokerages as early as 2019, even earlier than for domestic stocks. Korean brokerages also signed up with a number of US alternative trading systems that offered daytime trading of US stocks in Korean hours. The combination of a highly digitized, connected population with lower hurdles for access was a dramatic flow of Korean capital into US equities. To us, the case of Korea demonstrates what is possible if US markets were to become easily accessible 24/7 to everyone.

Second, corporates and institutions still face a number of inefficiencies around cash and capital in traditional off-chain markets. We focus on three examples here: trapped cash for settlement, inefficient collateral management, and manual corporate actions. While securities markets in the US have successfully moved from T+2 to T+1, this gap still involves the lock-up of capital to act as a backstop for trade settlement. If securities and payments were to settle atomically and instantly on blockchain, this could free up a huge amount of capital. Collateral markets are also still relatively inflexible today. Today, repo markets allow financial entities to borrow cash in exchange for lending collateral like US Treasuries. Repos are structured around overnight to longer-term financing and are constrained by end-of-day settlement cycles and operational windows. Digital assets held on a blockchain could arguably be posted instantly for much shorter periods, creating intraday repo possibilities and increasing the velocity of collateral. As a final example, NASDAQ highlights that processing corporate actions today alone "carries costs of more than $58 billion across the industry, driven by fragmented systems, manual workflows, and duplicative record-keeping." The programmability offered by digital assets would help address this as corporate actions could be programmed into smart contracts attached to the asset. In sum, there are clear incentives for agents – from corporate issuers to institutional asset managers – to embrace a tokenized asset environment.

Third, the tokenization train had already left the station but was at risk of being mediated via third-party offerings and fragmenting market liquidity. In the traditional space, we note that fund managers like Blackrock and Templeton had already begun issuing some of their funds like BUIDL and FBOXX on public blockchains. This involved making the funds accessible as digital tokens, but importantly the underlying assets themselves remained off-chain in traditional financial instruments. More importantly, the presence of blockchain technology has allowed for entry and innovation by new private players that have begun offering US assets as digital tokens on blockchain rails. New tokenized equity frameworks offer *representations* of real US stocks and ETFs, backed 1:1 by underlying traditional equities, held in regulated custody, and issued as on-chain tokens on many different blockchains. They are available to non-US clients in 100+ countries, and allow fractional purchases with as little as $1 in select US stocks and ETFs, some of which are available to trade 24/7. They do *not*, however, confer shareholder rights like voting or dividends in cash – instead, they offer price exposure to the underlying asset in an always-on format, via a *third-party* token. If tokenization was already underway, the incentive for traditional asset mangers and exchanges to be part of the process was becoming more evident – to avoid losing funds and liquidity to third parties. Rather than simply tokenizing *claims* on US securities, developments underway are now building the infrastructure to allow underlying US securities *themselves* to be issued, traded, and held as tokens.

As we discuss in the next section, asset tokenization is not a futuristic or frontier concept – it is live and is now happening at the heart of traditional markets, led by core players in the US financial ecosystem.

What is happening?

In a nutshell: tokenization is going mainstream, with adoption by core, traditional financial players. Securities depositories and exchanges under the aegis of regulatory guidance are building the market infrastructure to tokenize the US asset universe, with participation from asset managers and banks.

Regulatory efforts arguably kicked off in force with the launch of Project Crypto by the SEC in July 2025, alongside the White House's Digital Asset Report that called for laying the groundwork for the US to become the deepest and most liquid digital asset market in the world. One of the first tasks was to develop a taxonomy for crypto assets, where tokenized securities were importantly defined as *remaining* "securities". In January 2026, Project Crypto evolved into a joint inter-agency framework between the SEC and the CFTC that would allow for "coordination, coherence, and a unified approach to the federal oversight of crypto asset markets". Crucially, it is actions from regulatory agencies in the US that are driving progress, independent of pending efforts in Congress to pass the CLARITY Act.

The big green light for mainstream tokenization came in December 2025 when the SEC issued the DTC with a "no-action" letter which legitimized the idea that the same security could trade on two rails. DTC could therefore tokenize assets in its custody from US stocks, ETFs and bonds, and the digital version of these assets would have "the same entitlements, investor protections and ownership rights" as the traditional asset. This was followed up by a ruling in January 2026 which noted that the "format in which a security is issued or the methods by which holders are recorded (e.g., on-chain vs. off-chain) does not affect application of the federal securities laws". Assets can be held traditionally or in tokenized form (as a digital twin) and they will be treated the same, and will refer to the same security. In other words, **both traditional and tokenized architectures can now coexist, run side-by-side, and be interoperable, with the trust of DTC custody to back them.** While the passage of the CLARITY Act is still pending, the above SEC notifications and the commitment of Project Crypto to bringing US capital markets on chain means that the regulatory umbrella for tokenization has been sufficiently opened. This has been enabling of developments underway this year.

The progress at DTC remains crucial to track. DTC is the main central securities depository in the US – it is the venue where trades settle and clear, and the custodian where financial firms hold their assets. DTC has around $115 trillion of US assets under its custody. Tokenization at DTC has now begun. In July 2026, the DTC converted the first asset into tokens that were used for real world trades. 40 different financial firms participated. Amongst many examples, the SPDR S&P500 ETF Trust (SPY) – the world's first ETF – was tokenized. In another test, JP Morgan successfully posted tokenized assets to satisfy a margin requirement at the CME. The full set of tests that day can be found here. DTC intends to launch its tokenization service by October 2026, but full-scale tokenization will be a long process. The DTCC CEO has argued it could proceed over many years with markets "crawling, then walking, then running" and the pace to be set by market demand to hold assets as tokens. DTC is beginning by tokenizing the Russell 1000, ETFs, and US Treasuries before widening it out. Market estimates currently range from $2tn to $30tn for the size of the tokenized asset universe by the 2030s, but these estimates may well be revised higher as the infrastructure becomes operational.

US exchanges – NYSE and NASDAQ – are shifting to accommodate "always-on" trading. In current financial architecture, exchanges are the *trading* layer, where buyers and sellers meet and prices are discovered, while DTC is the *post-trade* infrastructure layer for settlement and custody, maintaining a central record of securities ownership after trades occur. The biggest development at US exchanges is that they are set to move towards round-the-clock trading. NASDAQ has also announced an intention to offer trading 23/5 trading by December 2025, while NYSE is aiming for 22 hours by later this year. As noted earlier, synthetic equities have already been trading outside market hours on permissionless blockchains and alternative trading systems, so increasing exchange hours will allow that liquidity to come back on exchange, and prevent tokenized equities from becoming off-exchange products that fragment liquidity.

Exchanges are also building new platforms and tools to enable a tokenized trading experience. The NYSE's new digital platform – being built in conjunction with Securitize – aims to not just offer 24/7 operations, but also instant settlement, fractional share trading, and stablecoin based funding. NASDAQ has also announced the launch of an "equity token design" that will allow public companies to leverage tokenization to program shareholder engagement from corporate actions to proxy voting. This will address the current condition of third-party digital tokens that offer investors economic exposure but without the underlying shareholder rights and corporate actions. NASDAQ aims to have full tokenization capabilities by 2027.

The first mover advantage

Tokenization is going mainstream in the US and this is happening now: as we have described above, the biggest exchanges and securities depositories are making significant upgrades to allow US asset markets to be traded and held on chain. While tokenization of the US asset universe could proceed over many years, the ship has left the harbour.

With the US moving to provide rapid regulatory clarity and support for the development of tokenized securities, and building the settlement infrastructure to support it, it stands to benefit from a first mover advantage. Not only might the US be setting the standards for what on-chain financial trading and settlement looks like, it could benefit from future network effects. Just as the US benefits from agency over networks like SWIFT and USD correspondent banking, it may exercise them over the future on-chain infrastructure. **If the majority of on-chain private money (stablecoins) is dollarized, and the majority of on-chain assets are US listed, this could reinforce USD payments and capital markets dominance.** And if access to capital is increasingly concentrated in the US, global companies too may be increasingly interested in listing and issuing in the US to access them. As a potential harbinger, it is perhaps notable that SK Hynix made the largest foreign debut in US markets with its $26.5bn ADR this summer, favouring US markets for their capital raising.

We look at some of the broader implications for markets from the interplay between the AI race and asset tokenization in Part 3.

See next page

Part 3: The future of the dollar, imbalances, and capitalism

In this final section, we look at how the AI race and approach to tokenization could impact the dollar, and how it might shape the future of global imbalances and capitalism itself.

Tokenization meets TINA

US equities already account for more than 50% of global equity market capitalization, and US Treasuries for more than half of OECD sovereign debt. For many investors, this gives US assets "There is No Alternative" status, popularly known as TINA: investors have no alternative but to hold assets which are such a large part of the global market portfolio. The position of US companies in the AI race thus far has further contributed to TINA, with investors compelled to have exposure to technology which could change the world. The ultimate outcome of the AI race will no doubt play an enormous role in whether the US' share of the global market portfolio grows or shrinks: it is not just how earnings and valuation multiples evolve, but also how terminal values of the corporate sector could change, and how debt sustainability could be impacted.

TINA is already playing a big role in attracting capital to the US, but tokenization could *add* to this: by making US assets more *attractive* and more *accessible.*

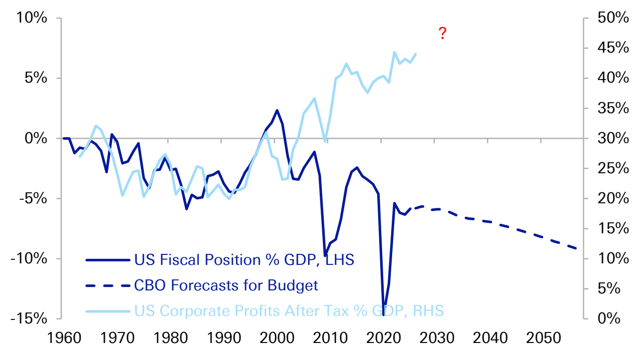
First, on attractiveness. The collateral implications of holding US assets on chain could be very attractive for investors. Fed Governor Cook noted in a speech that the most significant benefit of tokenization would be implications for capital and liquidity management for institutions. A tokenized US asset becomes more liquid and deployable, and capital and cash thus becomes more productive. In simple terms, you are more likely to hold a tokenized US equity or bond instead of cash, if you can easily repo your tokenized asset on chain when you need liquidity. If the value of US assets is going up, the cost of capital for US firms is going down. **The improved collateral velocity of tokenized assets could be another superpower for US assets, and another arrow in the quiver of US capital attractiveness.**

Second, on accessibility. We had argued that stablecoins could give anyone with an internet connection access to the dollar on chain. Asset tokenization could do the same for US assets**. In a world in which US equities and bonds are trading 24/7, with instant settlement and much lower costs and frictions, this could have dramatic knock-on effects on global behaviour, particularly for retail investors that could find access much easier.** We find it notable that NASDAQ openly acknowledged that "the investor base of public equities has the potential to expand as markets remain open across multiple time zones, making it increasingly important to have modern tools for public issuers to engage with investors worldwide." Tokenization need not also be limited to equities and bonds. One can imagine a world in which a person with even $1 in a place 10,000 miles away could own a tiny fraction of a tokenized Times Square on chain.

From an FX perspective, if tokenization succeeds in making US assets more attractive and accessible, this could help attract private capital flows. It is perhaps not surprising that the US is leveraging the power of technology – both AI and blockchain – to attract more foreign private capital at a time when US geopolitical leadership could be under pressure (see important examples here, here, and here) and crucially when US fiscal deficits have been widening. Both increase the need for alternative sources of private sector funding. **The US is arguably going "all-in" on private capital to offset public sector vulnerabilities.**

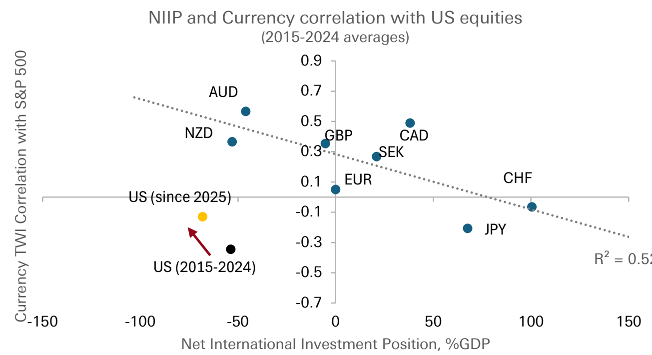
But lowering the hurdles for inbound capital, also exposes the US to capital going out. Tokenization would meaningfully increase capital mobility, in and out of US assets. The shift in US funding towards private sector, potentially shorter-term flows into equities – motivated largely by technology returns – will make the dollar riskier, more equity sensitive, and leveraged to the AI race. We have covered in earlier research how the dollar's historical properties of diversifying equity risk could be waning, with the dollar likely to become more positively correlated to risk. If for instance, AI proves uneconomical or the US starts to fall behind in the AI race, the dollar could be very exposed.

Figure 8: The US is increasing reliance on private capital - motivated by technology leadership - at a time of growing public sector vulnerability



Source : Deutsche Bank, Federal Reserve Bank of St Louis, CBO

Figure 9: The dollar is becoming a weaker diversifier for equity risk



Source : Deutsche Bank, Haver Analytics, Bloomberg Finance LP

Europe is on board, but China is taking a very different approach

How does the highly enabling regulatory environment towards tokenization in the US compare with the other major economic blocs, namely Europe and China?

Europe has been receptive to the opportunity presented by tokenization. Europe's DLT Pilot Regime introduced a regulatory sandbox for institutions to test the trading and settlement of tokenized securities on DLT as early as March 2023, even earlier than the US. ECB Governor Lagarde has noted that tokenization offers the prospect for Europe to "tackle the fragmentation that has held us back for decades." Note that Europe has over 30 central securities depositories across the union. An integrated market for tokenized assets could allow Europe to "leapfrog the fragmentation of existing legacy systems" (ECB). Europe has also embraced the role of tokenized securities as collateral, with DLT assets held in European CSD's eligible as collateral at Euroclear from March 2026. This puts digital assets on the same footing as traditional ones in Europe.

Where Europe sits apart from the US is perhaps in: the scale of its crypto ambitions, an explicit commitment to retaining a two-tier monetary system with central bank money as the final settlement asset, and the size of the AI economy itself. Europe is alive to the opportunities presented by tokenization for market unification – an important step to increasing Europe's capital market attractiveness, but the US arguably has more clearly stated ambitions to become "the crypto capital of the world". Moreover, Europe appears committed to keeping central bank money as the final settlement instrument between financial institutions even in a tokenized world, with initiatives like Pontes and Appia underlying this. In the US, it remains to be seen whether private stablecoins become the primary instrument of settlement, disintermediating central bank money, or whether the existing system of bank deposits and reserves is tokenized. Ultimately, the biggest divide between Europe and the US is in the AI economy itself – Europe has a much smaller footprint in data centres and independent AI labs, and thus may not be where the appetite to raise or deploy capital – even in a tokenized asset world – is most compelling.

Meanwhile, China is taking a very different approach altogether; while the US is embracing stablecoins and RWA tokenization, China is tightly controlling – if not restricting – them. In February 2026, the PBOC and CSRC along with eight other national organization issued a notice that updated China's 2021 ban on crypto-related activities. The notice defined RWA tokenization and tightened regulation around it – noting that such activity or intermediation of it was prohibited in China, except with regulatory approval on specific financial infrastructure (Xinhua). The notice also banned entities from issuing stablecoins pegged to RMB abroad, without regulatory approval. We note that Chinese regulators prevented Ant Group and JD.Com from proceeding with plans for yuan stablecoins in Hong Kong in October 2025. Instead, in January this year, PBOC began to allow payment of interest on e-CNY, retail CBDC deposits suggesting relative encouragement for controlled forms of central bank digital money. We look further at some of the drivers of this very divergent approach between the US and China.

Open markets versus open technology

The US is pursuing a hyper-capitalist approach in the AI race: throwing open the doors for private capital to fund large capex, and using the power of asset tokenization to supercharge it. We call this the open markets approach. Under the US model of shareholder capitalism, US companies prioritize profit maximization, keep AI model weights closed to allow companies to control pricing via subscriptions, and offer the carrot of return participation to shareholders to attract in global capital. The US then uses those huge sums of capital to make large capital investments to innovate and stay ahead of the competition. Importantly, the open markets approach is paired with a closed technology one.

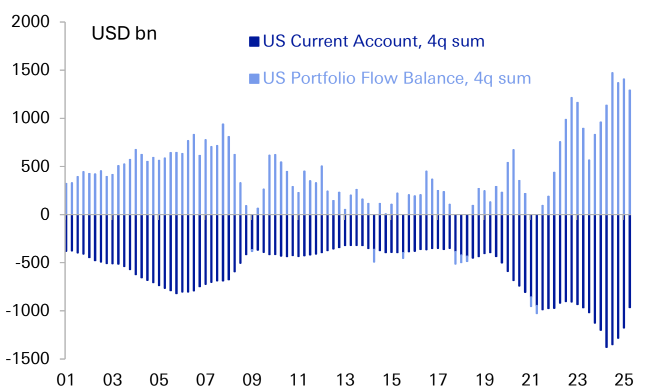
China is taking a very different approach in the AI race, pushing out lower-cost models with open-weights into the world. We call this the open technology approach, and it is paired with a closed markets one. Chinese companies are not only achieving similar model results, and charging clients less for tokens, they are offering their models with open-weights. Open weights allows users to download and customize models on their own servers, promoting global proliferation and usage among corporates. See more from the DB Research Institute on this here. China may be counting on the same penetration of its AI models around the world as it has achieved with manufactured goods. One can perhaps start to look at China's approach to AI as an extension of China's broader involution approach to manufacturing where state-based alignment combined with fierce domestic competition achieves low pricing, improving quality, rising market share and ultimately industrial dominance. That China is now trying this in the technology space, where the US has thus far remained a leader, bears close watching. The US' open markets are meeting China's open weights.

In sum, the US is taking an "open markets-closed technology" approach to the AI race, while China is taking an "open technology-closed markets" approach. This divergence is not just a reflection of different economic models, but underlying global imbalances, and very different needs for global funding.

A test for global imbalances and capitalism itself

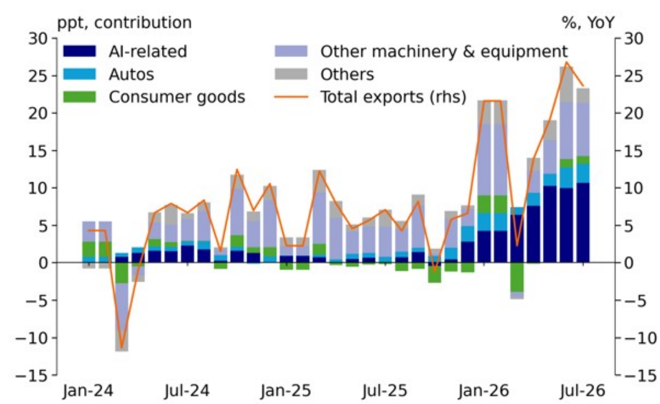
The US' open embrace of tokenization, compared to China's more closed approach to foreign capital is a reflection of underlying imbalances. The US' current account deficit combined with a fiscal deficit means the US requires foreign financing to fund incremental private capital expenditure, in the absence of domestic fiscal savings. In comparison, China is running a large and growing current account surplus, reflective of excess savings. China's trade surplus is in fact growing – with the AI cycle boosting AI-related exports. China does not need to turn to foreign capital because it can fund capital expenditures from abundant current account savings. This affords China the ability to maintain a more managed capital account, and explains the relatively restrictive approach to the high-speed blockchain rails that the US is so actively building.

Figure 10: The US is dependent on foreign capital flows to fund its deficits



Source : Deutsche Bank, Haver Analytics

Figure 11: Chinese exports are growing even faster as a result of AI



Source : Deutsche Bank, CEIC
Note: See "China Macro: Trade Imbalance and Rates Divergence" (31 Aug 2026)

The AI race could further reshape the state of global imbalances. As one potential path, imbalances could become more extreme. If China leverages AI to make industry even more efficient, adopting it for production and industrial automation, this could boost China's manufacturing dominance and good-driven surplus. One can imagine an alternative scenario in which the US current account improves, narrowing global imbalances. We have written in earlier research how AI revenue streams could help the US current account by around 1% over the coming decade via higher services exports. But this is crucially dependent on US AI labs retaining pricing power with private subscribers, and monetizing an international subscriber base. But China's low-cost, high-proliferation approach could cap the excess profits of US AI companies, and their ability to sell intellectual property for profit. How things evolve may come down to whether US and Chinese models become complements that coexist and serve different market segments, or whether they become substitutes in a winner-takes-all outcome. There is a lot at stake.

Ultimately, the AI race between US and China is more than a battle between AI models. It is a battle between *economic* models. The US is doubling down on capitalism: operating on the premise that increasing capital market openness will pull in the funding for large AI capital expenditures; in turn, this will help the US remain dominant in AI technology, allow corporates to maintain pricing power, and for the US to pull ahead in the race. Meanwhile, China does not need to leverage open markets to attract capital flows. China's industrial dominance and record trade surpluses mean it has ample and cheap sources of domestic funding. This allows China to keep a more closed market approach that need not prioritize profits firsthand, and allows it to compete with low-cost, open-weight AI offerings. If China's approach starts to undercut US profits, this could threaten the US model of shareholder capitalism, which has become critical not only to attracting funding for AI, but for growing US deficits. China's challenge is not only about technology, but the central maxim of US power: that free and open capital markets always deliver the winning innovation. The AI race is a big test for capitalism itself.

Appendix 1

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Deutsche Bank
Research Institute

The good, the bad and the ugly

September 03, 2026

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- As the summer holidays come to an end, we take a look at the three questions that are likely to occupy our time in the months ahead: Is the euro area economy proving resilient to the energy shock? Will the ECB stop at 2.50% or continue hiking until policy is restrictive? And will public finances and elections prove a destabilising combination?
- The European economy is proving remarkably resilient, with some elements of positive demand shock offsetting the negative energy supply shock. Expectations are improving, but some caution is still needed: the energy shock is not over and the AI trade deficit is widening.
- With respect to the ECB, a rapid normalisation of energy prices may be needed to stop the hiking cycle at 2.50%. That said, the inflation outlook is not yet consistent with an outright restrictive stance either.
- Budget season is about to begin, debt dynamics are vulnerable and elections imply political risk and market volatility. The degree of tightening of financial conditions will be a function of political choices, the resilience (or not) of growth and the restrictiveness (or not) of ECB policy.

Q1: Is the euro area economy proving resilient to the energy shock?

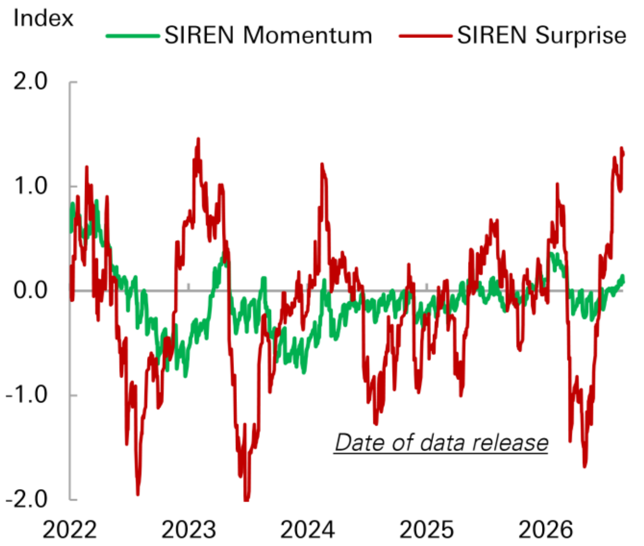
Strictly speaking, as a net energy importer the euro area should be highly exposed to the negative energy supply shock triggered by the Iran war. In any case, HICP inflation jumped to 3.0% in Q2, 1.2pp higher than expected at the start of the year. The household real income shock alone should have knocked more than 0.5pp off GDP.

And yet euro area GDP was not weak in Q2, it was surprisingly strong. At 0.44% qoq, GDP growth exceeded expectations. We don't have detailed components yet, but on the basis of available data household consumption grew despite the inflation shock.

It is possible that elevated household savings buffered the real income shock on the assumption that it would be fleeting. There was also a targeted and temporary easing of fiscal policy (lower energy taxes), but this was not substantial and only moderated the rise in inflation slightly. That said, the euro area fiscal stance was already easy for 2026. Not nearly as easy as 2022, and skewed heavily towards Germany. But the stance is easy, not tight.

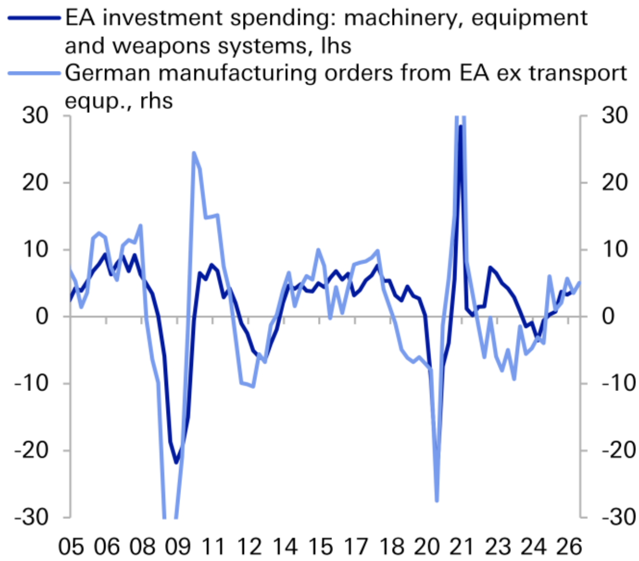
Most encouraging are the signals of rising investment spending: the BLS points to demand for loans for fixed asset investment, the flow of long-term loans to corporates has been a little volatile but hints at investment spending, and the rise in German industrial orders from elsewhere in the euro area has historically been consistent with a euro area investment cycle too.

Figure 1: Euro area activity data showing strong surprises and building momentum



Source: Bloomberg Finance LP, Deutsche Bank Research

Figure 2: Encouraging signals of a capex cycle building



Source: Eurostat, German Federal Statistics Office, Haver Analytics LP, Deutsche Bank Research

We need to be careful. We might be seeing the lagged transmission of the earlier ECB rate cuts. Private sector balance sheets are healthy, which would aid transmission. Maybe structural demand from AI investment and defence spending is offsetting higher energy costs and geopolitical uncertainty.

In short, maybe what Europe is facing is a positive demand story, not just a negative supply story. Having quickly dropped to 48.8 in April, the composite PMI rose to 52.1 in August and might have been higher were it not for heat-related headwinds in services.

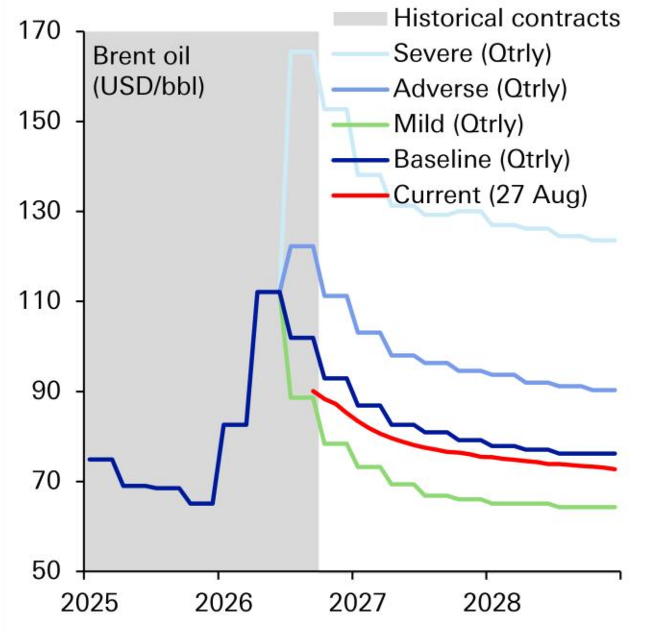
That said, there are reasons to be cautious about growth.

The energy shock is not over. Gas prices have reached a new high for the year, raising the prospect of an electricity-based production cost shock to add to the existing oil-based transport cost shock.

The energy shock might also be masked by temporary factors like stock-building. The shock in the Middle East went beyond energy into chemicals and metals. There was a threat to global supply chains. Fear of disruption might have triggered stock-building, which might explain why manufacturing has held up.

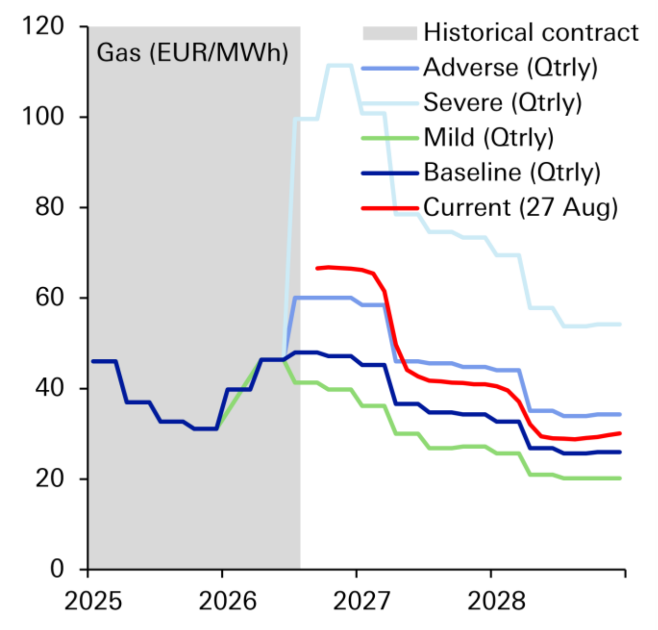
There is no evidence of domestic inventory building. But Europe might have benefited from temporary stock-building in its export partners. This might help explain export resilience.

Figure 3: Oil prices vs. ECB baseline and alternative scenarios



Source: Bloomberg Finance LP, Deutsche Bank Research

Figure 4: Gas prices vs ECB baseline and alternative scenarios



Source: Bloomberg Finance LP, Deutsche Bank Research

AI investment spending should keep rising. But the AI spending boom might be less beneficial for GDP than it appears, however, as Europe's AI-related trade deficit is widening as both AI-related exports weaken and AI-related imports grow.

Monetary policy is tightening too, which will restrain growth with a lag (see our latest review of upstream and downstream financial conditions here). Concerns about sovereign debt sustainability could generate additional growth-sapping risk premia.

Finally, competitiveness remains weak and the Second China Shock and increasingly direct competition with China could weaken European production and employment.

The bottom line is the economy has been remarkably resilient to the energy shock so far and our euro area GDP forecasts are probably too low at 0.5% in 2026 and 1.1% in 2027 (our German team have raised their 2026 forecast from 0.5% to closer to 1.0%).

Q2: Will the ECB stop at 2.50% or continue hiking until policy is restrictive?

Since early March we have been expecting the ECB to hike twice (Jun and Sep), raising the policy rate to 2.50%, the upper end of the range of neutral. This felt like good risk management. It signals the ECB commitment to price stability on the one hand without overly burdening economic growth in the context of a negative supply shock on the other.

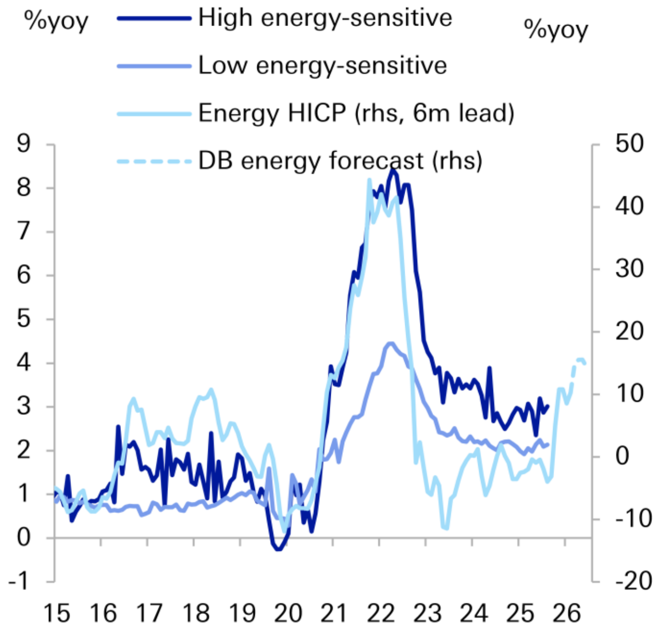
If growth was weakening amid a supply shock, inflation expectations were comprehensively anchored and data showed no risk of energy price pass-through, it would be easier to call into question the ECB's decision to hike rather than "look through" this energy supply shock.

In the round, the data, the ECB messaging (even this week Isabel Schnabel said the market "very well" understands the ECB reaction function, repeating what President Lagarde said in the July press conference) and the lack of resolution in the Middle East and still elevated energy prices mean a hike to 2.50% in September seems all but certain. Indeed, this is the impression from the July ECB Accounts. The question is, will the ECB stop there?

There are two scenarios in which the ECB hikes to 2.75%.

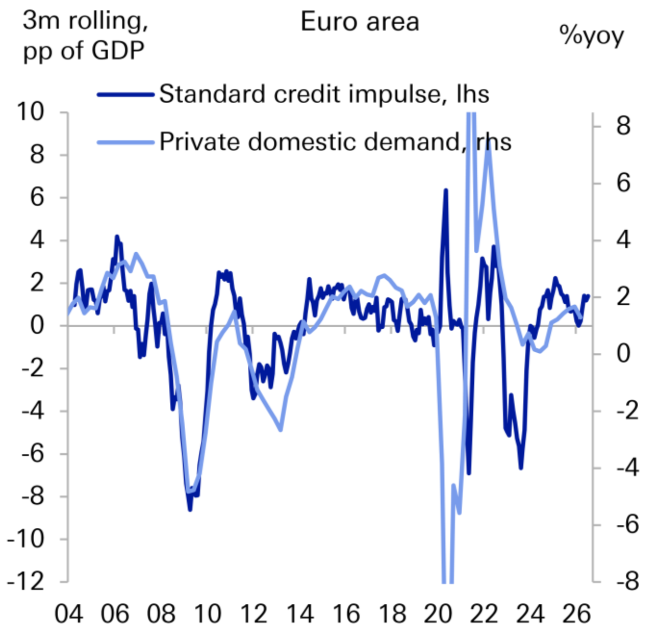
In one scenario, the ECB concludes that the upper end of the range of neutral is 2.75% rather than 2.50%. The persistent strength of credit growth in recent months despite higher lending rates and despite banks saying that lending standards are tightening might reveal that neutral is higher than expected. The scale of fiscal easing, even if led by Germany, could push r* higher. If this is right, the hurdle to raising policy rates to 2.75% might be lower than assumed and might not need strong or clear evidence of indirect or second-round inflation.

Figure 5: Maybe it's too soon to expect indirect energy effect to show up



Source: Eurostat, Haver Analytics LP, Deutsche Bank Research

Figure 6: The credit impulse is consistent with a robust domestic economy



Source: ECB, Eurostat, Haver Analytics LP, Deutsche Bank Research

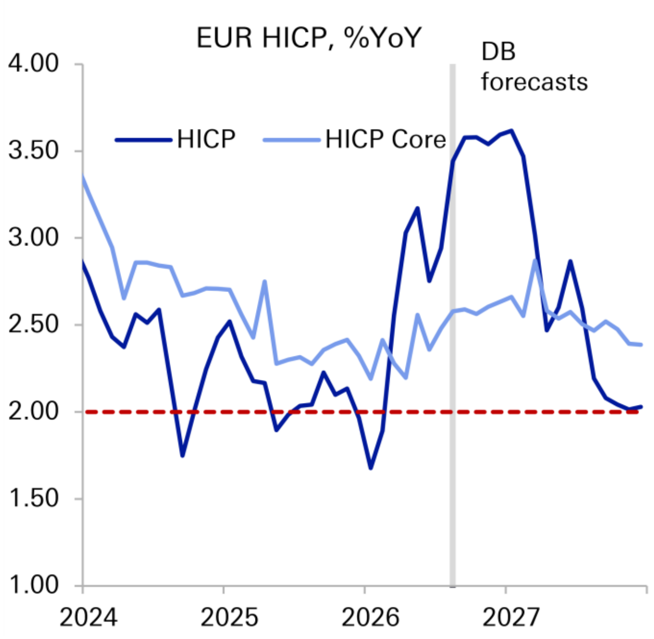
The argument is not one-sided. The strength of credit growth might reflect the lagged transmission of the earlier cuts rather than a higher neutral rate. In any case, our composite indicator of restrictiveness, based on a range of data that might reveal neutral, says that policy was still mildly restrictive in mid-2026 when the ECB started to hike. If that's right, it might be more difficult to argue 2.75% is still within the range of neutral.

The other scenario for getting to 2.75% (or higher) is the ECB decides that a restrictive policy stance is appropriate. In the ECB's current framework, this means moving from a "measured" tightening to a "forceful" tightening. This requires the ECB to believe that the inflation shock is moving from a 'large but not too persistent' shock to a 'more significant and sustained overshoot of the target'.

There are different routes to believing in a more significant and sustained overshoot of the inflation target.

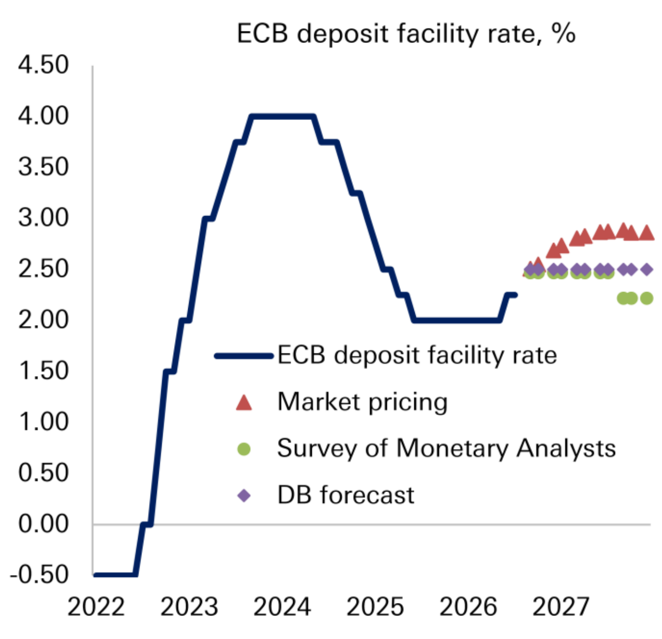
One would be if energy prices remain at elevated levels for longer than expected. Oil prices are in line with baseline, but gas prices are more consistent with the adverse scenario (at least in the near term). Given the lagged transmission of rising gas prices into inflation, this could add persistence to the headline HICP shock, lengthening the period during which energy inflation could spillover into core.

Figure 7: Headline and core HICP: DB forecasts



Source: Eurostat, Haver Analytics LP, Deutsche Bank Research

Figure 8: ECB policy rates: Market priced hikes to 2.75-3.00%



Source: ECB, Bloomberg Finance LP, Deutsche Bank Research

Another way would be if larger than expected indirect effects and in particular second-round effects materialised. So far, there is no compelling evidence of indirect effects and no evidence of second-round effects. However, both would only materialise with a lag. For example, it was six months before high energy-sensitive core HICP started rising in the 2022 shock.

A third way would be if stronger growth added persistence risk to inflation. The recent resilience of growth would add to this risk if it lasts. That said, if AI lifts productivity, the additional growth might not be as inflationary.

How the ECB staff forecasts change in September will be very important to the outlook for monetary policy. In June, the baseline forecast had core HICP averaging 2.5% in 2027 and staying at least 30bp above target through the year. Importantly, however, headline HICP was expected to be back down to the 2% target from Q3-27 and stay there through to the end of 2028.

It is important to remember the baseline staff forecast in June was conditioned on policy rates rising to 2.75%. But despite sticky core inflation, it is difficult to imagine monetary policy tightening much further if headline inflation is sustainably returning to target within a year.

As such, the near-term path for policy could depend critically on whether the Governing Council believes core inflation will converge down to headline in the medium term or headline converges up to core. The former is less hawkish, the latter is more hawkish.

Which of these two outcomes seems more or less likely to the Governing Council will depend on how it judges the risk of inflation persistence. This is a function of pricing power, bargaining power and purchasing power. Our view is there is some risk of persistence, but not as much as 2022 and not enough to warrant an outright restrictive policy stance. Not yet, at least.

Our 2.50% baseline might be too optimistic, but 3.00% feels too restrictive. 2.75% is a grey zone: a non-negligible possibility unless the situation in the Middle East is resolved quickly and energy prices normalise rapidly. Our work on policy rules shows that the ECB 'baseline' scenario macro forecasts are consistent with 2.75%. The 'milder' scenario forecasts are consistent with 2.50%.

There may not be any new policy signals from the ECB at the Jackson Hole symposium. Unlike in the past, there is no global policymaker panel. Isabel Schnabel is the ECB member on the official agenda – speaking about tokenized finance. However, markets will be on the watch for press interviews by ECB attendees.

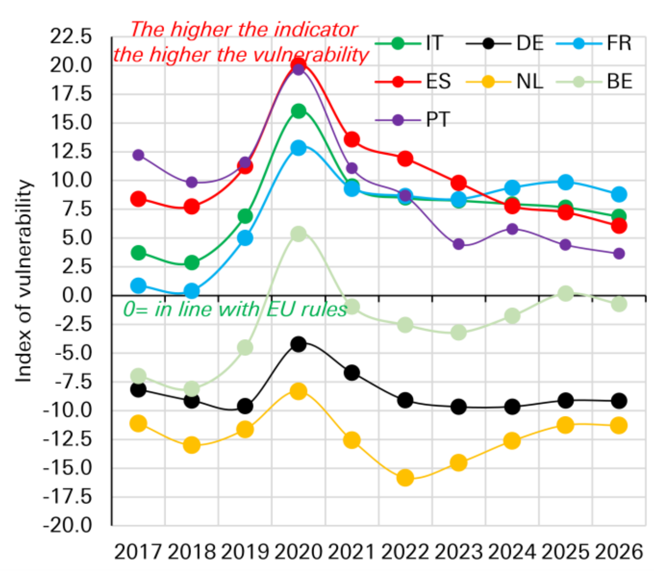
Q3: Will public finances and elections prove a destabilising combination?

The environment around European public finances is becoming more challenging for a number of reasons. Government bond markets are under pressure globally, driven currently by the US. The 'i minus g' gap at the heart of debt dynamics is becoming less favourable as the benefits of the low rates regime of the last decade expire, amplified by rising inflation risk, tighter monetary policy, rising term premia and generally low rates of economic growth.

At the same time, public debt to GDP ratios have not returned to pre-pandemic levels, fiscal vulnerability is elevated even when accounting for improved external balances, and European governments are facing numerous spending demands, from defence and energy to climate change and technology.

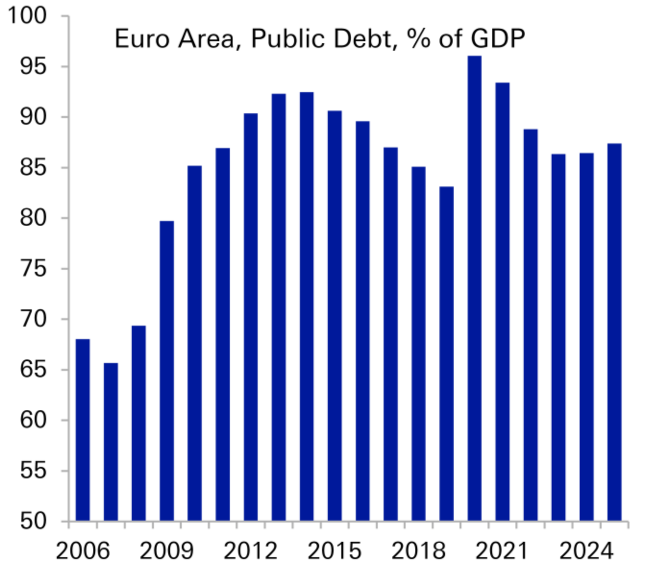
The combination of more vulnerable public finances and elections in several significant member states over the next 12 months creates the potential for tighter financial conditions. National elections lie ahead in Italy, Spain, Greece, Finland, Estonia, Poland and, perhaps most consequentially, in France. Financial conditions are well-behaved so far (the ECB's state-of-the-art Macro-Finance FCI has hardly tightened). The degree of tightening of financial conditions going forward will be a function of the fiscal policy choices made by governments, the resilience (or not) of growth and the restrictiveness (or not) of ECB policy.

Figure 9: DB fiscal vulnerability indicator incl. net external position – France most vulnerable



Source: European Commission, Deutsche Bank Research

Figure 10: Public debt remains high and above pre-pandemic levels



Source: European Commission, Deutsche Bank Research

With regard to France, the opinion polls say that Rassemblement National (RN) has a credible path to winning the Presidency with Marine Le Pen. If that happens, there is a good chance an early legislative election will be called too.

OAT markets have been on edge since President Macron's ill-judged legislative election in 2024 resulted in a deeply divided parliament with little capacity to drive strong fiscal consolidation. On the basis of what is known at the moment, there is a lack of a credible consolidation trajectory in the RN fiscal plans. Even under moderately optimistic assumptions about consolidation, France's debt ratio will likely remain broadly flat in the range of 120-125% of GDP over the next decade. A flat trajectory means vulnerability to any new negative surprises on growth, funding costs and/or deficits – and as we have said before, the deficit is under pressure from inflation (indexed spending), legacy policy decisions (stalling the pension reform), defence, ageing, etc.

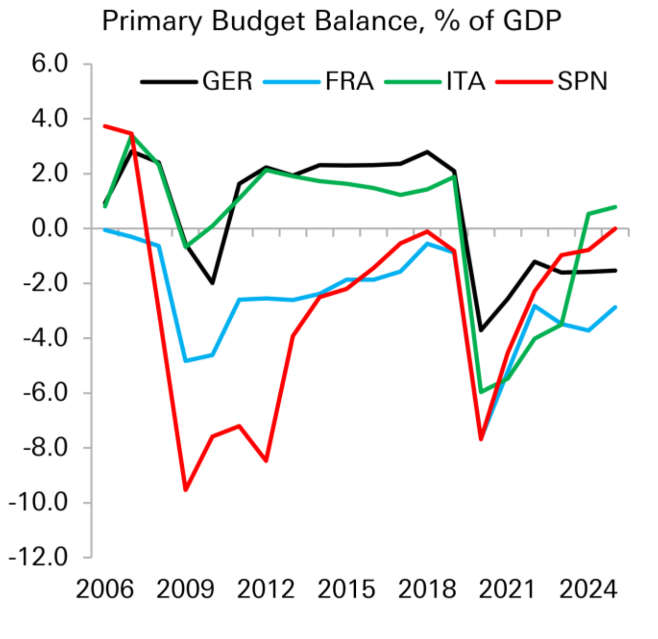
Much will depend on the course of French fiscal policy, particularly RN's fiscal policy. This could be revealed in three distinct stages over the year ahead.

- **The 2027 budget**. The incumbent Lecornu government needs to have a budget agreed by year-end. A deeply divided parliament and approaching election will make strong consolidation difficult. Importantly, however, RN is signaling that it won't oppose a budget. The alternative ("special law") would mean a less stable financial position in 2027. RN would prefer to have some breathing space if it is elected.
- **The election campaign**. This will reveal more about RN's fiscal plans. The limited details available so far imply a wider deficit. However, RN's fiscal

identity is unclear and RN President Bardella's admiration for Italy's market-friendly fiscal stance under PM Meloni suggests an understanding of the benefits of a stable bond market. The extent of the RN's intention to challenge Brussels will be important too. This could influence market confidence in the availability of European backstops (ECB TPI, ESM, etc).

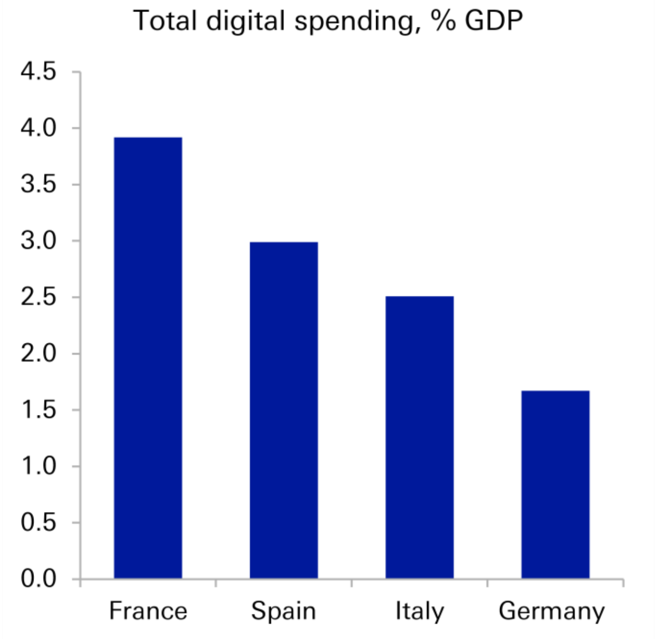
- **Post-election decisions**. Assuming an RN President would dissolve parliament in the spring to trigger legislative elections by the summer, the first task for the new parliament would be the preparation of the 2028 budget. This assumes a budget for 2027 has already been passed. If France is operating under a special law, a 2027 budget would take immediate precedence. These would reveal RN's fiscal plans and their consistency with EU rules.

Figure 11: France has the largest primary deficit among the Big4…



Source: European Commission, Deutsche Bank Research

Figure 12: …but France might have better growth prospects, helped by AI (digital investment spending)



Source: Eurostat, Deutsche Bank Research

Counterbalances. France will be the focal point for the market. As we have argued (here, here and here), the key theme for investors is not outright regime change, but a constant, volatility-inducing push and pull between RN's political objectives – shifting in places – and the fiscal, legal, and market constraints that would ultimately shape its capacity to govern. The push for political change will generate friction.

There are several buffers that could help keep the country off the more challenging paths.

- **The market is already pricing weak fundamentals**. French yields have been gradually adjusting to these weak fundamentals over the last couple of years. For example, the current yield already anticipated further credit downgrades.

- **Robust banking**. French banks are well-capitalised and unlike the crisis-hit periphery a decade and a half ago hold relatively little direct exposure to OATs. As a global safe asset, OATs are widely held abroad.

- **Backstops**. Unlike the sovereign debt crisis 15 years ago, there are backstops in place. For example, the ESM can lend to crisis-hit member states and the ECB can intervene in the market for eligible member states with TPI. If foreign investors offload OATs in scale, the French banks have some capacity. However, as bank exposure increases, the potential for a more destabilising sovereign/banking doom-loop would increase.

- **Economic growth.** There are several reasons why French economic growth could outperform its peers and help stabilise debt. One is AI. Among the larger member states, France is spending the most on digital investment and was the only EU member state to launch a notable AI model in 2025 (50 in the US). France is also relatively more exposed to defence production, and among the larger EU member states has relatively favourable demographics.

Appendix 1

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The Great Rebalancing (4/5): The connectivity chokepoint

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- Digital connectivity—the transmission of data through terrestrial fibre, mobile networks, subsea cables and satellites—has become a critical arena of geopolitical competition.
- The Internet's physical infrastructure is increasingly viewed as a source of strategic leverage, resilience and vulnerability.
- In the era of "infrastructure realism," the contest increasingly involves both states and hyperscalers such as Google, Meta, Microsoft and Amazon, whose AI and cloud ambitions are reshaping global connectivity infrastructure.

Digital connectivity has become a central arena of great-power competition. While states have long used communications infrastructure to project influence and control, in the age of AI, the contest to dominate digital networks has become a defining geopolitical frontier as countries and – increasingly – private actors look to win the AI race.[1]

This chapter in our Infrastructure Realism series examines the physical systems that enable digital connectivity: undersea cables and communications satellites. We assess their role on the Internet—the largest and most consequential connectivity network—and the strategic advantages and vulnerabilities they create for state actors.

Subsea cables are indispensable to the global economy. They carry more than 99% of intercontinental data traffic, making sabotage or prolonged disruption a direct threat to global Internet connectivity.[2] Satellites carry only a small share of international data, but they are crucial for last-mile access where fiber cannot reach, particularly for governments and defense users. As we go on to explore, the rise of communications-based satellites is also becoming integral to the global AI race.

Several incidents have underscored the criticality of connectivity infrastructure. In April, reports of Russian surveillance near UK cables and pipelines raised concerns.[3] In May, amid conflict in the Middle East, Iran threatened to sever undersea cables in the Red Sea. Iran's IRGC, also floated the idea, through state-affiliated media, of charging foreign operators infrastructure fees and licensing royalties for routes through Iranian territorial waters.[4] This would have significant implications for hyperscalers such as Meta, Google and Microsoft, which rely on these corridors for portions of their international traffic and cloud connectivity.

[1] The first instance of communications sabotage was during WW1, when the British cut Germany's undersea telegraph cables in the English Channel, forcing German telegraph traffic onto British-controlled networks that could be intercepted. And in the 21st century, autocratic states like Iran and Russia have imposed internet shutdowns to control their population / information access.

[2] The 99% does not account for total global internet traffic, given the vast majority of internet traffic is domestic and does not cross an ocean at all. Instead, the 99% applies to internet traffic that goes between continents. Mauldin, Alan. 2023. 'Do Submarine Cables Account for Over 99% of Intercontinental Data Traffic?'. In *Telegeography.Com*.

[3] Olivia Ireland, Thomas Copeland, and Frank Gardner, "UK Says Russia Ran Submarine Operation over Cables and Pipelines," BBC News, April 9, 2026, https://www.bbc.co.uk/news/articles/cre13qn9z7do.

[4] "IRGC Media Float Plan to Levy Fees for Subsea Internet Cables," Amwaj. Media, 2020, https://amwaj.media/en/media-monitor/irgc-media-floats-plan-to-levy-fees-for-subsea-internet-cables.

Section 1 explains how the Internet works and traces the physical journey of data, highlighting the points at which state disruption can occur. We give less emphasis to 5G, due to our definition of what constitutes the Internet's infrastructure. This is because while states—most notably China—have used 5G equipment to project influence, 5G is primarily an access network: it links mobile devices to the wider Internet but does not replace the role of subsea cables and satellites.

Sections 2 and 3 analyse subsea cables and communications satellites, detailing the strategic advantages and vulnerabilities of each ecosystem and how states are seeking to consolidate their influence. Finally, Section 4 examines the rise of private actors such as Google and Meta, which are reshaping global connectivity by building their own data corridors and controlling more of the Internet stack to meet rising demand.

1. How connectivity works

1.1 What is the Internet?

The Internet enables global communication, data exchange and access to information through a system of cables, satellites and wireless technologies. It is not a single entity but a common set of protocols that allows independently owned networks to communicate. In this sense, it is a "network of networks" through which data moves across billions of devices.

To move across the Internet, data is broken into small "packets" and routed across different systems using standard protocols. The primary ones are the Transmission Control Protocol (TCP), which ensures the reliability of the connection and data, and the Internet Protocol (IP), which acts as an addressing system to find and route packets to their destination.[5] Using these protocols, data travels through a global web of fiber-optic cables that connect major network hubs, forming what is known as the Internet Backbone. The process is as follows:

- A packet leaves the user's device and enters a local access network.[6] Over fixed broadband, the data packet passes through a Wi-Fi router and then to an Internet service provider (ISP). Over mobile broadband, 5G carries it to a radio base station and the mobile operator's core network before it reaches the wider Internet.
- From the ISP, the packet is passed from one junction (or router) to the next, crossing multiple networks toward its destination server.
- If the destination server is nearby, the packet may be exchanged at a local Internet Exchange Point (IXP). If it is distant, the ISP routes it onto the backbone, across a country or ocean, and into an ISP at the other end.
- The response takes a similar path back, with each router along the way making forwarding decisions based on the destination IP address.

How one network gets its traffic to the rest of the Internet, and whether it must pay for it, is where key corporations come in. There are two ways networks connect: 1) through transit, where a smaller network pays a larger one to carry its data across the globe, and 2) through peering, where two networks of similar size agree to exchange traffic directly at no cost. Within this system, a hierarchy of companies has emerged.

At the top are Tier 1 network operators, which can reach the entire Internet through settlement-free peering. This small group includes major telecom operators such as Lumen, AT&T, Tata Communications, NTT, Arelion, GTT and Cogent.

Below these are Tier 2 network operators, such as regional carriers and internet service providers like Comcast and Vodafone. These entities primarily purchase transit from Tier 1 providers to reach parts of the internet they cannot access via peering. Finally, Tier 3 network operators rely exclusively on paying for transit from Tier 2s to connect their customers. These are typically the local ISPs that provide Internet service to homes or offices. Therefore, a data packet from a user's laptop typically travels from a Tier 3 provider to a Tier 2, onto a Tier 1 backbone, across an undersea cable to another Tier 1 network, and then down through the tiers to its destination server.

1.2 How data physically travels

The Internet ultimately depends on physical infrastructure. As we said in step 1 above, the first leg is not always a cable, with mobile technologies like 5G carrying data wirelessly between the user's device and a nearby base station. Satellite systems can also provide a

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[5] Rus Shuler, "How Does the Internet Work?," Stanford.Edu, 2002, https://web.stanford.edu/class/msande91si/www-spr04/readings/week1/InternetWhitepaper.htm.

[6] Access networks connect the user's device or premises to the first point in an Internet or telecom provider's network. They can be wired or wireless, and in addition to 5G and Wi-FI mentioned above, other examples are fiber-to-home, cable broadband, and satellite broadband.

wireless connection over much longer distances. Beyond that access layer, however, both systems typically rely on terrestrial fibre and the wider Internet backbone. Most long-distance traffic travels as pulses of light through fibre-optic cables buried on land or laid across the seabed, and each glass strand is thinner than a human hair.

Terrestrial cables handle domestic and regional internet traffic, while undersea fiber-optic cables carry intercontinental internet and telecom traffic. When a user sends data across the globe, it travels through local terrestrial cables to a coastal landing station. This station then transmits the data across the ocean via a subsea cable to a corresponding landing station on another continent, which receives the data and sends it through local terrestrial cables to the destination server.

Satellites complement cables, providing connectivity in remote or disrupted environments where fiber-optic cables cannot reach. There are two different types: traditional satellites, which sit in a fixed, high-altitude orbit, offering broad coverage but with a noticeable delay due to the long signal path. A more recent approach, exemplified by services like Starlink, uses large constellations of thousands of satellites flying much lower to the earth to reduce latency and improve service.

1.3 How the Internet transitioned from a public good to a private advantage

Similar to crypto's arc discussed in this series (here), the Internet came to symbolise decentralisation during the globalisation and commercialisation of the 1990s and 2000s. Its ability to reroute traffic reinforced the perception that it was neutral and disintermediated. Yet the network depends on physical assets concentrated among a limited number of carriers—and increasingly hyperscalers—that control key digital corridors and possess extensive visibility over network traffic.

Conceived as a "network of networks," the Internet began as a US government initiative in the 1960s to facilitate information sharing among research universities.[7] Its predecessor, ARPANET (Advanced Research Projects Agency Network), became the first operational "packet-switching network" in 1969. Famously, its first attempted message – "LOGIN" – crashed the system after only the letters "LO" appeared, an event now widely regarded as the moment the Internet was born. Over the following decades, public researchers cultivated the Internet ecosystem by developing concepts like email, data packet switching, the World Wide Web service (WWW), and the protocols needed to create the Internet ecosystem, all with the goal of keeping the Internet open for all.

By the 1990s, the emergence of major internet exchange providers and the release of WWW software into the public domain enabled millions of people to use the Internet freely.[8] As private interest in commercializing the Internet grew, the US Congress passed legislation in 1993 that opened access to Internet applications like computing and networking to private enterprise, which had previously only been available to universities and government agencies. This led to thousands of firms entering the market to pursue commercial ventures using TCP/IP technologies.

By 1995, the government agency responsible for overseeing the Internet, NSFNET, was decommissioned and replaced by four privately operated Network Access Points in Washington, New York, Chicago, and California. This marked the Internet's transition from a government-funded academic experiment to a fully private enterprise. The dot-com bubble in the late 1990s and eventual collapse of Internet stocks in the 2000s cleared the way for the companies that today define the modern Internet, including Amazon (1994), eBay (1995), Google (1998) and Facebook (2004). The more recent roll-out of broadband,

[7] The Department of Defense created ARPA (the Advanced Research Projects Agency) in 1958 in an attempt to lead technological breakthroughs for the military. It was replaced by NSFNET as the main US research network for the Internet in the 1980s.
[8] The Internet is the global infrastructure of interconnected computers and networks, while the WWW is a service running on that infrastructure, using web browsers to access linked hypertext documents.

fiber-to-the-home, and 3G/4G/5G cellular networks has also helped make the Internet ubiquitous and increasingly mobile.

1.4 Why a decentralized system still creates concentration

No single entity owns the Internet, but a small number of companies and countries control the infrastructure on which it depends on. As a result, the Internet's ecosystem remains highly concentrated across few backbone carriers and transit providers, telecom-equipment suppliers, cable landing stations, and internet exchange points to name a few, thereby creating numerous chokepoints that different countries have raced to ensure a foothold in.[9]

Beyond telecom equipment, the physical layer of digital connectivity is also concentrated. Subsea-cable ownership spans telecom operators, consortia, state-backed entities and technology companies, yet control over capacity, construction, strategic routes and landing points is narrowing. Cable ownership is distinct from Tier 1 backbone status, but the boundary is blurred because operators such as AT&T and Tata also own subsea infrastructure or capacity.

Today, this landscape is also shifting as large content providers have become the dominant users of international bandwidth and increasingly fund, co-own or purchase large allocations of new subsea cable capacity. As we will explain more in section 4, this is flattening the traditional hierarchy, since under these private backbones, hyperscalers are reducing their reliance on traditional Tier 1 transit hierarchy by peering directly with consumer ISPs and even placing their own servers inside those ISP networks.

Figure 1: Notable chokepoints in the connectivity corridor

	Selected key players
Subsea cables (suppliers)	SubCom, Nippon Electric Company (NEC), HMN Technologies, Alcatel Submarine Networks (ASN)
Subsea cables (owners)	Hyperscalers (Google, Meta, Microsoft, Amazon) and Telecom carriers (Orange, Tata, Vodafone, Telstra, China Telecom, China Mobile)
Tier 1 Backbone Operators	Lumen, Arelion, AT&T, Tata Communications, Orange, NTT, GTT and Cogent.
Satellite LEO Constellations	Starlink, OneWeb/Eutelsat, Amazon Kuiper, China SatNet
Telecom Equipment / 5G	Huawei, Ericsson, Nokia, ZTE
Data centres / cloud	AWS, Microsoft Azure, Google Cloud, Alibaba Cloud, Huawei Cloud, Tencent Cloud, Baidu AI Cloud, Oracle Cloud

Source: Deutsche Bank Research, CAIDA.

2. Undersea cables: The invisible backbone of the internet

2.1 How undersea cables work

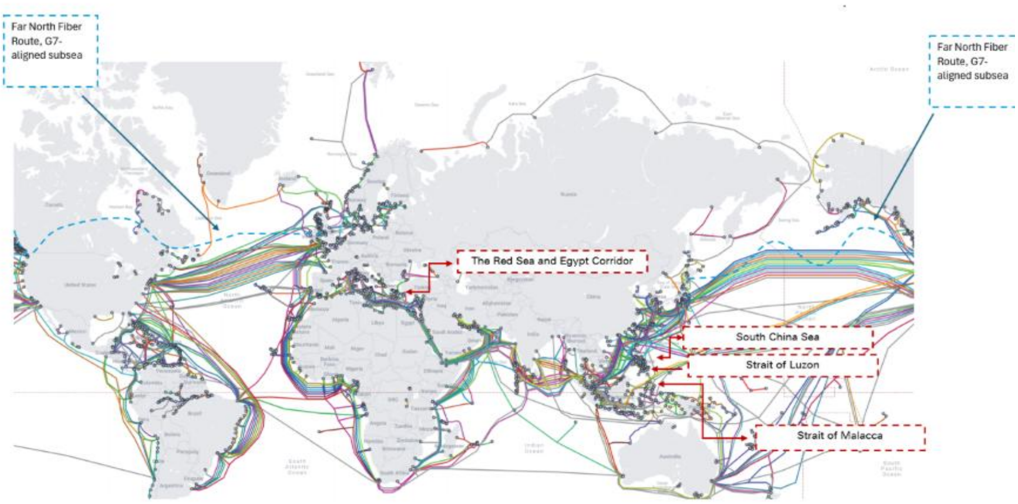
Subsea cables are long, fiber-optic conduits laid on the seabed that, in total, stretch for more than 1.8 million kilometers. There are approximately 500 active submarine cables around the world, and while many are barely thicker than a garden hose, yet they carry over 99% of all intercontinental data traffic. Their dominance is due to their high capacity, low latency and reliability, which make them ideal for transmitting vast volumes of data and information quickly. While most of the world's cables are laid by just four companies – SubCom (US), Alcatel Submarine Networks (France), Nippon Electric Company (Japan),

[9] The geopolitical case of China's Huawei is one example. As the company that provided many of the physical and software components needed for 5G, Huawei rapidly ascended to become a significant part of the Internet's underlying infrastructure, and thus a critical player in global digital connectivity. This later proved to be a problem during accelerating tensions between China and Western countries, as many nations sought to disentangle themselves from Huawei's supply chains in their telecom networks.

and HMN Technologies (formerly Huawei Marine Networks; China) – it is US hyperscalers who now finance and control most of the new cable capacity.[10]

Today, subsea cables are the foundation of the global telecommunications industry because they transfer data and voice traffic more cheaply and quickly than any alternative. Yet for much of the 20th century, the bulk of transoceanic traffic was primarily carried by satellites. The transition only began in 1988 with the installation of TAT-8, the first trans-oceanic fibre-optic cable, which linked the US, the UK and France. This marked the point at which submarine cables began to outperform satellites in volume, speed and cost-effectiveness of data and voice communications.[11] Consequently, almost all transoceanic telecommunications are now routed via the submarine cable network, with satellites handling only around 3% of total traffic now, according to ESCAEU.[12]

Figure 2: Global submarine cables, with the most geopolitically contentious corridors



Source: TeleGeography, www.submarinecablemap.com, Deutsche Bank Research.

2.2 Why undersea cables are strategic

Beyond serving as the Internet's backbone, these underwater cables are essential for global trade, finance, cloud computing, and government communications. For example, submarine cables can carry around 375mn simultaneous telephone conversations on 1 fiber optic strand.[13] The SWIFT network also relies on cables for transmitting financial data to more than 8,300 member financial institutions in 195 countries.[14] It is also estimated that $10trn worth of financial transactions travel through these subsea cables every day.[15]

A 2026 study found that undersea cable disruption also had massive macroeconomic consequences for global economies.[16] Specifically, a typical cable disruption often triggers an immediate 2ppt decline in GDP capita growth for an economy. There is also evidence to show that the impact continues to accumulate over time; within three to six

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[10] The number of systems by supplier is ranked by ASN (23), Subom (9), NEC (8), and HMN Tech (6). Data taken between 2021 – 2025. Kieran Clark, "Submarine Telecoms Industry Report," SubTel Forum, October 30, 2025, https://subtelforum.com/submarine-telecoms-industry-report/.

[11] Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.

[12] "Submarine Telecommunication Cables." n.d. www.Escaeu.Org. Accessed July 7, 2026. https://www.escaeu.org/articles/submarine-telecommunications-cables/.

[13] Submarine Telecommunication Cables." n.d. Www.Escaeu.Org. Accessed July 7, 2026. https://www.escaeu.org/articles/submarine-telecommunications-cables/.

[14] Swift data

[15] "Securing Europe's Subsea Data Cables." 2024. Carnegie Endowment for International Peace. 2024. https://carnegieendowment.org/research/2024/12/securing-europes-subsea-data-cables?lang=en.

[16] Joël Cariolle. Digital Disasters: The Macroeconomic Costs of Submarine Cable Breaks. 2026. ⟨hal-05544990⟩

years, the affected country's economy diverges, falling 9 ppts below its non-disrupted neighbors. These disruptions also reduce overall economic efficiency, innovation and labor productivity, followed by a tightening of financial conditions as domestic credit for non-financial private agents' contracts and cross-border banking positions weaken. The study also noted that in countries with fewer international cable relations and/or without internet exchange points, disruptions widen the GDP per capita gap immediately and substantially over the medium run, compared to nearby non-disrupted economies that instead experience offsetting gains consistent with reallocations.[17]

These findings highlight a broader point: the resilience of subsea cable infrastructure is not merely a telecommunications issue, but a broader precondition for economic stability and the functioning of modern financial systems.

The strategic importance of these networks is further amplified by the geography of the routes themselves. Because international data traffic is concentrated within a limited number of physical corridors, damage to key cable systems could have magnanimous consequences. One example that illustrates this point is to consider a high-frequency trader in London pulling live market data from a server on Wall Street in New York. The request would be transmitted as packets across terrestrial fibre networks in the UK, where it enters a low-latency transatlantic cable system. Examples of such routes include EXA Infrastructure's Express system, which forms part of the low-latency route between New York and London or Aqua Comm's AEC-1 linking New York and Ireland, and then on to the UK.

Many of these transatlantic cables pass through the GIUK Gap in the Atlantic, a geopolitically sensitive maritime corridor between Greenland, Iceland, and the UK. The passage is contentious because it is a critical chokepoint for NATO maritime surveillance and because it is also increasingly a primary route for Russia's Northern Fleet to access the Atlantic. So this concentration of critical data infrastructure within a militarily significant area creates a potential vulnerability.

As such, subsea cable infrastructure security has become a national priority for several countries, including the EU, US, Japan, India and Australia.

2.3 Undersea cable vulnerabilities

The criticality of subsea cables is heightened by the fact that they can be vulnerable to damage, outages and cyber risks, such as natural or man-made hazards, and hazards to other infrastructure including the landing stations and IT systems that support the cables. This systemic risk is amplified because many global networks rely on a limited number of routers, hubs and suppliers.

Crucially, there is often no "Plan B" when a major cable is disrupted, and there are a reported 150-200 cable faults recorded annually.[18] The challenge is threefold: first, repairs are immensely costly, estimated at $500,000-$1mn per incident for a subsea fibre optic cable, and $10 -$100mn per incident for a subsea power cable.[19] Second, there are only ~40 specialist cable repair vessels in the entire world, owned by a handful of companies like SubCom, Orange Marine, and HMN Technologies. These vessels operate on a first come first served basis within the consortium's zone agreement. And while the repair ships can mobilize within a 24-hour period, repair time frames vary according to fault location, while the requirement to secure repair permits within territorial or exclusive economic zone waters can be burdensome and significantly delay starting operations.[20] Finally, these cables are often concentrated in physical chokepoints – such as along

[17] Joël Cariolle. Digital Disasters: The Macroeconomic Costs of Submarine Cable Breaks. 2026. ⟨hal-05544990⟩
[18] International Cable Protection Committee.
[19] Qiu, Winston. 2025. "Statistics on Subsea Cable Fault and Repair - Submarine Networks." Submarinenetworks.Com. June 28, 2025. https://www.submarinenetworks.com/en/nv/insights/statistics-on-subsea-cable-fault-and-repair.
[20] Constable, Mike, Burdette, Lane, Mauldin, Alan. "The Future of Submarien Cable Maintenance: Trends Challenges, and Strategies." June 2025.

shallow straits or continental shelves – that are also areas of geopolitical strife, making conducting any reparations difficult in this environment.

While direct evidence of deliberate cable sabotage has been hard to prove thus far, several incidents of damage to critical undersea infrastructure sites have caught the attention of governments.[21] The US Congress, in its proposed Strategic Subsea Cables Act of 2026, noted damage to cables connecting the Norwegian mainland with the Norwegian archipelago of Svalbard (2022), multiple cuts to cables in the Taiwan Strait (2023, 2025), and disruptions to cables in the Baltic Sea (2024, 2026) as key examples of this growing threat.[22]

Perhaps the more immediate concern for individual users was the threat that emerged earlier this year, when Iran threatened to cut its undersea cables in the Red Sea. Since these routes carry around 17-25% of global internet traffic, the ramifications of such an act would be severe.[23] Even before this threat, subsea cable operators in the Red Sea have been plagued with numerous problems. Yemeni territorial disputes and Houthi attacks on shipping lanes have obstructed the deployment of several new cables – including the 2Africa cable, which is almost complete, and other cables such as the IEX, Africa-1, and Raman, which are still awaiting deployment. The geopolitical strife in the corridor has led to recent reports that all cable projects there have been "delayed indefinitely."[24]

Geopolitical risk is rising, but most cable damage remains accidental. The International Cable Protection Committee estimates that fishing gear and ship anchors account for roughly 70–80% of faults.

2.4 State actors responding

Governments have long regarded submarine telecommunications networks as critical infrastructure requiring strong protection (for example, ACMA, 2007).[25] More recently, they have also recognized the geopolitical value of controlling these corridors. In 2015, China set a goal of capturing 60% of the fibre-optic cable market through state-backed firms under its Digital Silk Road initiative.[26] HMN Technologies (formerly Huawei Marine) is one of four companies that collectively dominate global cable manufacturing.

However, China's influence is primarily concentrated in the Global South, as Chinese firms have been largely excluded from most Western-connected systems since the US launched its 2020 Clean Network program to counter China's influence across telecoms and data. Under the Clean Network, Chinese telecom and cable firms were shut out of lucrative bids for cable-laying projects, most notably when HMN Technologies lost the SEA-ME-WE 6 project (running from France to Singapore via the Suez) in 2023 despite its competitive pricing. aChina has pivoted to financing new cable projects connecting Africa, Asia and the Middle East, such as its flagship PEACE cable (Pakistan-East Africa-Europe) and EMA (Europe-Middle East-Asia) project, which is backed by three state-owned telcos: China Telecom, China Mobile, and China Unicom in partnership with Egypt and Saudi Arabia.

Over the last few years, the US has consolidated its toolkit to protect its subsea cable security. In February of this year, the US Congress introduced the bipartisan Strategic

[21] In 2025 Western media discovered that engineers at China's Lishui University in Zhejiang had filed a patent for the creation of a submarine cable cutting device back in 2020. While it was argued it was needed for emergency situations, the announcement raised suspicion amongst Western media, which uncovered the patent applications in January 2025.22 And in April 2026, the South China Morning Post reported China had successfully tested cable cutting 11,483 feet (3,500m) undersea.

[22] Joe Wilson. 2026. US Strategic Subsea Cables Act. https://joewilson.house.gov/sites/evo-subsites/joewilson.house.gov/files/evo-media-document/undersea-cables-signed_0.pdf.

[23] Iran Signals a New Threat to the Gulf's Internet | WIRED Middle East

[24] "Stalled Subsea Cable Projects Threaten Middle East Digital Ambitions." 2026. FinancialTimes. Financial Times. June 25, 2026. https://www.ft.com/content/5e6ce736-3a48-406a-933a-ed215370bd16?syn-25a6b1a6=1.

[25] Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.

[26] "The Digital Silk Road: A Growing Priority for Beijing as Its Tech Champions Expand Overseas." 2024. Merics. March 27, 2024. https://merics.org/en/tracker/digital-silk-road-growing-priority-beijing-its-tech-champions-expand-overseas.

Subsea Cables Act. A core proposal of the act requires the President to impose sanctions on foreign actors who intentionally damage critical undersea infrastructure, including subsea fiber optic cables.[27] If passed, this act will build upon the US's 2020 Undersea Cable Control Act, which focused more on preventing adversaries from accessing technology to build and operate undersea cables. Together, the two bills aim to deny countries like Russia and China the ability to conduct what a sponsoring senator referred to as "shadow warfare" against US critical infrastructure in subsea cables.[28]

Finally, a landmark development to watch is the recent US Federal Communications Commission (FCC) vote to extend cable surveillance to terrestrial endpoints, such as cable landing stations and the hardware housed within them that converts the subsea optical fiber signals. The FCC has identified these endpoints as chokepoints vulnerable to data interception, physical sabotage, and cyber espionage. Under the new rules, hyperscalers and tech firms employing AI infrastructure could receive expedited licensing for their terrestrial hardware, while foreign rivals would need to apply for individual licenses and face the possibility of denial. This is a potential win for the hyperscalers, although the pool of eligible hardware suppliers may significantly shrink, creating short-term bottlenecks in cable landing equipment manufacturing.[29]

Europe has historically had a strong presence in the subsea cable industry, but this position is being challenged by competing US and Chinese interests and the rise of US hyperscaler control. France remains a leading global supplier through its ownership of Alcatel Submarine Networks and Orange Marine, while Italy's Elletra holds a strong mid-tier position.

The EU is attempting to carry out a coordinated approach towards cable security via its 2025 EU Action Plan on Cable Security, but implementation has been slow. In June of this year, however, the European Commission finalized a €5.8mn grant to establish the first two regional cable hubs in the Baltic Sea and the Mediterranean Sea, and a separate €40mn fund to increase its submarine cable repair capacity.[30]

Finally, US-allied attempts to find a geopolitically secure "Fourth Route" away from volatile transits in the Red Sea and South China Sea have led to the creation of the Far North Fiber express route. This will be the first long-haul submarine cable connecting Europe and Asia exclusively through Allied-sovereign waters. It is expected to cost $1.17bn and should be ready for service by the end of 2026.[31] The project is owned by US Alaskan company Far North Digital, in conjunction with its Canadian affiliate True North Global Networks, but it has since expanded into a consortium with companies from Japan and Finland as well. Alcatel Submarine Networks will supply the cables.

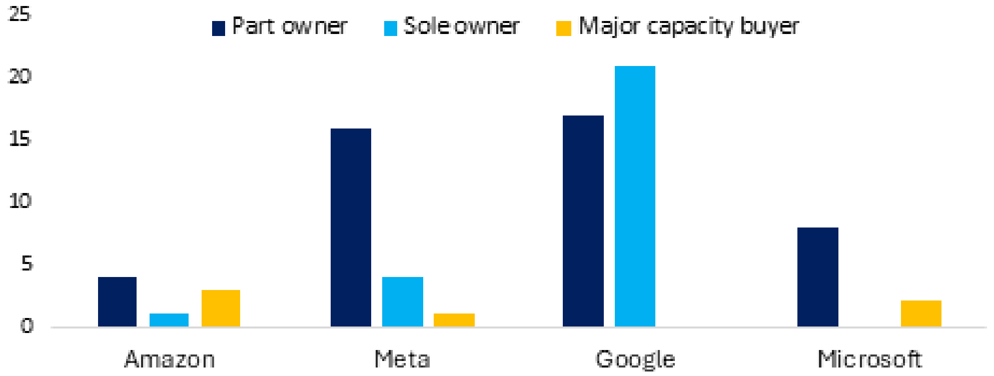
All of this strategic interest has translated into unprecedented subsea cable infrastructure investment, estimated to have averaged over $2bn per year in new construction over the last 9 years.[32] Most new investment is forecast to be directed toward trans-pacific routes.

The most consequential response, however, is coming from the private sector, which controls most of the infrastructure. AI is sharply increasing bandwidth demand and prompting content and cloud providers—notably Google through its Pacific Connect

[27] Qiu, Winston. 2026. "US Introduces Strategic Subsea Cables Act of 2026 — Senior Industry Analyst Briefing." Submarine Networks. April 14, 2026. https://www.submarinenetworks.com/en/nv/insights/us-introduces-strategic-subsea-cables-act-of-2026.
[28] "Wilson and Meeks Introduce 'Strategic Subsea Cables Act Of 2026.'" 2026. Representative Joe Wilson. March 27, 2026. https://joewilson.house.gov/media/press-releases/wilson-and-meeks-introduce-strategic-subsea-cables-act-2026.
[29] Qiu, Winston. 2026. "FCC Overhauls Subsea Cable Licensing: SLTE Licensing and Fast-Track Process for US Tech Giants." Submarine Networks. June 26, 2026. https://www.submarinenetworks.com/en/nv/insights/fcc-subsea-cable-licensing-slte-fast-tract-process.
[30] Ernst, Lauren. 2026. "EU Launches Regional Cable Hubs To Protect Critical Undersea Infrastructure." Ocean News & Technology. June 23, 2026. https://oceannews.com/news/subsea-cable/eu-launches-regional-cable-hubs-to-protect-critical-undersea-infrastructure/.
[31] "Far North Fiber - Submarine Networks." 2026. Submarinenetworks.Com. 2026. https://www.submarinenetworks.com/en/systems/trans-arctic/far-north-fiber.
[32] Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.

Initiative and Meta through Project Waterworth—to accelerate investment.[33] This private dominance means resilience will depend as much on international coordination with corporate owners as on national policy.

Figure 3: Number of submarine cables owned by content providers



Source: Telegeography, Deutsche Bank Research..

3. Satellites: Beyond terrestrial networks

Like subsea cables, satellites form a part of the Internet's physical layer. Throughout much of the 20th century, satellites carried the bulk of all transoceanic digital connectivity. The launch of Telstar, the first operational telecommunications satellite in 1962, was soon followed by other satellites that helped commercialize the international communications and broadcast industry.

Today, however, traditional satellite communications (satcom) complement rather than replace cables. Satellites provide broadcast and communication services to sparsely populated regions not served by terrestrial infrastructure and act as a strategic back-up for disaster-prone regions. While subsea cables deliver more consistent, high-capacity links between major population centres, satellites are a cost-effective solution for governments seeking to connect regions with low population densities.[34]

The renewed focus on satellite communications is driven less by their current share of global traffic than by their strategic potential. Two forces are decisive. First, the war in Ukraine and US-China technology competition are prompting sovereign investment in space-based communications. Second, rising AI workloads are intensifying demand for bandwidth and encouraging the deployment of satellite mega-constellations. We examine the implications in greater detail in the next chapter.

For these reasons, communications-based satellite networks and space-based connectivity represent the next major frontier in the contest for control over digital connectivity.

3.1 How satellites support connectivity

There are three main types of satellites in operation: communications, Earth-observation, and radiofrequency- based. This section focuses on communication satellites, which provide services ranging from radio, TV, telecommunications, and Internet by receiving signals from Earth and retransmitting them to a ground station elsewhere, creating a communications channel between distant regions that were previously unable to communicate with one another.

[33] The State of the Network: 2026 Edition. 2026. Telegeography.
[34] Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.

Traditional satellite communications (also referred to as "satcom") rely on large geostationary (GEO) satellites positioned about 35,800 kilometres above Earth. Systems such as the early Intelsat 1 (1965) and the more recent Jupiter 3 (2023), provide broad, fixed coverage with high availability and favourable ground-terminal geometry.[35] GEO assets have historically supported predictable business models because of their long operating lives, typically around 15 years, and high launch costs. Their strategic value is clear, but so is their limitation: distance creates latency. The farther a satellite is from Earth, the longer data takes to travel, which is one reason subsea cables continue to carry most global connectivity.

In addition, GEO satellites also suffer from increasingly longer development cycles and higher integration complexity. As a result, both state and private actors are turning their interest towards low-earth orbit (LEO) satellites instead. LEO satellites orbit between 160 and 1,500km above the Earth's surface, enabling a more seamless user experience for applications like live streaming, video calls, remote desktops, etc. Because they circle the planet up to 16 times a day, LEO constellations can provide ubiquitous coverage to nearly any point on Earth. SpaceX's Starlink was the harbinger of the LEO satellite model, but several LEO constellations have emerged in response to Starlink since then, including Amazon LEO, OneWeb (now part of Eutelsat), Telesat Lightspeed, and China's Guowang and Qianfan projects. In fact, in July 2026 Amazon announced it had filed an application with the US FCC to deploy a constellation of more than 5,100 satellites that would provide direct-to-device connectivity by 2028.[36] Doing so will directly challenge Starlink Mobile, a subset of Starlink comprised of around 650 satellites that beam mobile data to phones in remote regions on earth, and which has recently also announced plans to launch "thousands" more satellites by end of 2028.[37]

3.2 Why satellites are strategic: the arrival Starlink

Although subsea cables carry the bulk of internet traffic, communications-based satellites are also integral to the global economy. They provide connectivity to remote and underserved regions that undersea cables cannot reach, while also enabling communications for mobile platforms such as ships, aircraft and military assets. Indeed, industries such as aviation, rail, shipping and logistics depend on satellites for navigation, route optimization, and cargo tracking.

Perhaps to an even greater extent than cables, satcom has become strategic to a country's military infrastructure. Satcom companies are developing partnerships to cater to civilian and government forces. For example, Canada's Telesat (c. 1969) announced a strategic partnership with the Canadian Armed Forces to develop and deliver a multifrequency, Arctic military satellite communications capability via its forthcoming Lightspeed constellation, which the company targets to launch in late 2026.[38] Meanwhile, established satcom companies with significant government -related revenue streams, like Viaset, SES and Eutelsat, have seen strong growth over the past year. Our DB equity analysts estimate that existing GEO communications contracts alone account for around $50bn-60bn in defence-related spending (excluding non-US allies).[39]

The greatest harbinger has been SpaceX's Starlink, which has provided encrypted communications, imagery and geolocation services for the Ukrainian military against Russia since 2022. In addition, Starlink's "direct-to-device" business model that connects smartphones directly to satellites threatens to collapse the traditional, hierarchical

[35] Yu, Edison, Roshan Ranjit, Scott Deuschle, Christophe Menard, and Laura Li. 2026. "State of the Private Markets: Rise of Micro-GEO Satellites." Deutsche Bank Research.
[36] "Amazon Targets Musk's Starlink with Satellite Constellation for Mobile Services." 2026. FinancialTimes. Financial Times. July 27, 2026. https://www.ft.com/content/83e93ce0-1ebd-4b04-a70c-82ff7b64a5f4?syn-25a6b1a6=1.
[37] Kan, Michael. 2026. "Starlink Mobile Will Span 'Thousands' of Satellites By 2028." PCMag UK. August 3, 2026. https://uk.pcmag.com/networking/166530/starlink-mobile-will-span-thousands-of-satellites-by-2028.
[38] Yu, Edison, and Laura Li. 2026. "Telesat: Focus Increasingly Shifts toward Defense." Deutsche Bank Research.
[39] Yu, Edison, Roshan Ranjit, Scott Deuschle, Christophe Menard, and Laura Li. 2026. "State of the Private Markets: Rise of Micro-GEO Satellites." Deutsche Bank Research.

Internet infrastructure, where cables traverse terrestrial data and comms-based satellites take it on its last leg.

Starlink's ability to bypass the conventional hierarchy of connectivity provided vital communications for Ukrainian forces and civilians at a time when conventional communications like fiber lines, cellular towers and power grids were damaged early in the war. This highlighted how a private company could become essential to a nation's military capability during a war. The gravity of this power was demonstrated again in February 2026, when SpaceX and Ukraine introduced a terminal verification system that effectively blocked Russia from using Starlink military terminals, after it was revealed that Russia had acquired Starlink terminals via third-party countries and grey markets to conduct strike drones.

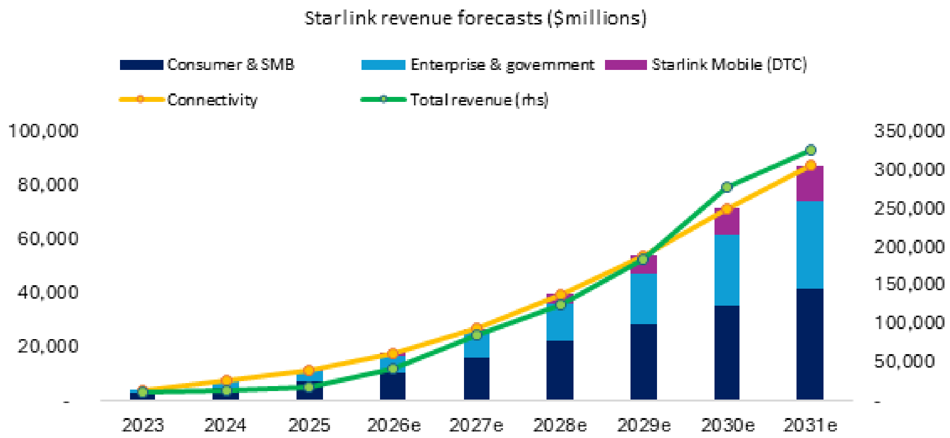
The entry of Starlink in 2019 has shifted the goalposts and urgency for more communications-based satellites. Bringing affordable, high-speed internet to those that lack access to traditional broadband, Starlink's true purpose is to help deploy Musk's goal of AI compute at a massive scale. In fact, Musk confirmed that Starlink is also designed to support a project called Starmind, which aims to place massive data centers in space. He recently filed applications with the FCC for a constellation of space-based AI data center satellites through this project. These space AI data centers will compute in space but then use lasers to offload the processed data directly into the Starlink constellation, which will beam the results back to Earth with low latency.

By using Starlink to create a reliable and robust orbital network used for Earth, Musk hopes to eventually extend the satellite internet service to the Moon, Mars, and beyond. Our equity analysts place Musk's aims for orbital data centers within a 5-year horizon and human voyage and colonization in Mars and off-world within 10-year horizon.

Starlink has upped the game for the rest of the world considerably. First, by its sheer dominance in the number of satellites it operates and in the numbers of subscribers it now has on its network. Starlink now accounts for 54% of all operational satellites in orbit,[40] and our equity analysts estimate Starlink's consumer broadband subscribers will grow to around 17mn by the end of 2026.[41] Second, its ability to simultaneously increase speeds while cutting hardware prices has made it very hard for anyone else to compete. Finally, Starlink's entry into LEO also made it harder for pure-play GEO industry players like Viaset and Intelsat to focus solely on consumer broadband, although as mentioned above there is a growing market opportunity for these players to enter the government services market. We will go more in depth on Starlink and its parent company SpaceX in our final chapter on Cloud and Data Centers.

[40] "10,400+ Starlink Satellites in Orbit — Live Count (2026) | Orbital Radar." 2026. Orbital Radar. May 31, 2026. https://orbitalradar.com/how-many-starlink-satellites.
[41] Yu, Edison, Bryan Kraft, Ross Seymore, Benjamin Black, Laura Li. 2026. "SpaceX: Apex of Civilization Ambition." Deutsche Bank Research.

Figure 4: Starlink connectivity revenue to reach $18bn in 2026 and $88bn by 2031



Source: Deutsche Bank Research equity research, company filings.

3.3 Satellite vulnerabilities

Because communications-based satellites are so critical, their vulnerability to deliberate attacks is a matter of strategic concern. Satellites are susceptible to attacks such as jamming (when the uplink or downlink frequency is filled with noise so the signal cannot get through) and spoofing (transmitting fake signals) of their radio waves. Countries like Finland, Estonia and Poland have all reported GNSS disruptions associated with alleged Russian activities in the Baltic region, which has affected numerous services such as commercial aircraft and maritime navigation. Other examples of disruption include cyberattacks on satellites via their ground stations and control networks, as well as physical and cyberattacks on the orbiting satellites themselves. Notable incidents include evidence that Russia was conducting widespread jamming of Starlink GPS signals soon after their invasion of Ukraine in 2022. In 2021, Russia also conducted an anti-satellite weapon test that destroyed an inactive satellite and generated a massive field of orbital debris.

Figure 5: Major satellite networks in orbit

Constellation	First launched	In orbit	Total planned
Starlink (US)	May 2019	10,877	42,000
Amazon Leo (US)	Oct 2023 (prototype satellites); production deployment began Apr 2025, initial commercial service expected in late 2026	390+	7,700
OneWeb (UK/Eutelsat)	Feb 2019, merged with Eutelsat in 2023.	654	Gen-1 complete, additional 440 satellites planned with Airbus
Guowang (China)	December 2024	177	12,992 (~13,000) filed with the ITU
Qianfan (China)	August 2024	238	Over 15,000 satellites approved for the final system

Source: Satellitemap.space, ChinainSpace, CNBC.

3.4 How state actors are responding

A rapid, high-stakes military space race looks to be underway between the US, China, and Russia, focused on establishing, securing, and defending satellite communication (satcom) constellations, particularly those in Low Earth Orbit (LEO).

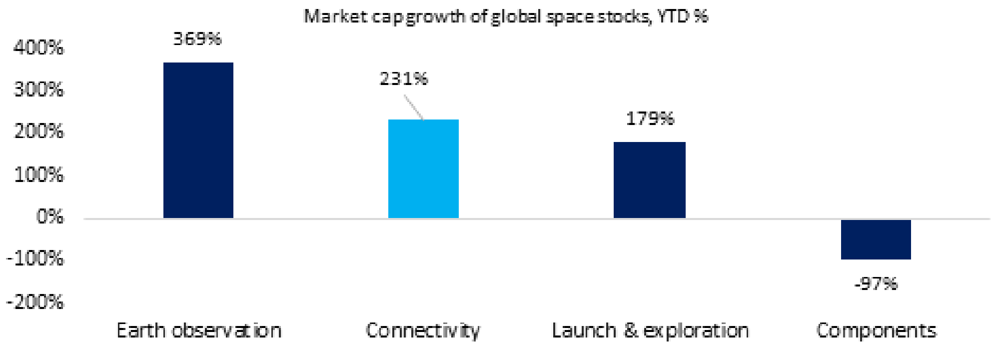
While US companies currently dominate the satcom landscape, they face a growing challenge from other countries, especially China. China is developing its own satcom ecosystem through two key initiatives: the state-run Guowang network, aimed towards China's domestic telecom and national security, and the Shanghai-backed Qianfen constellation, which is aimed at the commercial and export markets. Together, these projects aim to deploy between 13,000 and 15,000 satellites.[42] Guowang is designed primarily to connect rural areas and government institutions, while Qianfan is more commercially focused, aiming to provide in-flight connectivity (IFC) and direct-to-device (D2D) communications, with active pilots underway in Brazil, Malaysia, and Thailand.

Both constellations are in their early stages, with fewer than 200 satellites each currently in orbit. In December 2025, China also submitted filings with the International Telecommunication Union (ITU) to reserve orbital slots and spectrum for nearly 200,000 additional satellites – a move to secure future resources, even though China does not yet have the launch capacity to deploy such a number.[43] To facilitate more launch activity, China has also implemented new regulatory guidelines to facilitate IPOs for firms developing reusable rockets.

The EU is also attempting to reduce its dependence on US private operators by building IRIS, a government-controlled constellation of approximately 290 satellites, with services targeted to begin in 2029. However, progress on the project remains slow. Other notable sovereign interests include the UK's stake in OneWeb and Canada's investment in Telesat's Lightspeed project.

Driven by these strategic imperatives, the market for communications-based satellites is growing, as they become critical infrastructure for consumers, enterprises, and governments. Deutsche Bank equity analysts estimate the satcom market grew by over 35% YoY to nearly $26bn in 2025, mostly driven by consumer broadband and government contracts.[44] This growth is expected to continue, with market revenue projected to reach $65-70bn by 2030.

Figure 6: Oversized demand for comms-based satellites has led to a rally in space connectivity stocks



Source: Bloomberg Terminal, Deutsche Bank Research. Returns up to August 2026. Connectivity stocks: AST SpaceMobile, APT, Beijing BDStar, China Satcom, Echostar, Eutelsat, Globalstar, Gogo, Iridium, JSAT, SES, Space42, Telesat, Thaicom, Viasat.

[42] Yu, Edison, Bryan Kraft, Roshan Ranjit, Lee Horowitz, and Laura Li. 2026. "Space & Satellites: Market Growing and Becoming Critical Infrastructure." Deutsche Bank Research.
[43] "China ITU Filing to Put ~200K Satellites in Low Earth Orbit While FCC Authorizes 7.5K Additional Starlink LEO Satellites – IEEE ComSoc Technology Blog." 2026. Comsoc.Org. January 13, 2026. https://techblog.comsoc.org/2026/01/13/china-itu-filing-to-put-200k-satellites-in-orbit-fcc-authorizes-7-5k-additional-starlink-leo-satellites/.
[44] Yu, Edison, Bryan Kraft, Roshan Ranjit, Lee Horowitz, and Laura Li. 2026. "Space & Satellites: Market Growing and Becoming Critical Infrastructure." Deutsche Bank Research.

4. The new power map of connectivity

4.1 The rise of private control

While cloud and space data centers will be the focus of our next chapter, this section zooms in how connectivity supply chains are being affected by the AI build-out.

In the connectivity model set out in Section 1, hyperscalers own the destination of data: the cloud, where applications, websites and AI models run. Historically, these companies have relied on telecom operators for transmission and concentrated investment on data centers and cloud services. That division is now eroding, as hyperscalers are increasingly acquiring, funding or building the underlying infrastructure, including subsea cables and communications satellites—that carry data to and from their platforms.

AI's data intensity is accelerating this vertical integration. In 2016, traditional backbone providers carried most international bandwidth; today, content and cloud networks account for nearly three-quarters of demand.[45] Owning connectivity allows hyperscalers to lower long-run unit costs, scale capacity with cloud demand and reduce dependence on third-party networks. It also gives them greater control over the economics and governance of the data chain.

This trend is evident in new cable construction, where private companies are starting to swallow up market share. For example, of the $14bn projected investment in new submarine cables entering service between 2025 and 2027, driven largely by content providers such as Google and Meta rather than the traditional internet backbone providers.[46] While traditional telecom operators still account for the majority of existing builds, this new investment demonstrates the steady and growing influence of large-scale cloud and content companies in shaping global connectivity.

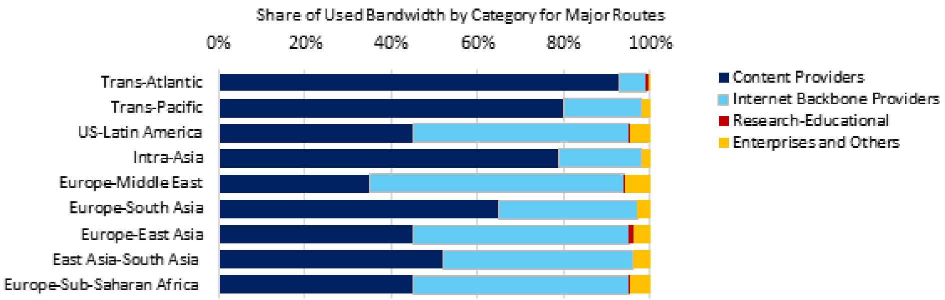
The same dynamic is beginning to emerge in the satellite ecosystem, where hyperscalers and space technology companies are exploring space-based data centers as a potential long-term solution to the terrestrial constraints that increasingly accompany the buildout of AI compute capacity.

To make orbital computing viable, companies need communications infrastructure capable of moving massive amounts of data between space and Earth, creating a strategic rationale for greater ownership of satellite networks and related connectivity assets. A notable entrant is Jeff Bezos' Blue Origin, which announced TeraWave in January 2026, a multi-orbit communications network comprising 5,280 satellites in LEO and 128 satellites in MEO. Blue Origin plans to begin deploying the constellation in 4Q27 and is targeting enterprise, data-center, and government customers. DB analysts estimate that the constellation could require ~$18bn of capex.

Although Blue Origin has not explicitly positioned TeraWave as infrastructure for orbital data centers, its high-capacity, optically interconnected network could eventually provide part of the communications layer required to connect space-based compute systems with terrestrial networks. In addition to SpaceX and Blue Origin, Amazon is building its LEO communications constellation, while Nvidia-backed Starcloud has filed for regulatory approval to deploy orbital data-center satellites. Starcloud also plans to launch AWS Outposts hardware aboard its second satellite in October 2026, providing an early test of AWS infrastructure operating in space.

[45] The State of the Network: 2026 Edition. 2026. Telegeography.
[46] Ibid.

Figure 7: Content networks represent around three-quarters of all global bandwidth demand



Source: Deutsche Bank Research. Data inferred from Telegraphy's 2026 State of the Network Report. Represents used bandwidth as of year-end 2024.

4.2 Effects of the AI disruption

The result is a sharp shift in control over global connectivity. Traditional carriers are increasingly becoming minority consortium members or implementation partners, while hyperscalers lead funding and technical design. The Internet's decentralised architecture increasingly coexists with concentrated private ownership of critical connectivity infrastructure.

The consequences are substantial. By owning subsea cables and satellite networks, hyperscalers are increasingly able to build private connectivity corridors linking their data centres, reducing dependence on traditional telecom operators. This vertical integration bypasses—and may ultimately compress—the tiered, multi-provider structure that has defined telecommunications for decades.

This disruption has forced traditional telecom operators to either merge to survive or retreat into more niche services. By shrinking the telecommunications market, the remaining players can better afford the massive capex required for increasing broadband demand. Notable deals in recent years include $7.3bn merger between US-based Viasat and British satellite firm Inmarsat in 2023, SES's acquisition of Intelsat for $3.1bn in 2024, and the $8bn Rocket Lab/Iridium[47] consolidation in 2026. By creating these multi-orbit groups, they have been able to concentrate on key segments of mass-market connectivity.

Europe is perhaps at the greatest disadvantage in this new connectivity map, finding itself crowded out by US-led private innovators and state-backed Chinese companies. The European Centre for Development Policy Management argues that the growing concentration of subsea-cable ownership and operational control among US hyperscalers is creating structural dependencies that could erode Europe's digital sovereignty. [48] This concern is particularly acute for Europe, where ownership and operational control of critical subsea infrastructure is increasingly concentrated among non-EU actors. Unlike the United States, which exercises oversight through institutions such as the FCC, Europe lacks a unified governance framework for the subsea cable ecosystem, raising concerns about long-term digital sovereignty.[49]

The same risk exists in space, where Starlink dominates the global satellite network. As the CEO of the French telecoms operator, Bouygues Telecom, warned in an interview earlier this year, Europe needs to "get some sovereignty" or else it risks having its

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[47] Deutsche Bank and/or its affiliates is acting as advisor to Rocket Lab Corporation on the potential acquisition of Iridium Communications, Inc..

[48] Pearson, Sasha. n.d. "Troubled Waters: Europe's Subsea Telecommunications Network." Ecdpm. https://ecdpm.org/work/troubled-waters-europes-subsea-telecommunications-network.

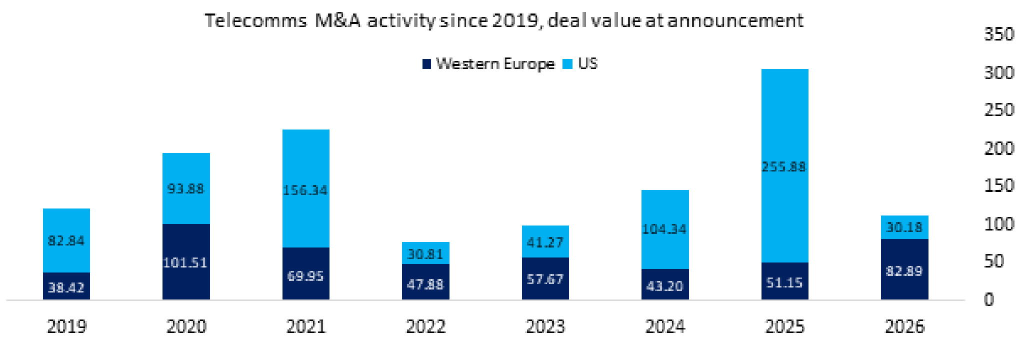
[49] Ibid.

connectivity controlled by a non-state actor like Starlink.[50] Reflecting these pressures, in June 2026, Bouygues itself partnered with Orange and Free-illiad Group to acquire and break up French telecommunications company SFR for €20.35bn.

In the space arena, the EU has attempted to address this problem with its proposed Space Act, which aims to cultivate European champions. However, this has created friction with the US, whose FCC has publicly warned that Washington would retaliate if Europe imposed protectionist measures that favored its own satellite companies over American ones.[51] In the meantime, the formidable presence of Starlink continues to drive a new wave of private communications-based satellite initiatives seeking to compete.[52]

States therefore face a strategic dilemma. AI and cloud adoption are directing more traffic through hyperscaler-owned networks, increasing dependence on private technology companies even as governments must guarantee resilience and control in a crisis. The US is best positioned because its firms dominate the industry. European governments have responded by buying stakes in, or taking control of, strategic telecom assets, including Italy's acquisition of TIM Sparkle and Spain's 10% stake in Telefónica.

Figure 8: Aside from 2022, where activity dropped sharply amidst the backdrop of Russia Ukraine, the telecoms M&A industry has seen massive growth



Source: Deutsche Bank Research, Dealogic. Data only draws upon publicly available deal value at announcement data. Data up to July 2026.

5. Conclusion

Digital connectivity emerged from an ideal of neutrality and disintermediation. The Internet was designed to share knowledge across research institutions, while cables and communications satellites expanded international exchange. Today, connectivity is an arena of state and corporate power. Control over data flows is becoming as consequential as control over payment infrastructure, and the race for AI leadership is intensifying the strategic value of the networks that carry information.

Subsea cables are uniquely fragile because their criticality is matched by limited substitutability. The parallel with payments is instructive: excluding a country from SWIFT can amount to financial siege; severing its cable links can impose a digital one. Yet payments have emerging alternatives—including CIPS, SPFS and stablecoins—whereas subsea cables have few practical substitutes. A replacement cable cannot be laid during a crisis, so a major disruption could simultaneously impair payments, markets, military command-and-control and basic communications.

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[50] Ohlen, Elsa. 2026. "'Europe Doesn't Realize How Dangerous It Is': Telecoms CEO Warns of U.S. Dominance in Satellites, AI." CNBC. May 21, 2026. https://www.cnbc.com/2026/05/21/starlink-spacex-satellites-telecoms-europe-dangerous-bouygues-ceo.html.
[51] Ranjit, Roshan, Robert Grindle, and Keval Khiroya. 2026. "European Satellites: Levelling Up." Deutsche Bank Research.
[52] Yu, Edison, and Laura Li. 2026. "The Final Frontier: Space 101." Deutsche Bank Research.

The most important actor to watch may not be a state but a company: SpaceX. Starlink has reshaped satellite communications on cost, scale and performance, compressing the traditional hierarchy of space connectivity. Its ambitions now extend into markets historically served by terrestrial telecom and cable providers, in both advanced and emerging economies.[53] The implications for governments, defence and incumbent telecom operators will be significant.

In the era of infrastructure realism, power lies less in the number of cables or satellites built than in control of the network's junction points and architecture—from the ocean floor to low-Earth orbit. As with dollar-based payments and Taiwan's semiconductor capacity, concentration in cable landing rights and orbital slots creates a new class of systemic risk. The central question is no longer who wins the cable race, but who controls the rails through which global value flows—and on whose terms.

The final part of our series turns to data centres, moving the chokepoint further up the technology stack—from minerals and cables at the physical layer, through chips at the fabrication layer, to the infrastructure that stores and processes data at scale.

We would like to thank Edison Yu, Laura Li, Benjamin Black, Raymond Wong and Yi Xiong for their valuable insights and comments to this paper.

[53] Yu, Edison, Bryan Kraft, Ross Seymore, Benjamin Black, Laura Li. 2026. "SpaceX: Apex of Civilization Ambition." Deutsche Bank Research.

Appendix 1

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The Next 90: Your Geopolitical Risk Radar

Sep–Nov 2026

September 2026

Dr. Helen Belopolsky | Miha Hribernik

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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The Next 90: Your Geopolitical Risk Radar | Sep–Nov 2026

A geopolitical calendar mapping the next 90 days

SEPTEMBER

For a detailed breakdown of key events, issues and themes in September, please see: *Latest Geopolitical Monthly*

Early	**US/Mexico**	Next bilateral round of USMCA review talks
1	**G20**	Meeting of Finance Ministers and Central Bank Governors
1	**SCO**	Shanghai Cooperation Organisation (SCO) Heads of State Summit in Bishkek
6-20	**Germany**	Regional elections
8	**UN**	Opening of the 81st UN General Assembly
8	**Canada/US**	Canadian retaliatory tariffs on the US are set to enter into force
12–13	**BRICS+**	18th BRICS+ summit in India
13	**Sweden**	General election
15	**EU**	European Parliament plenary session
18-20	**Russia**	Legislative and regional elections
22-28	**UN**	UN General Assembly debate
24	**US/China**	Xi-Trump Summit in the US
27	**France**	Senate election
30	**US**	Key government funding deadline

OCTOBER

3	**Latvia**	General election
4	**Brazil**	General election
5-28	**Canada**	General election in Quebec, municipal elections in British Columbia, Ontario and Manitoba
9-17	**Czech Republic**	Senate and municipal elections
12-18	**IMF/World Bank**	Annual meetings of the IMF and WB in Bangkok
15-16	**EU**	European Council meeting
19	**Canada**	Alberta independence referendum
23	**Morocco**	General election
24	**Slovakia**	Municipal and regional elections
25	**Bulgaria**	Presidential election
27	**Israel**	Legislative election
30-31	**G20**	G20 Foreign Ministers' Meeting in Atlanta
TBD	**EU/China**	Ministerial meeting in Beijing; deadline for "tangible results" on trade talks
TBD	**China**	Fifth Plenum of China's Central Committee

NOVEMBER

3	**US**	Midterm elections (House, Senate, gubernatorial and state legislatures)
4	**South Africa**	Municipal elections
7	**New Zealand**	General election
9	**EU**	Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings
9-20	**COP31**	COP31 in Türkiye
10	**US/China**	Expiry of the 2025 Busan agreement
10-12	**ASEAN**	49th ASEAN Summit in Metro Manila
18-19	**APEC**	APEC Economic Leaders' Meeting in Shenzhen
28	**Taiwan**	Local elections
28	**Palestinian National Authority**	Legislative election
TBC	**US/North Korea**	Potential meeting between Donald Trump and Kim Jong-un

Note: The calendar contains selected key events and is not intended to provide an exhaustive list.

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This Month in Geopolitics: September 2026

Key events, themes and trends to watch in the month ahead

September 2026

Dr. Helen Belopolsky | Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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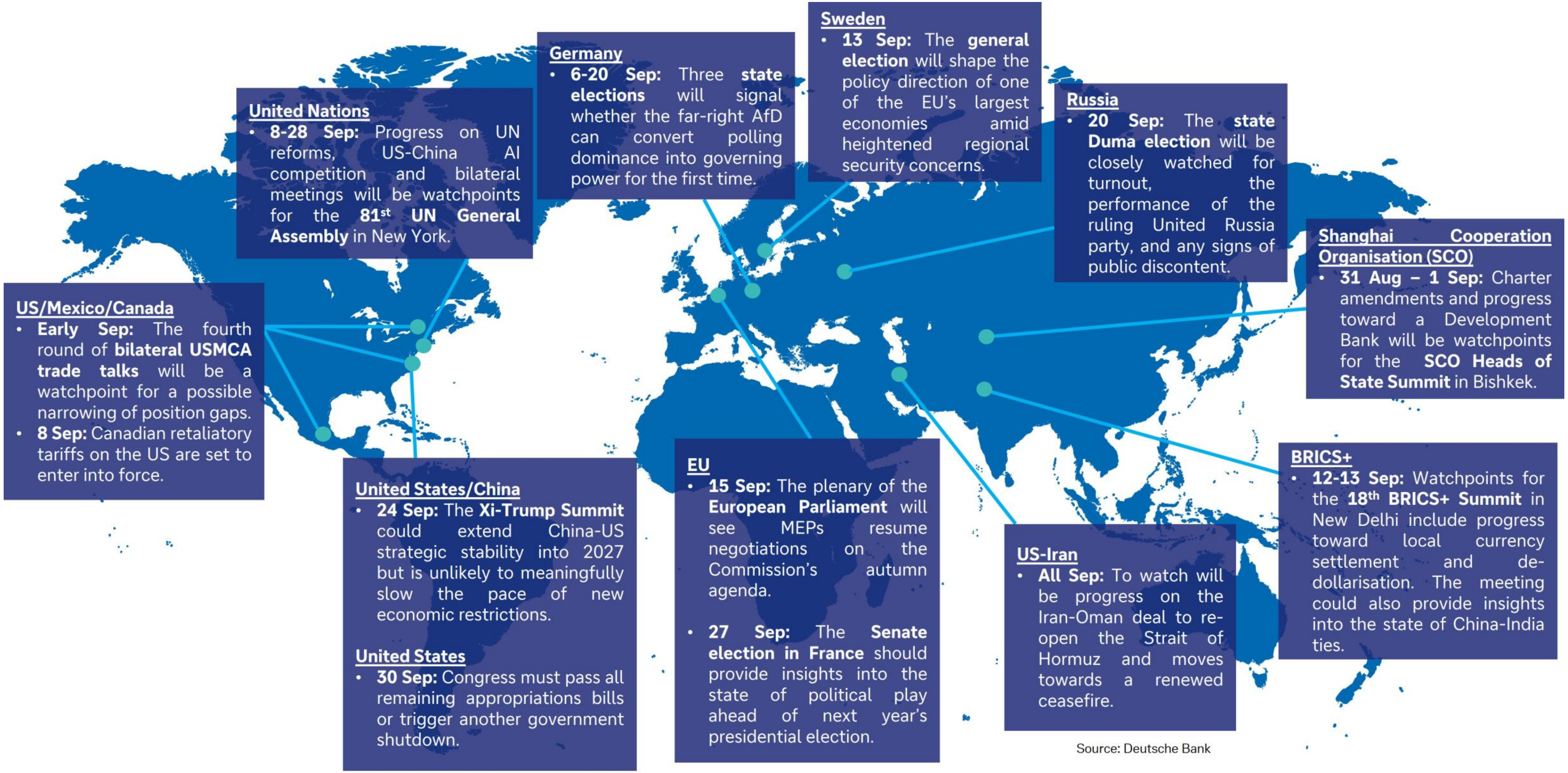
At a Glance: Geopolitical events to watch in **September 2026**

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides.



United Nations
- **8-28 Sep:** Progress on UN reforms, US-China AI competition and bilateral meetings will be watchpoints for the **81st UN General Assembly** in New York.

Germany
- **6-20 Sep:** Three **state elections** will signal whether the far-right AfD can convert polling dominance into governing power for the first time.

Sweden
- **13 Sep:** The **general election** will shape the policy direction of one of the EU's largest economies amid heightened regional security concerns.

Russia
- **20 Sep:** The **state Duma election** will be closely watched for turnout, the performance of the ruling United Russia party, and any signs of public discontent.

US/Mexico/Canada
- **Early Sep:** The fourth round of **bilateral USMCA trade talks** will be a watchpoint for a possible narrowing of position gaps.
- **8 Sep:** Canadian retaliatory tariffs on the US are set to enter into force.

Shanghai Cooperation Organisation (SCO)
- **31 Aug – 1 Sep:** Charter amendments and progress toward a Development Bank will be watchpoints for the **SCO Heads of State Summit** in Bishkek.

United States/China
- **24 Sep:** The **Xi-Trump Summit** could extend China-US strategic stability into 2027 but is unlikely to meaningfully slow the pace of new economic restrictions.

United States
- **30 Sep:** Congress must pass all remaining appropriations bills or trigger another government shutdown.

EU
- **15 Sep:** The plenary of the **European Parliament** will see MEPs resume negotiations on the Commission's autumn agenda.
- **27 Sep:** The **Senate election in France** should provide insights into the state of political play ahead of next year's presidential election.

BRICS+
- **12-13 Sep:** Watchpoints for the **18th BRICS+ Summit** in New Delhi include progress toward local currency settlement and de-dollarisation. The meeting could also provide insights into the state of China-India ties.

US-Iran
- **All Sep:** To watch will be progress on the Iran-Oman deal to re-open the Strait of Hormuz and moves towards a renewed ceasefire.

Source: Deutsche Bank

In the spotlight for September 2026: US-Iran war and China-US Summit

US-Iran war (all of September)

August brought a mix of escalation and small progress in the conflict: US strikes on Iranian oil infrastructure and an ongoing blockade pushed fuel shortages inside Iran as President Trump ramped up economic pressure. Iran and Oman outlined a proposal for a temporary shipping corridor through the Strait of Hormuz, with talks ongoing toward a permanent arrangement.

It appears both the US and Iran are waiting for the other side to blink though a new framework agreement isn't beyond the realm of possibility as backchanneling continues. Economic pressure on Iran is ramping up with President Trump's "economic D Day" threat and fuel shortages within Iran adding to the regime's challenges. Political pressure on the Trump administration is likely to ramp up moving closer to the mid-terms. To watch will be whether the Iran-Oman shipping corridor talks move from proposal to an actual operating channel, which could signal a genuine de-escalation amid a war that seems to keep grinding on with no military victory able to drive resolution. In addition, the US' latest round of sanctions targeting networks moving Iranian oil through China, Hong Kong and the UAE, will be a point of friction at the Trump-Xi summit. We should prepare for more volatility in the coming months as neither side is prepared to lose face.

Trump-Xi Summit (24 September)

The summit could extend China-US strategic stability into 2027 but is unlikely to meaningfully slow the pace of new economic restrictions. During the run-up to the summit, investors should watch for any sign of progress in three areas: the Board of Trade to support non-sensitive commerce, AI dialogue and a possible extension of the Busan agreement beyond November. The meeting could also deliver new purchase commitments. **While relations remain stable, there is little evidence to indicate a major breakthrough**. Recent reporting suggests stalled progress in some areas, such as on a new Board of Investment, in line with our assessment that national security concerns will limit progress in this area.

Investors should also be prepared for both sides to unveil new restrictions or retaliatory measures ahead of the summit, as Beijing and Washington work to strengthen their negotiating positions. US economic curbs against Iran's economic partners will be a watchpoint. New restrictions would add to a growing list of measures adopted this year (see table), which will likely expand further in the coming years. There is little prospect that the September summit can resolve structural competition in areas such as critical minerals, semiconductors or AI.

Beyond summitry: China-US relations remain competitive in 2026	
January	Washington approved **limited Nvidia H200 chip sales to China**, signaling a more transactional approach to tech controls.
February	The US launched a **major critical minerals effort**, including new bilateral frameworks, financing tools and a 56-country coalition, aimed at reducing dependence on China.
April/May	**China codified new anti-derisking regulations,** which allow Beijing to penalize companies that cut Chinese suppliers or comply with foreign sanctions/export controls. China separately ordered the unwinding of the **Meta-Manus AI** deal, signaling scrutiny of US investments in sensitive China-linked tech firms. **New US Congressional bills**, such as the MATCH Act, seek to tighten semiconductor equipment controls, close loopholes and pressure allies to align with US restrictions. **US sanctions on Chinese refiners** tied to Iranian crude triggered China's first formal use of blocking measures.
June	The Pentagon included several major Chinese tech companies on its **Chinese Military Companies** list. In response, China introduced **export controls** on 10 US companies, and restricted procurement from another 46 corporates. The US banned China-linked **Polestar** from selling cars in the US from model year 2027 onward due to national security concerns.
July	**China's new outbound investment regime** covers the overseas deployment of Chinese technology, data, talent and capital, especially in AI, chips, batteries, EVs and solar. The **US imposed tariffs under a Section 301 forced labour investigation** on over 60 markets, including China (12.5%). Washington also banned the import of new **Chinese humanoid and quadruped robots** and power inverters.
August	The US announced tariffs of up to 100% on imported **drones and components**, primarily impacting Chinese exporters. A White House **report alleged the transshipment** of China-linked goods into the US through more than 40 countries. The US also added a record 43 Chinese companies to the **UFLPA Entity List**; China retaliated by **imposing countermeasures** on 7 US compliance, audit and testing firms. US sanctioned dozens of mainland and HK entities over Iran China **tightened drone-related exports to the US**, potentially constraining access to critical minerals, rare earths, advanced electronics and other dual-use components. In late August, Beijing approved **limited sales of H200 chips** in China.
November	The **2025 Busan agreement** is set to expire but will likely be extended, delaying the introduction of new bilateral economic restrictions.
December	The US Congress will adopt **annual must-pass defence legislation**, which frequently includes provisions aimed at China.

Note: The table provides a non-exhaustive list of selected key developments.
Source: Deutsche Bank Research

Key events, issues and themes to watch in **September 2026**

Listed in chronological order

Shanghai Cooperation Organisation (SCO) Heads of State Summit (31 August – 1 September)

Charter amendments and development bank progress will be key watchpoints. The summit in Bishkek will bring together leaders from the SCO's 10 member states: China, Russia, India, Pakistan, Iran, Belarus, Kazakhstan, Kyrgyzstan, Tajikistan and Uzbekistan. Founded in 2001, the SCO aims to promote security and economic cooperation across Eurasia, but its practical impact has been limited given its relatively loose structure and member states' diverging geopolitical priorities.

However, the Bishkek summit could offer significant watchpoints, including amendments to the SCO's Charter and possible progress toward the establishment of an SCO Development Bank. The latter would give the SCO a dedicated mechanism to fund regional infrastructure. A dedicated development bank could also support local currency settlement – potentially boosting efforts by member states such as China and Russia to reduce their reliance on the US dollar.

Next bilateral round of US-Mexico USMCA review talks (Early September)

The fourth round of talks is unlikely to deliver a breakthrough but should show whether US-Mexico position gaps are narrowing. 1 July marked the start of the US-Mexico-Canada (USMCA) trade deal's mandatory six-year joint review process whereby the three countries are required to assess whether to extend the agreement for another 16 years, review it or allow it to move into annual review mode. The US has declined to confirm the renewal of USMCA "in its current form".

We expect USMCA negotiations to remain difficult and protracted. In early September, the US and Mexico will hold further negotiations focused on resolving outstanding issues (e.g., automotive rules of origin, steel/aluminum, economic security, and issues around agriculture, labour, environment and alleged China-linked transshipment). Auto talks will be a key watchpoint. Washington is reportedly requesting 50% US-made content in North American vehicles, while Mexico's counter-proposal would exempt Canadian and Mexican parts and levy only non-North American content – potentially lowering US vehicle prices.

Source: Mike Dudin on Unsplash

Key events, issues and themes to watch in **September 2026**

Listed in chronological order

German state elections: Saxony-Anhalt (6th), Mecklenburg-Vorpommern (20th) and Berlin (20th)

The three state elections in September will close out Germany's 2026 "super election year" and are together being read as a stress test of whether the far right AfD can convert polling dominance into governing power for the first time. In Saxony-Anhalt, the AfD is polling around 41%, well ahead of the Chancellor Merz's CDU's roughly 23-27%. The polling puts the AfD within reach of an absolute majority – a level of power no AfD branch has held in any German state. In Mecklenburg-Vorpommern, the AfD leads at 34-38% support, well ahead of the SPD (19-27%), which has led the state since the late 1990s. Berlin, by contrast, remains genuinely fragmented; the CDU sits around 19-22%, with the AfD, Greens, and Die Linke behind. These elections are seen as a test of the AfD's rise and whether that puts strain on Chancellor Merz's federal coalition.

Canadian retaliatory tariffs on the US (8 September)

On 8 September, Canada will enact retaliatory tariffs matching US levies (15-50%) across roughly 700-900 tariff lines, roughly C$27.6bn of US imports (including steel, dairy, appliances, electronics, etc.). Aluminum duties will double to 50%. Despite Prime Minister Carney suggesting "we are at war," Ottawa says it remains open to a deal. But September could still see further escalation as both sides seek to gain leverage before renewed negotiations.

UN General Assembly's 81st Session (8-28 September)

The event will bring together heads of state and government from more than 190 countries at a time of significant geopolitical volatility. The session – officially opening on 8 September and followed by a General Debate during 22-28 September – will oversee the final stages of appointing the next UN Secretary-General. The 81st UNGA will be crucial for advancing the "Pact for the Future" (the political vision) and the UN80 initiative (the institutional delivery mechanism), the aim of which is to make the UN more efficient and "fit-for-purpose" by reviewing mandates, improving efficiency, and undertaking structural reforms. Within the reform agenda, the Global Digital Compact has become a flashpoint for US-China competition over AI governance. UNGA debates will be critical in shaping global norms for AI, pitting the US vision of open, values-based innovation against China's model of state-centric control. Numerous bilateral meetings between world leaders will take place alongside formal UN proceedings.

Source: Bernd Dittrich on Unsplash

Key events, issues and themes to watch in **September 2026**

Listed in chronological order

18th BRICS+ Summit in India (12-13 September)

Watchpoints include signs of a deepening China-India diplomatic thaw and BRICS+ progress toward local currency settlement. India's official agenda includes cooperation and resilience, development finance, trade finance and global institutional reform. To watch will be whether President Xi's expected attendance – marking his first visit to India in seven years – signals an acceleration in the ongoing China-India diplomatic thaw.

The summit could also revive discussion of local-currency settlement, BRICS Pay (an initiative to facilitate cross-border payments in BRICS currencies) and the BRICS 'Unit', a prototype gold-backed settlement instrument. Nevertheless, progress toward cooperation on de-dollarisation will likely remain gradual given member states' diverging geopolitical priorities.

Sweden's general election (13 September)

The election will shape the policy direction of one of the EU's largest economies and a key NATO member amid heightened security concerns in the Baltic region. The governing Moderate Party-led centre-right coalition, supported by the Sweden Democrats will compete against the opposition Social Democrats and their centre-left allies. The campaign is expected to focus on crime, immigration, cost of living, economic competitiveness, energy security and defence following Sweden's accession to NATO. The outcome will influence Sweden's approach to defence spending, support for Ukraine, migration, energy policy and industrial competitiveness, while providing an important indicator of broader political trends across Northern Europe.

European Parliament plenary session (15 September)

The first plenary after the summer recess is expected to be a key legislative session as MEPs resume negotiations on the Commission's autumn agenda. Particular attention will focus on implementation of the European Defence Readiness 2030 agenda; follow up to the Competitiveness Compass, including measures to reduce regulatory burdens and strengthen the Single Market; and progress on the Economic Security Strategy, notably outbound investment screening and export controls for sensitive technologies.
The Parliament is also expected to debate additional support for Ukraine, including use of frozen Russian assets, the next package of sanctions, and progress on Ukraine's EU accession.

Source: Frederic Köberl on Unsplash

Key events, issues and themes to watch in **September 2026**

Listed in chronological order

Russian state Duma elections (18-20 September)

While the elections are not expected to be either free or competitive, they remain an important political event as the Kremlin seeks to reinforce domestic legitimacy. The vote will be closely watched for turnout, the performance of the ruling United Russia party, and any signs of public discontent despite tight political controls. Internationally, the results will be scrutinized for indications of the regimes confidence and willingness to sustain military operations in Ukraine, while any voting in occupied Ukrainian territories is likely to draw strong international condemnation. While the results of the elections are a forgone conclusion for President Putin's United Russia party, Mr. Putin appears to be avoiding a possible mass mobilization announcement in light of battlefield losses in Ukraine until after the elections.

France Senate election (27 September)

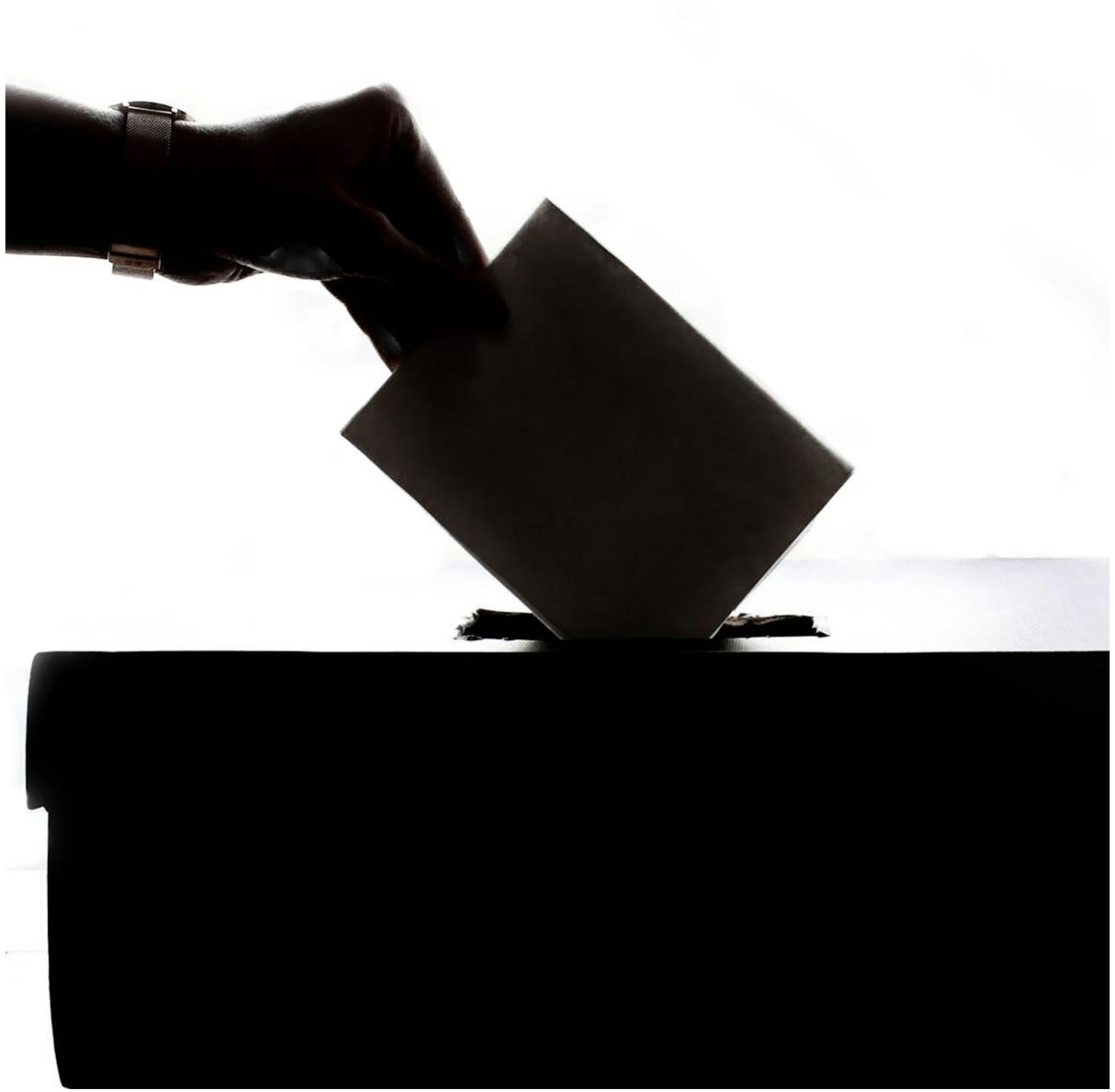
The vote should provide insights into the state of political play ahead of next year's presidential election. 178 out of the Senate's 348 seats will be contested in the indirect election to the Upper House of France's parliament. The electoral college mainly consists of local officials. While the election's outcome on state policy will likely be relatively limited, the outcome could provide signals ahead of the French presidential election in April 2027.

A key development to watch is whether the far-right National Rally Party can secure enough seats to form its first parliamentary group in the Senate. Marine Le Pen, the party's presidential candidate, continues to hold a significant lead in recent opinion polls.

Key US government funding deadline (before 30 September)

30 September marks the end of the US federal fiscal year, and it's the "hard" deadline by which Congress must pass all remaining appropriations bills or trigger another shutdown. This is a real risk given that 2026 has already seen two shutdowns tied to disputes over immigration enforcement funding, including a record 76-day partial DHS shutdown.

Congress still has yet to reconcile FY2027 appropriations work before the fiscal year closes. It's also political loaded timing – landing right before the 3 November midterms. How Congress handles this deadline (smooth passage versus another standoff) could shape voter sentiment, especially on issues like immigration enforcement that have already driven this year's funding fights.



Source: Element5 Digital on Unsplash

Appendix 1

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Something else Warsh said at Jackson Hole

August 31, 2026

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Executive summary

Kevin Warsh used his first Jackson Hole address as Chairman to make an inflation-forward case, and the market heard it. Yields on two-year treasuries jumped 11 basis points on the day and the implied probability of a September interest rate hike jumped from 35% to 58%.

The longer-term implications for sovereign debt were less talked about. This piece looks at some of the deeper issues behind the megatrend of sovereign debt – particularly given Wash's interesting comment that "Trends matter most". Indeed, our megatrend framework identifies sovereign debt as the most concerning negative trend facing developed economies over the medium term.

Yet, our proprietary data shows that people are more interested in the equity market than bond market. Unusually, that preference extends even to periods of economic uncertainty. Indeed, 28% of Americans would buy equities during periods of economic uncertainty while only 24% would buy bonds. In the UK, the same picture exists with figures of 25% and 18% respectively.

We have previously shown that since interest rates normalised in 2022, the number of Vix spike events per month has risen 51% versus the pre-covid level. More shocks are arriving into a market that is less willing to own the traditional shock absorber.

The question is whether the potential productivity gains from AI will substantially alleviate the negative weight of the sovereign debt burden in many developed countries. Our tech megatrend indicates this is possible, and so does history. However, for the optimistic scenario to eventuate, the positive impacts of AI on the economy need to accelerate further. That leaves the medium-term distribution of outcomes for sovereign debt wide and unusually asymmetric.

Something else Warsh said ...

The market reaction last Friday was focussed on the here and now. Warsh argued that "the responsibility for 65 months of sustained, elevated inflation sits squarely with the central bank", and remarked that "we have work to do" whilst pointing to PCE inflation running at 3.7% over twelve months and 4.1% over six. Markets moved the September probability materially higher and two-year yields jumped 11bps.

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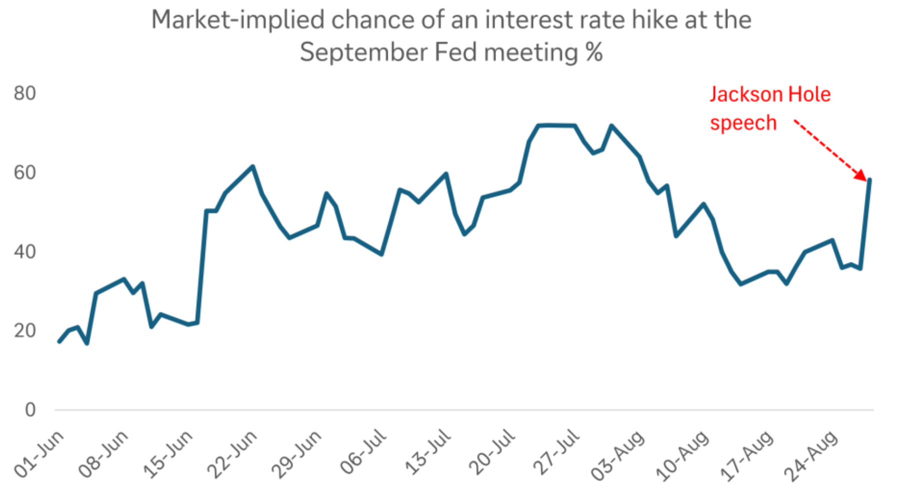
Figure 1: An inflation-forward speech spiked the chance of a September rate hike



Source: Bloomberg Finance L.P., Deutsche Bank Research

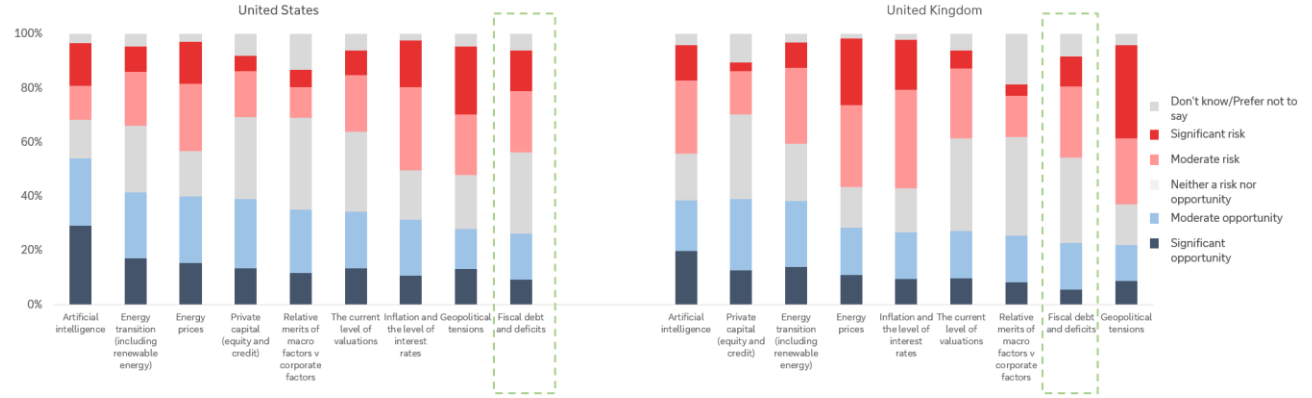
Something else Warsh said received very little coverage – "Trends matter most". It was easy to interpret this as referring to the current trajectory of short-term inflation. However, it is possible, and even likely, that he was referring to the Fed's decision making around sovereign debt as a whole – across the short, medium and long term.

Indeed, the current situation of US sovereign debt and its trend is one of the key reasons why Treasury Secretary Bessent has previously said he wants to target the 10-year yield. The current buyback programme falls into line here. The key thing here is the credibility of the fiscal trajectory because, as future shocks arise, the way the bond market performs will affect broader portfolios.

Our worry is how investors view bonds

A proprietary survey from our dbDataInsights team found that the average person in the US and UK sees more opportunity in equities rather than bonds. First, when investors assess fiscal debt against other assets and themes, they see a lot of risk and fewer opportunities. The sell-off in bonds over the last 4 years (following a 40-year bull market) has certainly contributed.

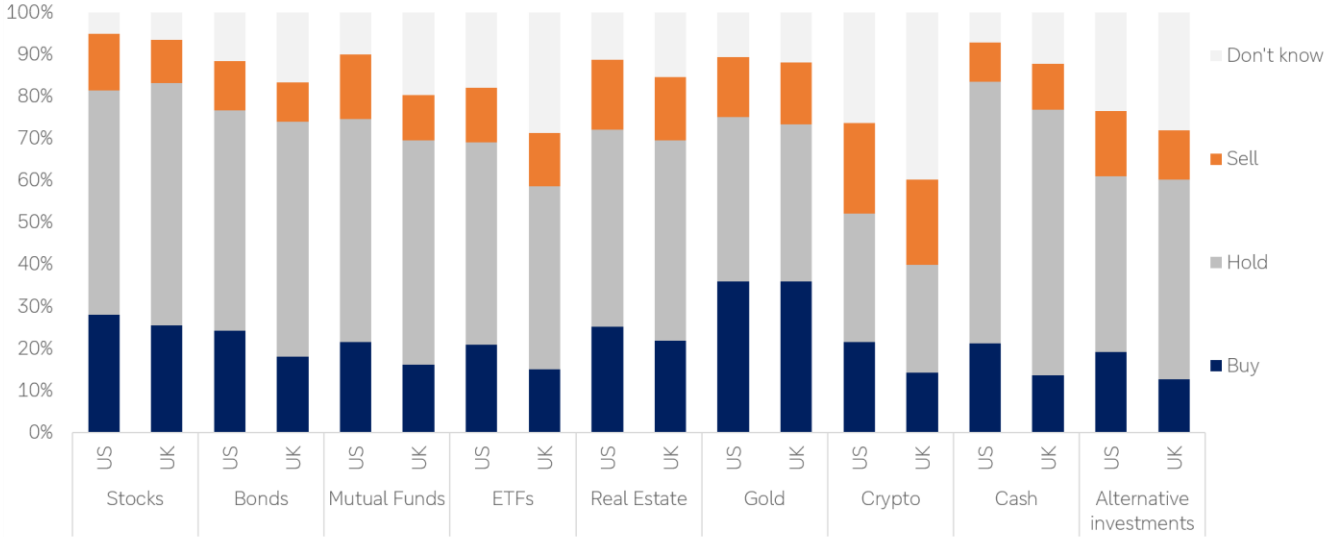
Figure 2: Survey question: “For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026"



Source: dbDataInsights

Second, in times of economic uncertainty, investors are more likely to want to buy stocks than they are to buy bonds. This is a reversal of the textbook relationship between the two:

Figure 3: Survey question: 'In any period of economic uncertainty, what do you think is the best strategy for each of the following investment asset categories?”



Source: dbDataInsights

This tilt away from bonds is not necessarily irrational. Since 2022, investors have found that bonds can lose significant value at the same time that equities fall. Our own work on haven assets shows that, since 2020, the combination of six traditional haven assets (gold, dollar, Swiss franc, yen, 10yr treasuries and bunds) has failed more often than it has worked, and it failed specifically during the four big shocks of the decade (Covid, rate rises, tariffs, Iran).

The worry into the medium term is whether investors continue to see the bond market this way as sovereign debt positions across most developed countries continue to worsen. That will affect investor portfolios given that bonds are supposed to be a counterpoint that rallies when everything else falls and avoids investors being forced to sell risk assets into weakness.

Why a less-loved bond market is a problem: uncertainty is rising

We have shown previously that since interest rates normalised in 2022, the number of VIX spike events per month has increased by 51% compared with the pre-Covid level. This follows from our broader megatrend framework that tells us that as interest rates have normalised in the 2020s, megatrends have re-emerged as a defining force in economies and markets for the first time since the 2008 financial crisis.

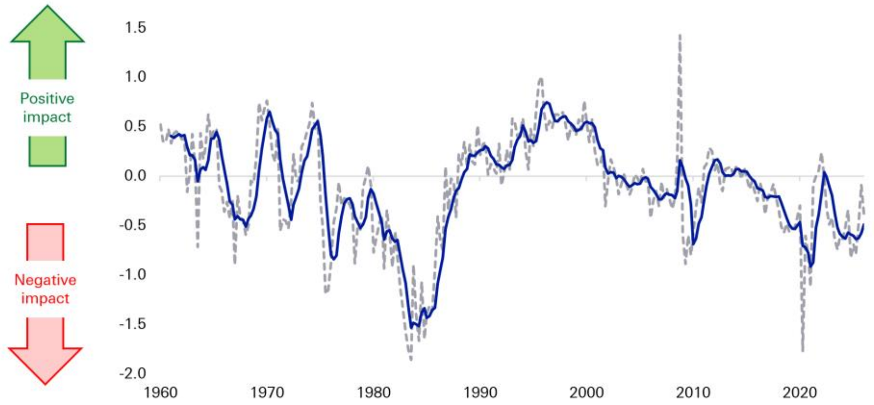
So, we have more shocks arriving, a haven complex that has been unreliable through the shocks we have already had, and an investor base that says it would rather buy equities than bonds when things become uncertain. That is a fragile configuration. It is also one in which the sovereign debt megatrend can do damage. That is, not through a gradual repricing, but through unpredictable sovereign debt sell-offs, particularly triggered by exogenous shocks and investor emotion.

Sovereign debt is the most concerning megatrend in our framework

Our megatrend model tracks six global forces: technology, sovereign debt and deficits, geopolitics and globalisation, domestic politics and social discontent, demography, and energy disruption and transition. We use almost 100 data series aggregated into a quarterly indicator for each megatrend, all standardised so they can be compared and combined.

Of the six, sovereign debt is the one we are most concerned about. Indeed, our sovereign deficits indicator has been trending down for three decades. The deterioration is being compounded by demography: working-age populations are shrinking across most of the G20, which simultaneously weakens the revenue base and strengthens the claim on the spending side.

Figure 4: Our sovereign debt megatrend indicator and its impact on economies and markets



Source: Bloomberg Finance L.P., Haver Analytics, Factset, Deutsche Bank Research

In terms of where debt is heading, in the US, the CBO has public debt rising from around 100% of GDP to 120% by 2036, with deficits pushing over 6% of GDP. Most other developed economies face a version of the same problem. Aside from declining demographics, the deterioration in sovereign debt in interconnected with domestic politics and social discontent, and geopolitics.

Even our optimistic scenario only reaches neutral

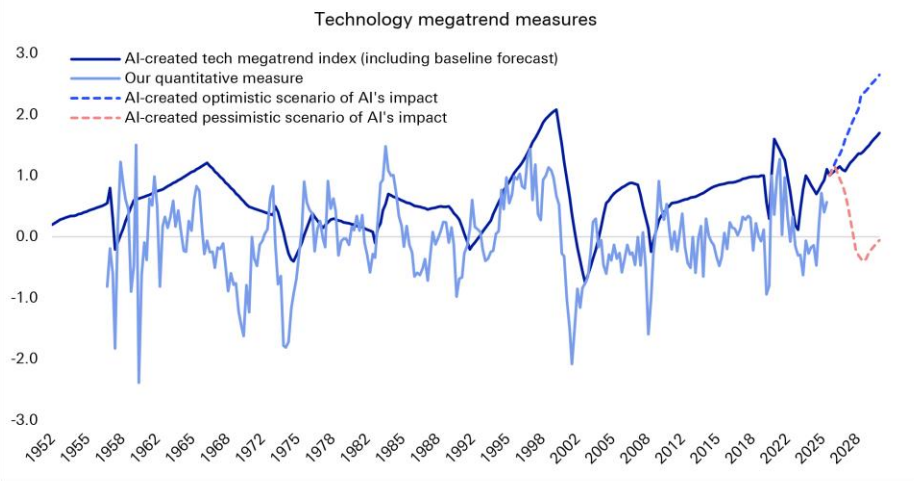
We ran the sovereign debt megatrend forward to 2030 under baseline, optimistic and pessimistic scenarios. In the optimistic case — where AI adoption accelerates, productivity growth lifts materially, government spending contributes positively to GDP, and global demand for treasuries holds up — the indicator moves from having a clearly negative impact on markets and economies to having a broadly neutral one.

None of our scenarios move the sovereign debt megatrend materially into positive territory. That asymmetry should shape how the medium-term risk is positioned. The best realistic case for sovereign debt over the next five years is that it stops being a drag.

Can technology rescue us? Warsh asked the right questions

Technology is the one megatrend currently making a clearly positive contribution to global economies and markets. And our analysis points to AI being significantly more impactful than the internet of the 1990s. Indeed, whenever our technology indicator has accelerated to a peak, productivity growth has accelerated sharply behind it: to 3-4% for sustained periods in the 1990s. Our baseline has the indicator exceeding its 1990s peak by 2030, which implies productivity growth potentially above that range.

Figure 5: Should our tech indicator continue to climb as we expect, the implied productivity boost will go some way towards offsetting the worsening sovereign debt position.



Source: Bloomberg Finance L.P., Haver Analytics, Factset, Deutsche Bank Research

That is the escape route from the debt overhang and Chair Warsh's remarks included several references to this. He asked, "Will the application of AI cause a significant, sustained rise in productivity across the economy? He also pointed to the conversation between AI being "complementary or competitive to labour" He also asked, "Will the next generation of AI models demand even greater capital intensity, or will the models themselves help devise a capital-light solution?"

We would draw attention to the capital-intensity question in particular, because it has a bearing on sovereign debt. In the future, a capital-hungry AI build-out may compete with the government for savings at exactly the moment when the government's borrowing requirement is rising. The same technology can therefore either compound the sovereign debt problem in the near term or relieve it, depending on which way the question resolves.

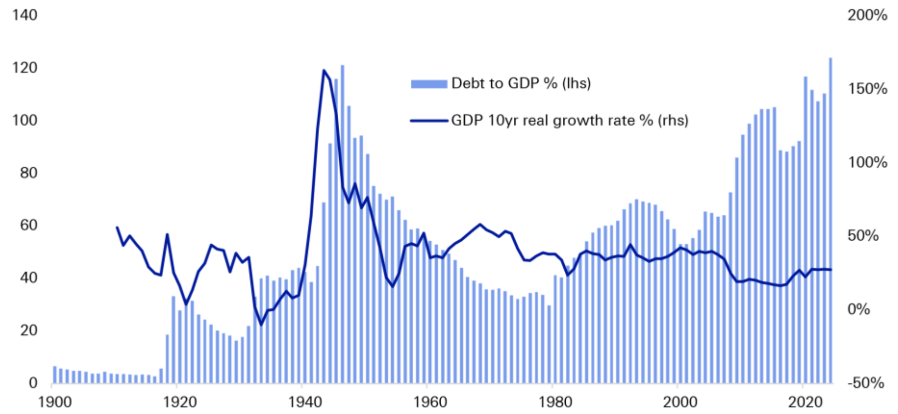
Warsh is waiting for his task forces before committing to answers, and has said explicitly that their recommendations "will come later and have no bearing on decisions we make in the current policy conjuncture".

History says a bad sovereign debt position can be overcome

None of this is a prediction of crisis, and history is on the side of the optimists more often than the mood suggests.

The clearest parallel is the "Long Depression" in the US in the mid-1870s, which followed a financial crisis and investor fear at the overbuilding of railroads — a capital-intensive general-purpose technology financed enthusiastically and then abruptly distrusted. There was little government intervention and unemployment likely topped double digits. Yet towards the end of that decade the benefits of the railway technology began to feed through to the wider economy, several other trends including sovereign debt moved in a positive direction at the same time, and the result was startling economic growth. The post-war experience is the second example: the US emerged from the Second World War with debt above 100% of GDP and then delivered thirty-five years of deleveraging alongside relatively strong growth.

Figure 6: US debt-to-GDP and GDP growth rate



Source: Fineon, Bloomberg Finance L.P., Haver Analytics, Factset, Deutsche Bank Research

The lesson from both episodes is that high debt can be overcome when the other megatrends pull in the same direction. After WWII, globalisation expanded quickly, domestic politics were comparatively stable, social discontent was low, and energy technology was a net positive. The debt burden was carried by other things going well.

Today, things are different. Technology is positive force on the medium-term outlook for the global economy and energy modestly so. However, the remaining four megatrends — sovereign debt, demography, domestic politics and geopolitics — are having a negative effect. Technology, therefore, has a lot of work to do.

What this means for portfolios

We would draw four conclusions for medium-term positioning.

1. Investors spent Friday trading the Fed but the more durable question is what compensation is required to fund a debt stock heading towards 120% of GDP, especially if the average person continues to see equities as more attractive than bonds, even in bad times.

2. Bonds and (safe) currencies can be risk assets at different times: We have flagged before that haven assets could become risk assets, and that treasuries and the dollar in particular will become more volatile if the sovereign deficits megatrend deteriorates further. The haven correlation data since 2020 shows that hedging becomes materially harder when the hedge is itself exposed to the trend being hedged.

3. Sovereigns are increasingly being judged on demographics and fiscal credibility: Yields of countries with ageing populations and no credible fiscal plan for the resulting dependency ratios will that come under increased volatility.

4. If AI delivers, the sovereign debt problem becomes manageable and the equity market carries the benefit. If it does not, there is no obvious hedge. That argues for genuine exposure to the productivity outcome, but also for scepticism about business models that require capital intensity to keep rising indefinitely. Warsh's capital-light question cuts both ways.

What we are watching

Three key sovereign deficit variables we are tracking (among others):

1. The contribution of government spending to GDP. A potential bright spot: this indicator has recently risen sharply. If that continues alongside greater AI adoption, there is a path to sovereign deficits becoming less of a drag.

2. The US 10-year yield versus the G7 average. Treasury yields have been higher than the G7 average for most of this century, but the gap has begun to shrink. Continued convergence would signal that global demand for treasuries remains strong, which in turn supports the resilience of the US twin deficits.

3. The fiscal policy uncertainty index. Turned deeply negative in 2024 and 2025, below the Covid lows. Subjective, but an important lead indicator for short-term market views — particularly during an exogenous shock, where it is the transmission channel into fear-based sell-offs.

Appendix 1

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China - Modernizing the monetary policy framework

August 26, 2026

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Executive Summary

The **path to modernizing China's monetary policy framework has reached an important point.** The announcement by PBoC Governor Pan Gongsheng in June, introducing overnight reverse repo operations and stating the intent to narrow the interest-rate corridor, is a key step in this journey. **What is the ultimate destination, and what else does the PBoC need to do to get there?**

China currently operates a **hybrid monetary-policy framework** in which both policy rates and monetary/credit aggregates play important roles. The objective is ultimately to move away from a framework where monetary/credit aggregates serve as intermediate policy targets toward one where policy rates increasingly serve as the primary policy signal, with liquidity-management tools continuing to support policy implementation and transmission.

Key takeaways:

— Over the past decade, China's monetary policy framework has evolved toward greater use of price-based tools, although quantitative measures continue to play an important role.

— The objective is for policy rates to play a larger role in signaling the monetary stance, while liquidity-management and structural tools help with policy implementation.

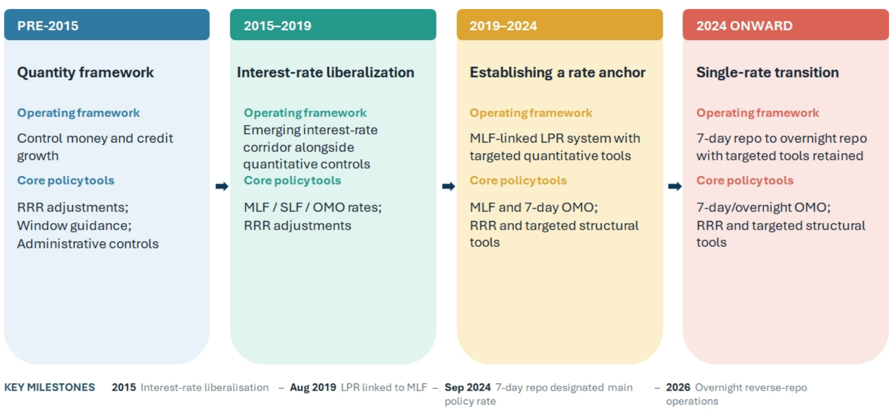
— Key reform priorities include strengthening policy expectations, reinforcing the interest-rate corridor, deepening bond-market liquidity, improving credit transmission and increasing exchange rate flexibility.

— Improving the monetary policy framework alone may not be enough. Achieving more effective economic management will also require supportive fiscal policies, better coordination across policymakers and timely measures to stabilize growth.

A decade of monetary policy modernization

Over the past decade, China has undertaken one of the most important monetary policy framework reforms among major economies. As interest-rate liberalization progressed, financial markets deepened and the economy became more complex, a framework centered mainly on quantitative controls became less effective. Policymakers therefore began shifting toward a more market-based system in which prices, rather than administrative controls, play a larger role in transmitting monetary policy.

Figure 1: China has gradually strengthened its price-based monetary policy framework over the past decade

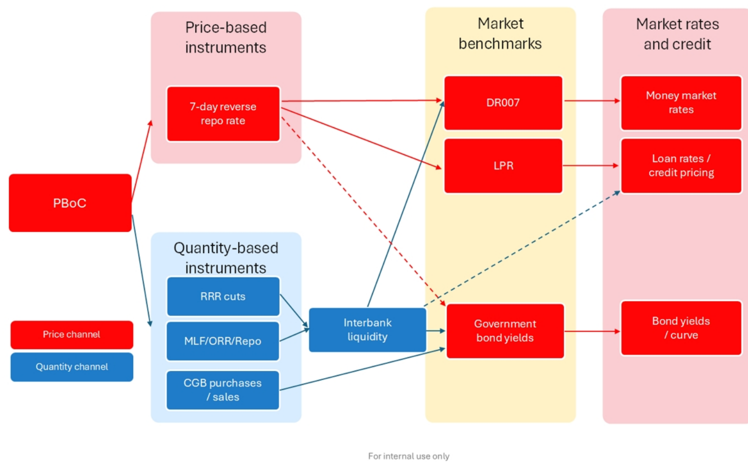


Source :Deutsche Bank Research

China's framework today looks much more interest-rate-based than it did a decade ago. Benchmark lending and deposit rates have largely been liberalized, the Loan Prime Rate has become a more important reference for bank lending rates, and the 7-day reverse repo rate now serves as the primary policy rate. At the same time, **this remains a hybrid system**. Today, the PBoC's policy toolkit can be broadly divided into four categories:

- **Policy rate**: the 7-day reverse repo rate serves as the primary policy rate and anchors short-term money-market rates.
- **Short- to medium-term liquidity tools**: Reverse repos, outright reverse repos (ORR), and the Medium-Term Lending Facility (MLF), which influence interbank liquidity conditions and bank funding costs.
- **Long-term liquidity tools**: Reserve requirement ratio (RRR) adjustments and outright purchases or sales of government bonds, which affect the structural liquidity position of the banking system.
- **Structural monetary policy tools**: Relending, rediscounting and other targeted facilities designed to channel credit towards strategic sectors and policy priorities (see Appendix B for the full list of liquidity tools).

Figure 2: China's monetary framework remains a hybrid system in transition



Source : Deutsche Bank Research

Some of the key building blocks of a modern interest-rate framework are now in place, with the 7-day reverse repo rate serving as the primary policy anchor and market-based interest rates playing a larger role in policy implementation than in the past. **The objective of China's monetary policy modernization is not the elimination of quantitative or structural tools, but the establishment of interest rates as the primary mechanism through which monetary policy influences financial conditions and economic activity.**

Figure 3: China's monetary policy differs from that of DMs more in objectives and implementation than tools

Dimension	PBoC	Fed	ECB	BoE	BoK
Primary objective	Price stability, growth support, credit conditions and financial stability all play important roles.	Inflation and employment	Price stability	Price stability with support for growth/employment	Price stability
Policy toolkit	Policy rate and liquidity operations	Policy rate and balance sheet	Policy rate and balance sheet	Policy rate and balance sheet	Policy rate and liquidity operations
No. of annual public communication	Quarterly monetary policy reports and occasional official speeches	Minutes and votes published, over 400 official speeches per year	Meeting accounts published, over 400 official speeches per year	Minutes and vote splits published, over 100 official speeches per year	Minutes and voting records published, ~10 official speeches per year
Communication style	Flexibility-focused. Quarterly reports and qualitative guidance	Highly transparent and forecast-driven	Rules and forecast-based	Forecast and minutes-driven	Relatively transparent by Asian standards
Forecast publication	Limited forecast disclosure	SEP projections	Staff forecasts	Detailed MPR forecasts	Growth and inflation forecasts
FX consideration	Exchange-rate stability remains an important consideration	Little direct role	Limited	Limited	Important but secondary

Source :Deutsche Bank Research; Note: Though financial stability is an important consideration for all central banks, it is not treated as a primary objective here due to the lack of an explicit and measurable target

Against this backdrop, an important question is whether the modernization of the monetary-policy framework has translated into stronger policy transmission. While there are signs of progress, the evidence remains mixed. Since January 2022, three of the seven policy rate cuts have been followed by a meaningful decline in lending rates within the subsequent month.

As a result, the next phase of monetary policy modernization is likely to focus on strengthening the channels through which policy rates influence financial markets and economic activity. This suggests that future reforms are likely to center on areas such as expectations formation, financial-market depth, interest-rate pass-through and the broader role of market signals within the monetary policy framework.

The next chapter of monetary policy modernization

What is China's likely destination?

The evidence above suggests that China's monetary-policy framework has entered a new phase of development. The key question for the next stage we believe is therefore not whether China will reduce its reliance on liquidity-management tools, but whether policy rates will increasingly displace monetary and credit aggregates as the primary guide for monetary-policy implementation and transmission. We believe the answer is yes. Over the past decade, policymakers have established many of the key building blocks of a more market-based framework, including a clearer policy-rate structure, greater reliance on market pricing and a reduced emphasis on monetary aggregates. As a result, the next phase of reform is likely to focus less on creating new instruments and more on strengthening the channels through which policy signals affect financial markets and economic activity.

Importantly, this does not imply convergence towards a conventional developed-market model. Structural and quantitative tools are likely to remain part of the framework, while financial stability, exchange-rate stability and development objectives will continue to influence policy decisions. Rather, modernization is likely to involve gradually increasing the role of interest rates as the primary mechanism through which monetary policy influences financial conditions, while improving the effectiveness of policy transmission within China's broader policy framework.

Moreover, China's evolving macroeconomic environment may reinforce the need for quantity-based and structural policy tools. As policy rates remain structurally lower amid accommodative policy stances, China may increasingly face constraints similar to those experienced by other major economies near the effective lower bound. In such an environment, conventional interest-rate adjustments may become less powerful, increasing the importance of balance-sheet policies, liquidity facilities and targeted credit-support measures. Rather than disappearing, quantity-based tools may therefore become a permanent complement to interest-rate policy, much as they have in many developed economies since the Global Financial Crisis.

Priorities for the next phase of monetary policy modernization

Simplifying policy objectives to strengthen market expectations. As China's monetary policy framework continues to evolve, policy communication and market expectations are likely to play an increasingly important role in shaping financial conditions. Modern monetary policy increasingly operates through expectations of future policy settings rather than current policy rates alone and the central of that is a clear set of policy objectives by the central bank. In China, the PBoC operates under a broad set of objectives, including growth, price stability, financial stability, exchange-rate stability and broader development goals, providing policymakers with flexibility. Continued development of the framework and clear communication of policy intentions could further support the role of expectations in financial markets.

Greater clarity around policy objectives would strengthen the expectations channel. This would not require adopting a formal inflation-targeting regime or a Fed-style dual mandate. A more pragmatic approach could be to streamline the PBoC’s mandate around three primary objectives. While the relative emphasis differs across jurisdictions, most major central banks operate with a similar set of objectives including inflation, growth or employment as well as financial stability.

Aligning the PBoC's mandate more explicitly around these objectives could improve policy transparency without requiring a wholesale adoption of a Western monetary-policy framework. A more focused mandate, supported by clear and consistent communication, would make the PBoC's reaction function easier for markets to interpret and enhance the signaling power of policy-rate changes.

Strengthening the corridor to improve short-term transmission. A second priority is improving the operating framework itself. China has made considerable progress in establishing the 7-day reverse repo rate as the primary policy rate, while Governor Pan has also signaled a possible future transition towards a more overnight-rate-based framework. At the same time, the interest-rate corridor continues to evolve as policymakers refine the monetary-policy operating framework. Although temporary repo and reverse repo operations were introduced in 2024 and reformed in 2026, they have not yet become a significant part of day-to-day operations. As a result, their role in guiding money-market conditions and supporting policy implementation will remain an area of interest for market participants.

In the United States, market rates such as SOFR generally trade close to policy rates within a well-established floor system. By contrast, China's corridor remains more of an evolving guide than a tightly binding operating framework. Future reforms are therefore likely to focus on narrowing the corridor, strengthening overnight liquidity operations and improving the predictability of short-term liquidity management.

A further area of development is the broader financial-stability architecture underpinning monetary operations. Recent discussions around emergency liquidity facilities for non-bank financial institutions suggest policymakers increasingly recognize the need for stronger liquidity backstops as financial markets become more market-based. A more comprehensive liquidity-safety framework could improve market resilience and support the implementation of monetary policy during periods of stress.

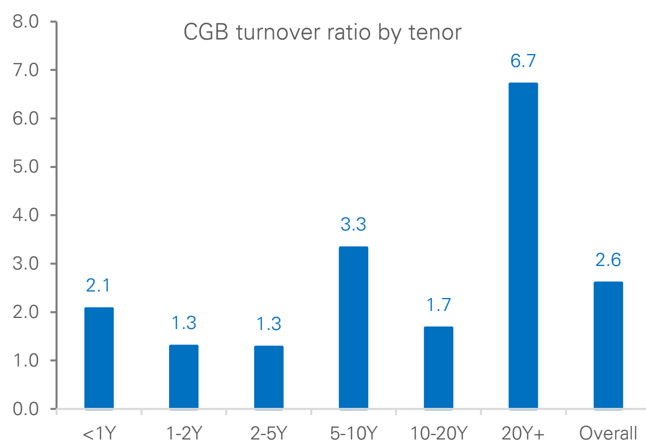
Deepening secondary bond markets to improve yield-curve transmission.China has built one of the world's largest government bond markets over the past two decades. While commercial banks continue to hold a significant share of outstanding government bonds, policymakers have also encouraged broader participation by institutional and foreign investors. As secondary-market activity continues to evolve, deeper liquidity and more active trading could further enhance the informational content of market interest rates and the role of the yield curve within the broader monetary-policy framework.

A deeper and more active government bond market would strengthen the transmission of policy expectations along the yield curve. Potential reforms include broadening participation beyond commercial banks, developing government-bond derivatives and hedging markets, improving repo and securities-lending infrastructure, strengthening market-making arrangements and enhancing transparency around issuance and liquidity operations.

These reforms would improve monetary policy transmission while strengthening the role of government bond yields as benchmarks for broader financial-market pricing.

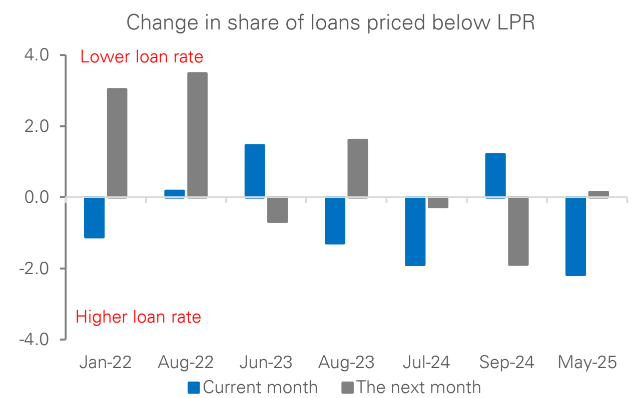
Improving liquidity circulation and credit transmission. Another area of development relates to the credit channel and the interaction between financial conditions and economic activity. While policy easing has supported ample liquidity conditions in recent years, the impact on borrowing, investment and growth can vary depending on broader economic and business conditions. China's financing system remains predominantly bank-based, with around 80% of corporate financing coming from loans, a higher share than in many major economies. Policymakers have therefore placed increasing emphasis on broadening financing channels and supporting the development of debt and equity markets alongside traditional bank lending. Consistent with this objective, PBoC Governor Pan Gongsheng in 2023 highlighted efforts to expand financing channels for private enterprises across debt, equity and bank lending. Going forward, the evolution of credit transmission is likely to reflect not only monetary-policy developments, but also broader economic conditions and structural reforms.

Figure 4: A more active bond market could support price discovery and transmission



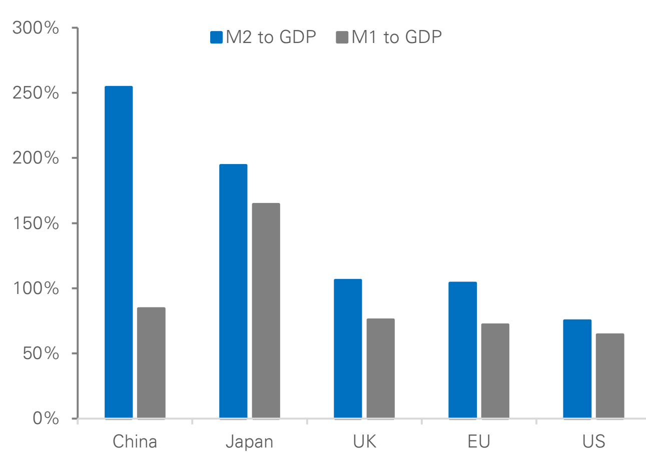
Source : Deutsche Bank Research, Wind

Figure 5: Policy-rate cuts have not always been followed by immediate declines in loan rates



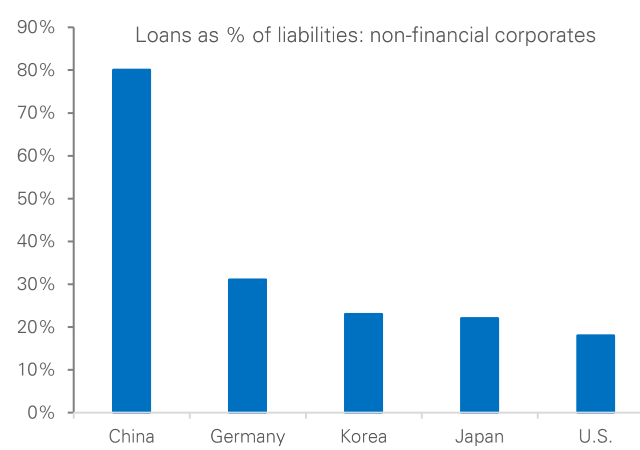
Source : Deutsche Bank Research, Wind

Figure 6: Liquidity and economic activity can diverge over time



Source : Deutsche Bank Research, Wind

Figure 7: Chinese corporates remain heavily reliant on bank-loan financing



Source : Deutsche Bank Research

Exchange-rate developments and RMB internationalization. Another area of development concerns the interaction between monetary policy, exchange rates and international financial flows. China's monetary-policy framework operates within a broader set of policy objectives, resulting in transmission dynamics that differ from those in many developed-market economies. As RMB internationalization advances and financial-market development continues, the relationship between monetary policy, capital flows and exchange-rate developments is likely to evolve further. This process will remain an important area to watch as China's financial system becomes increasingly integrated with global markets.

A modern framework, not a Western one. Ultimately, in our view, monetary policy modernization in China should not be viewed as a process of convergence towards a developed-market model. Rather, it is a process of adapting market-based monetary policy tools to China's own institutional environment. The likely end state, we believe, is a framework with a clear policy rate, deeper and more responsive financial markets, stronger policy transmission and greater transparency, while retaining broader policy objectives and strategic credit-support mechanisms. In other words, China is unlikely to become a Chinese version of the Federal Reserve. Instead, we think it is more likely to develop a modern interest-rate framework with distinctly Chinese characteristics.

Beyond monetary policy reform: A broader policy agenda

The modernization of China's monetary policy framework has brought it closer to the operational practices of many developed-market central banks. The PBoC has increasingly emphasized price-based policy instruments, strengthened its policy-rate framework, and improved the signaling and implementation of monetary policy. These developments represent an important step in the evolution of China's monetary-policy framework and should help further strengthen the role of market-based interest rates over time.

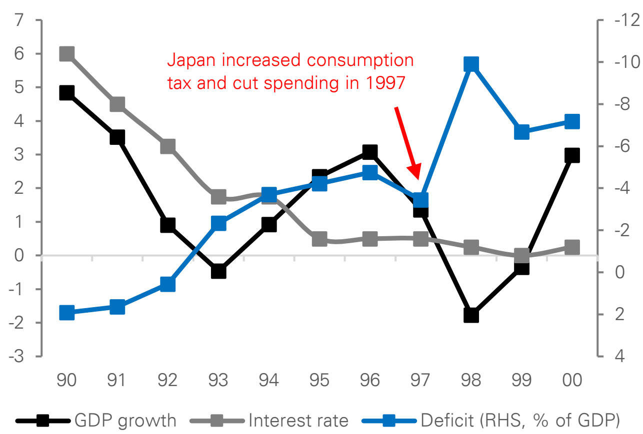
At the same time, monetary policy operates within a broader economic and financial environment. Factors such as credit demand, developments across different sectors of the economy, and ongoing adjustments in the property market can all influence how monetary-policy actions are transmitted through the economy. As a result, the evolution of monetary-policy tools is only one part of the broader policy landscape. Experience from developed economies similarly suggests that monetary policy is often most effective when supported by favorable macroeconomic and financial conditions.

More broadly, China's economic objectives are pursued through a combination of monetary, fiscal, regulatory and structural policies. We therefore view monetary-policy modernization as one component of a broader policy agenda rather than a standalone initiative. Going forward, the interaction between monetary policy and other policy measures will remain an important factor shaping economic and financial outcomes.

Monetary policy and fiscal policy interaction. The effectiveness of monetary policy is often influenced by the broader macroeconomic policy environment, including fiscal policy. Japan's experience in the late 1990s illustrates the importance of policy interaction. Despite very low interest rates and repeated monetary easing, fiscal consolidation in 1997 coincided with renewed economic weakness and a prolonged period of low inflation.

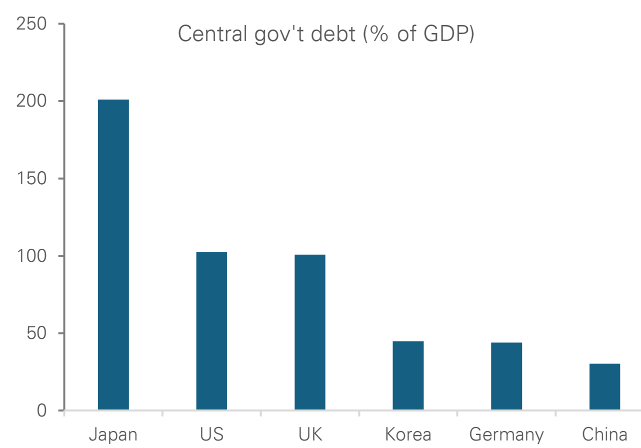
In China's case, sustaining monetary policy modernization, and potentially an easing cycle in the coming years, will need more forceful fiscal support. In recent years, policymakers have gradually expanded fiscal support while emphasizing the role of the central government in supporting economic stability. With a comparatively strong central-government balance sheet and a low-interest-rate environment (Figure 13), fiscal policy retains scope to play an important role in supporting broader macroeconomic objectives. More broadly, the interaction between monetary and fiscal policy is likely to remain an important feature of China's macroeconomic framework in the years ahead. More proactive fiscal support could help stabilize the property market, strengthen the social safety net, and support household consumption. Importantly, stronger fiscal support would complement monetary easing by generating demand for credit and investment opportunities, thereby improving the effectiveness of monetary policy itself.

Figure 8: Japan shows uncoordinated policies weaken transmission



Source : Deutsche Bank Research, Haver

Figure 9: China's central-government balance sheet remains among the strongest in the world



Source : Deutsche Bank Research, Haver, Wind

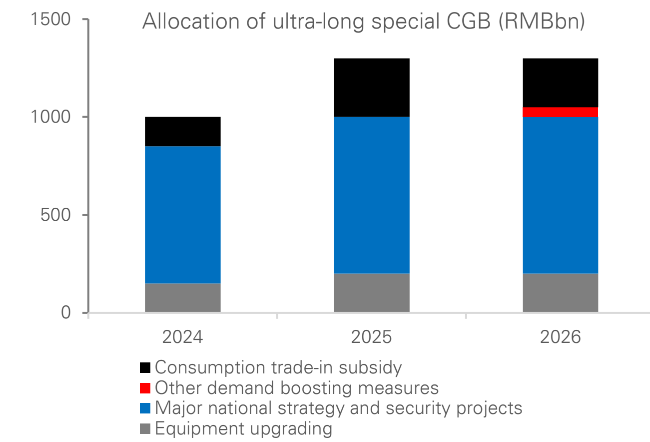
Better balancing long-term strategic priorities with near-term economic needs. China's policy agenda has increasingly emphasized long-term priorities such as technological innovation, advanced manufacturing and the development of future industries. Between 2023 and 2026, a significant share of funding raised through ultra-long-term special government bonds was directed toward these strategic objectives, highlighting policymakers' focus on enhancing long-term productivity growth and competitiveness.

At the same time, long-term development and cyclical stabilization can be mutually reinforcing. A stable macroeconomic environment can support employment, household income, business activity and investment, helping to create favorable conditions for longer-term economic upgrading. In this sense, cyclical conditions and structural transformation need not be viewed as competing objectives.

International experience suggests that strategic development policies and macroeconomic stabilization measures can coexist. South Korea, for example, has combined support for priority sectors such as semiconductors with broader macroeconomic measures aimed at maintaining economic stability during periods of cyclical weakness. More generally, a stable economic environment can provide an important foundation for investment, innovation and industrial upgrading over time.

Looking ahead, the interaction between China's long-term development objectives and broader economic conditions will remain an important theme for policymakers and markets. The pace of structural transformation, developments across key sectors of the economy, and broader macroeconomic conditions will all play a role in shaping the environment in which longer-term strategic goals are pursued.

Figure 10: Most incremental fiscal resources were allocated to strategic sectors



Source :Deutsche Bank Research, Government Work Report

Figure 11: Korea's experience shows that strategic upgrading and cyclical growth can go hand in hand

	China		Korea	
	Name	Amount	Name	Amount
Strategic and industrial policies	Major national strategy and security projects (2024–2026)	RMB2.3tn (1.6% of annual GDP)	K-Chips Act (2023)	Tax incentive (not direct spending)
	Policy financing tool to support specific strategic sectors	RMB1.3tn (0.9% of annual GDP)	Semiconductor Ecosystem Support Package (2024)	KRW26tn (1% of annual GDP)
			Chip fund (2026)	KRW5tn (0.2% of annual GDP)
Cyclical growth policies	Consumption trade-in subsidies (2024–2026)	RMB700bn (0.5% of annual GDP)	Budget expansion to boost economic activity (2025)	KRW20.2tn (0.8% of annual GDP)
	Childcare subsidies (2025–2026)	RMB180bn (0.1% of annual GDP)	Supply-shock stimulus (2026)	KRW17.3tn (0.6% of annual GDP)

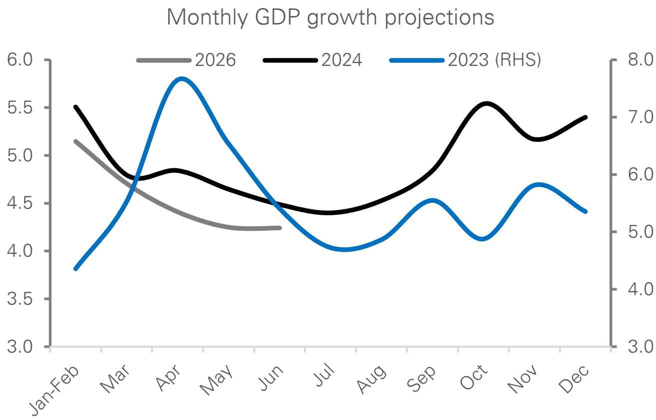
Source : Deutsche Bank Research
Note: China's strategic-sector allocation excludes investment funded by local government special bonds, which amount to around 3% of GDP annually

Proactive policy support and greater policy consistency. Expectations play an important role in shaping the effectiveness of macroeconomic policy. As such, the consistency and coordination of policy measures can influence household, business and market expectations, alongside broader economic outcomes. China's post-pandemic recovery has exhibited recurring fluctuations in growth momentum, with periods of policy support often coinciding with firmer economic activity. While policy measures have helped support activity, the consistency and persistence of policy support remain important considerations in anchoring expectations and strengthening confidence.

International experience also points to the importance of policy consistency. Japan's experience during the 1990s and 2000s demonstrated how expectations can be influenced by the interaction between monetary, fiscal and broader economic policies. Following 2013, a more coordinated policy framework was accompanied by improvements in growth, inflation and market sentiment, although external factors and global developments also played an important role.

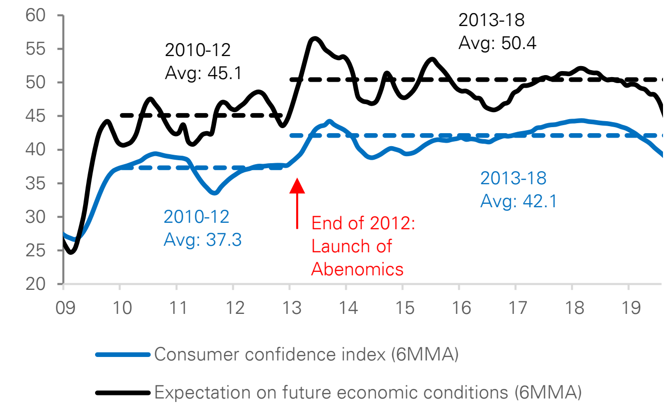
More broadly, we view the modernization of China's monetary-policy framework as an important enabling reform. A more market-based framework may further strengthen policy implementation and the role of interest rates within the financial system. At the same time, economic outcomes will continue to reflect a wide range of factors, including fiscal policy, structural developments and the broader macroeconomic environment. In this regard, the evolution of China's policy framework is likely to be shaped not only by monetary-policy reforms, but also by how different policy initiatives interact in support of longer-term economic objectives.

Figure 12: Growth cycles have highlighted the importance of policy consistency



Source : : Deutsche Bank Research, CEIC
Note: Monthly GDP growth is estimated by interpolating official GDP data based on our monthly nowcast.

Figure 13: Abenomics triggered an immediate and lasting recovery in consumer confidence



Source :Deutsche Bank Research, CEIC

Market Implications

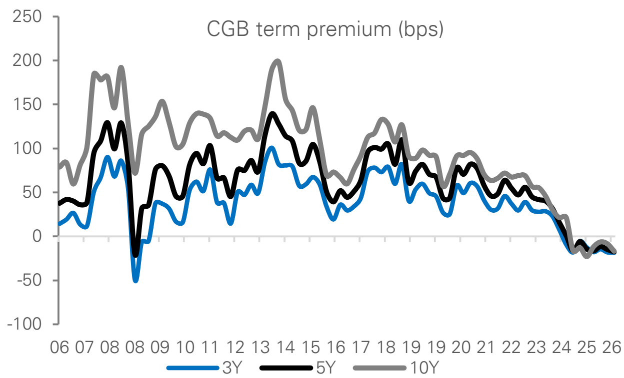
If China's monetary-policy reforms continue, they could gradually reshape the country's bond and currency markets. Over time, policy signals may become clearer, interest-rate changes may transmit more effectively through the financial system, and markets may become deeper and more responsive.

A steeper yield curve for China Government Bonds (CGBs). China's government-bond yield curve has become much flatter in recent years. This reflects both expectations that policy rates will remain low for some time and a decline in the extra compensation investors typically demand for holding longer-term bonds, known as the term premium. If monetary-policy modernization improves transmission and is supported by broader reforms, the yield curve could become steeper over time. Short-term yields could remain low if policy easing continues, while long-term yields could gradually rise as economic activity, confidence and inflation expectations strengthen.

Near-term easing, medium-term repricing higher for interest rate swaps (IRS). China's medium-term interest-rate expectations have moved lower in recent years as growth and inflation have moderated. Further policy easing could keep rates low in the near term. Over time, however, if reforms help support investment, lending and economic activity, expectations for future growth may improve. In that scenario, medium-term interest rates could gradually move higher and market pricing could become more balanced across different maturities.

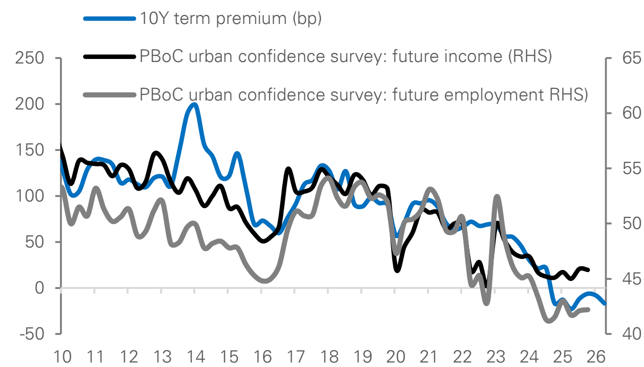
Greater volatility in RMB exchange rate. The RMB remains less volatile than many major currencies. If China continues modernizing its monetary policy framework and allows greater exchange-rate flexibility over time, the RMB could move more freely in response to economic conditions and market forces. This would likely result in greater two-way currency fluctuations while improving price discovery and supporting the development of deeper hedging and foreign-exchange markets. Given China's importance to the regional economy, greater RMB flexibility could also contribute to increased volatility in other Asian currencies that are closely linked to China through trade and financial flows.

Figure 14: CGB term premia have been declining across tenors



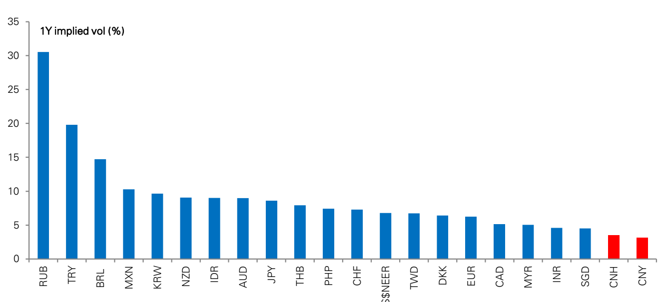
Source : Deutsche Bank Research, Bloomberg Finance LP

Figure 15: Stronger economic expectations could boost the term premium



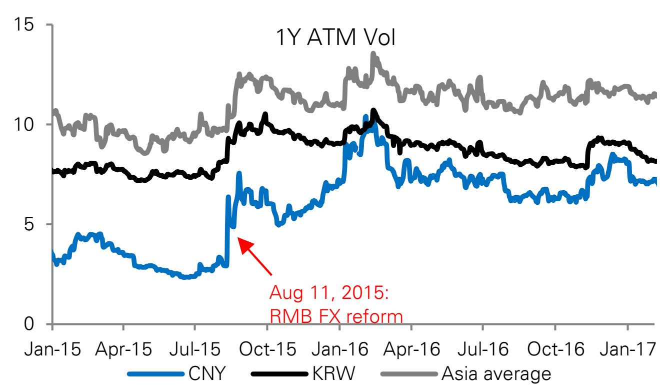
Source :Deutsche Bank Research, Bloomberg Finance LP, Wind

Figure 16: Rising RMB volatility could lift volatility across Asian FX markets



Source : Deutsche Bank Research, Bloomberg Finance LP

Figure 17: RMB FX regime change could have a significant impact on regional volatility



Source : Deutsche Bank Research, Bloomberg Finance LP

Appendix 1

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AI at 70: 14 lessons from a lifetime of boom and bust

August 2026

Adrian Cox

Deutsche Bank Research Institute

With deep dedication.

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Artificial intelligence marks 70 years since Dartmouth Summer Research Project

“Context is all you need”: clues from the past for what comes next

Seventy years ago today, the curtain came down on the summer workshop that gave birth to artificial intelligence. Dwight Eisenhower was in the White House, the Suez Crisis was gripping Britain and Elvis Presley topped the US charts with “Don’t Be Cruel” and “Hound Dog”.

The Dartmouth Summer Research project brought together a dozen or so academics convinced that every aspect of intelligence could in principle be described precisely enough for a machine to simulate it. Emboldened by the techno-optimism of the age, John McCarthy, who coined the name AI, wrote that “we think that a significant advance can be made… if a carefully selected group of scientists work on it together for a summer.”

Since then, AI has lived a lifetime of summers and winters, punctuated by stunning breakthroughs and false dawns. As it is now the cornerstone of a new investment and valuation boom, the past 70 years are a treasure trove of clues for what to expect in the future.

To paraphrase the research paper that sparked the current boom in generative AI in 2017, “Attention is all you need”, you could say “context is all you need” to understand what is to come. The signals are there, from non-linear growth to infrastructure constraints and valuation booms and busts.

Some will say “this time is different”. Maybe. But the following charts give a flavour of what could happen if it isn’t. To quote the UK No. 1 single from 70 years ago, sung by Doris Day: “Whatever Will Be, Will Be (Que Sera, Sera)”.

14 lessons from a lifetime of boom and bust

1. AI progress is non-linear
2. Progress exceeds Moore's law
3. Today's leading tech may not be tomorrow's
4. The pipeline matters
5. Cheaper AI will likely boost spending
6. Software revolutions need hardware
7. Hardware is often a bottleneck
8. Globalisation has made tech cheaper, and more at risk
9. Tech revolutions take time to pay
10. Consumers have adopted AI faster than businesses
11. The biggest valuation booms are often tech booms
12. This time is different – and will need to be
13. Early leads do not guarantee long-term winners
14. Long-term trends tend to persist

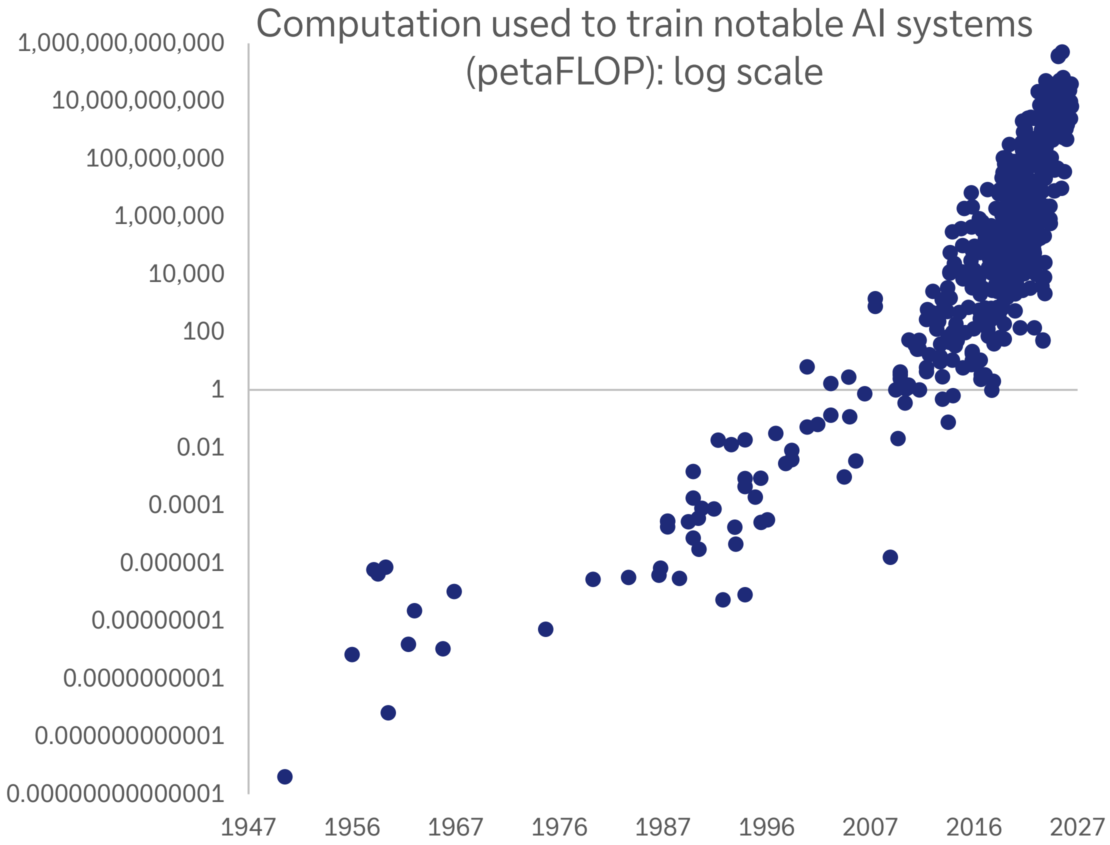
A Proposal for the

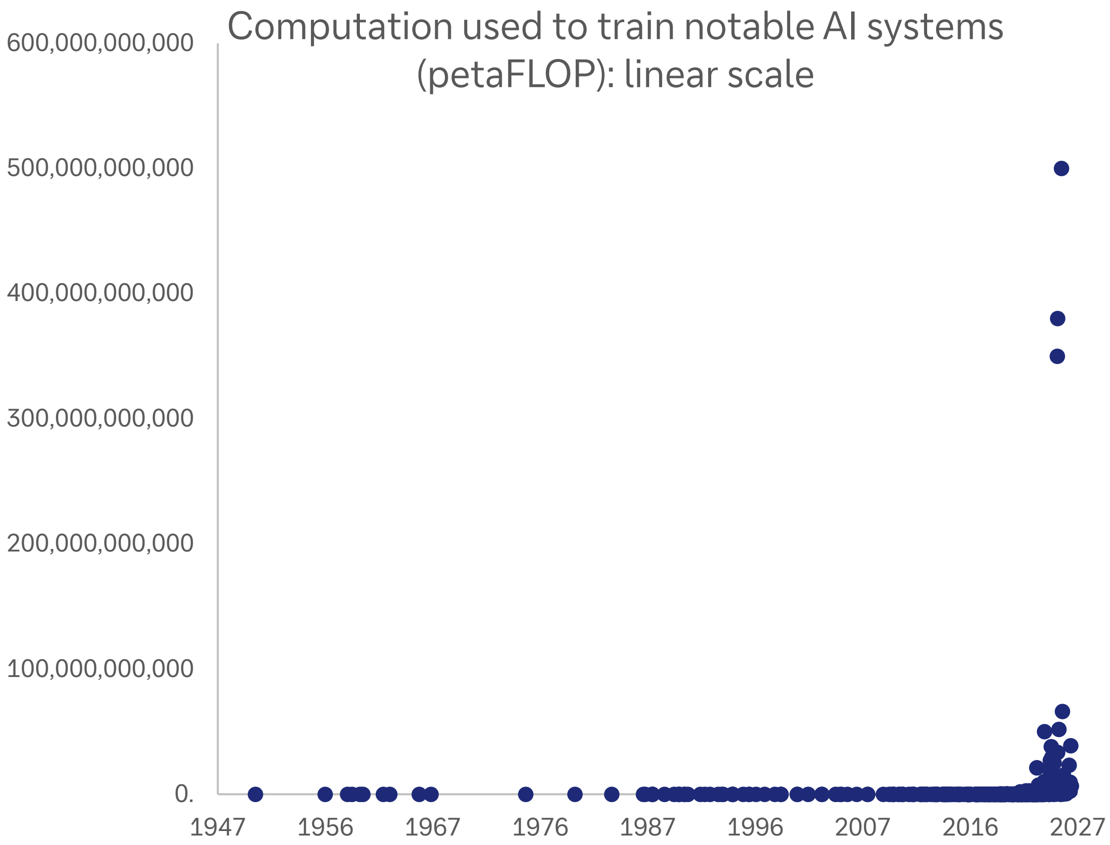
DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so pre-

1. AI progress is non-linear – and exponential growth is almost invariably underestimated

Compare what AI charts usually look like (on the left) vs what they actually represent (on the right)





Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research. Note: One FLOP (floating-point operation) represents a single arithmetic operation using floating point numbers; one petaFLOP is one quadrillion FLOPs

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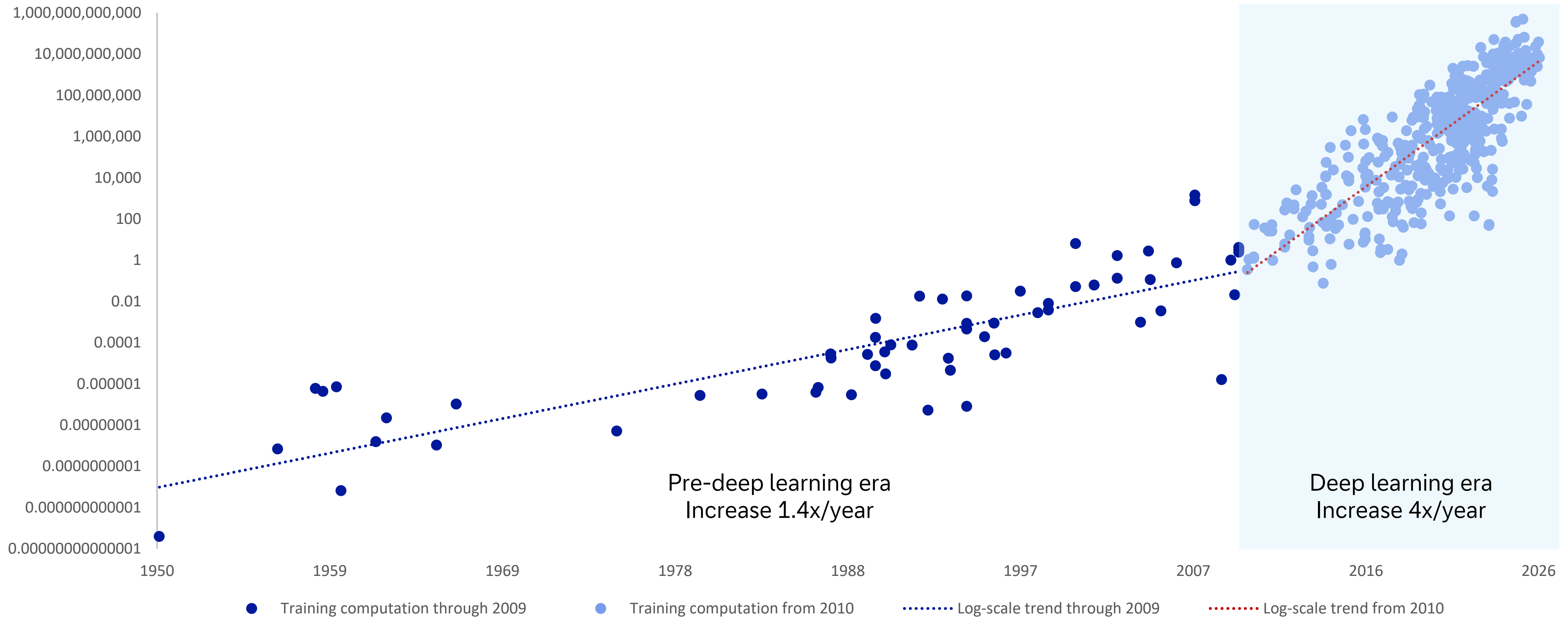
2. The pace of progress exceeds Moore's law when many factors are improving at once

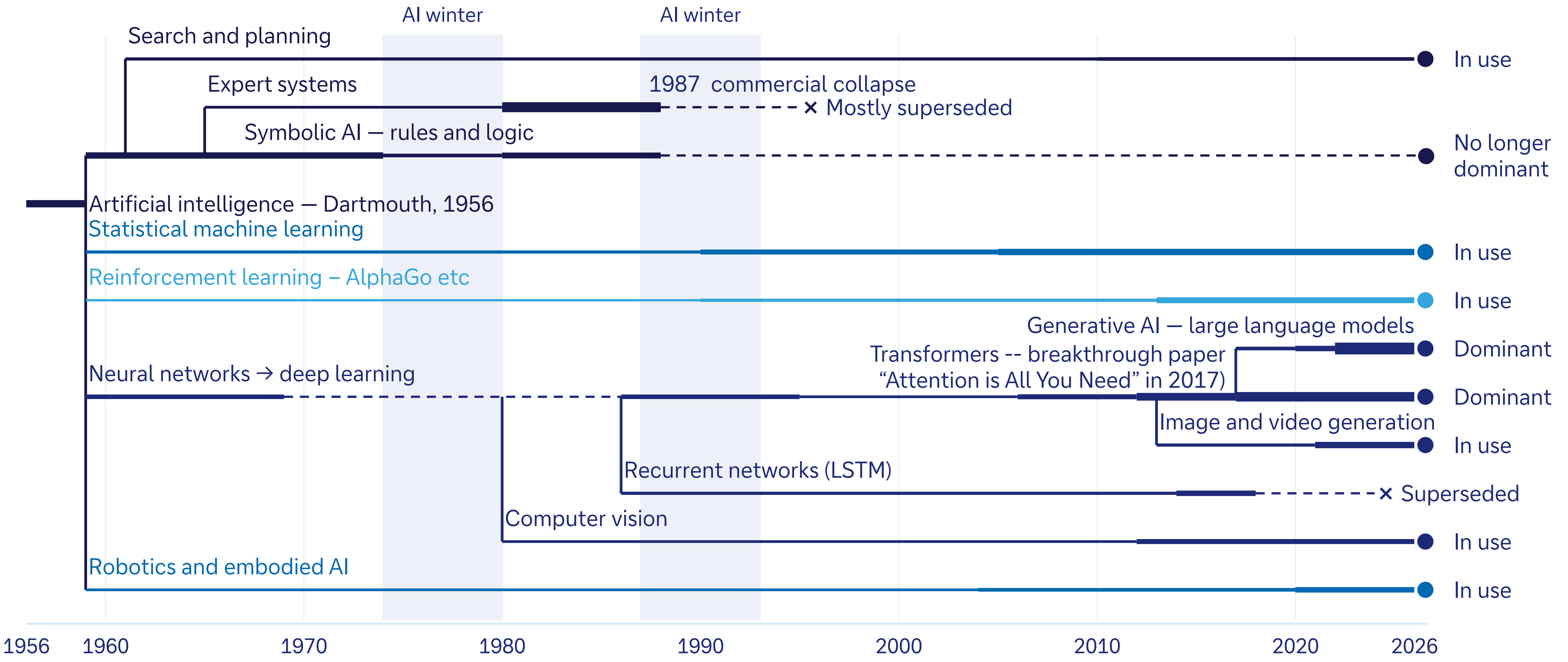
Conventional computing capability has doubled every 18-24 months but bigger and better systems, memory and algorithms have accelerated capabilities far beyond that

Computation used to train notable AI systems (petaFLOP): log scale



Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research

3. Today's leading tech may not be tomorrow's
AI has an ever-evolving family tree; large language models may give way to, eg, world models



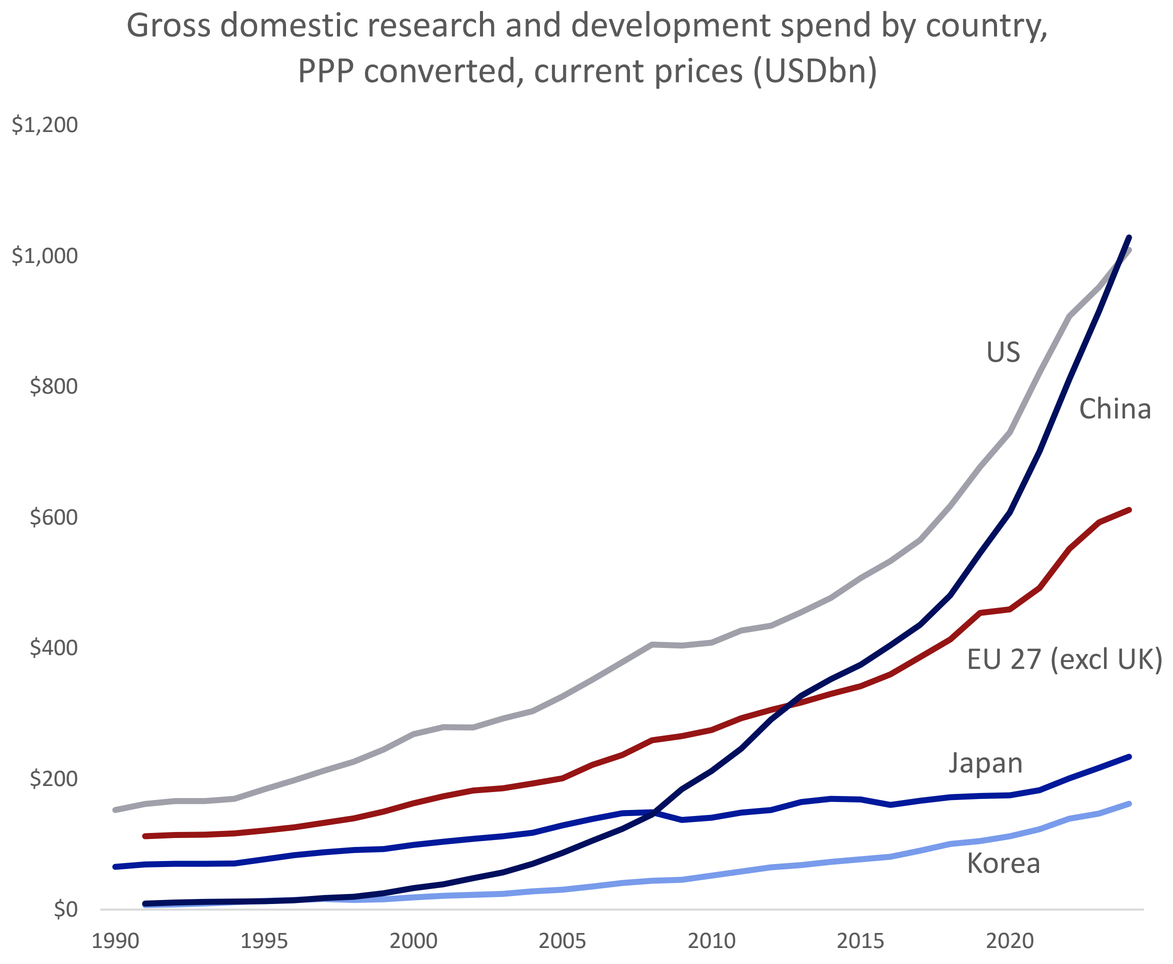
Colour = lineage · Thicker line = more activity · Shaded = AI winters · × = superseded · ● = still in use

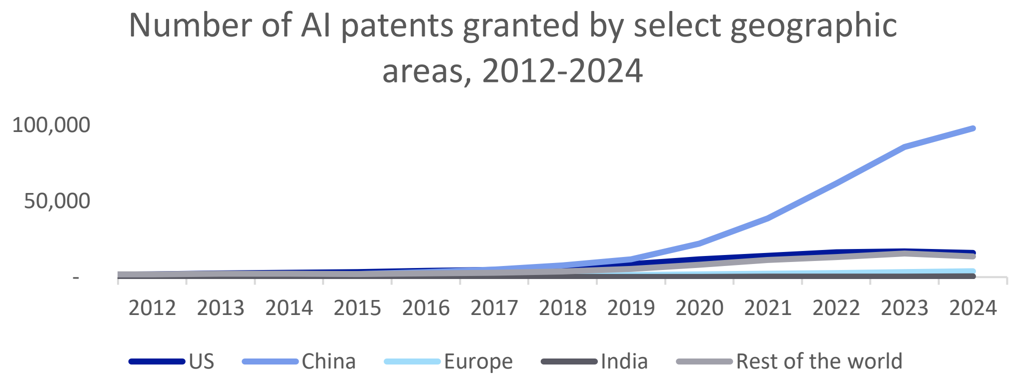
Dates mark impact, not first invention.

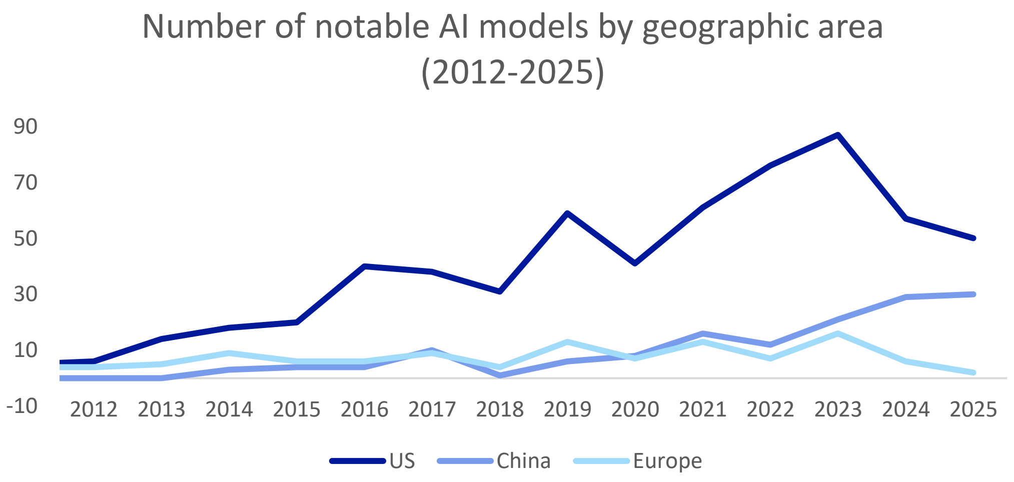
Source: Deutsche Bank Research

4. The pipeline matters: China's overnight success with DeepSeek was years in the making

Success of Chinese models reflects a multi-year push in R&D, overtaking the US in 2024





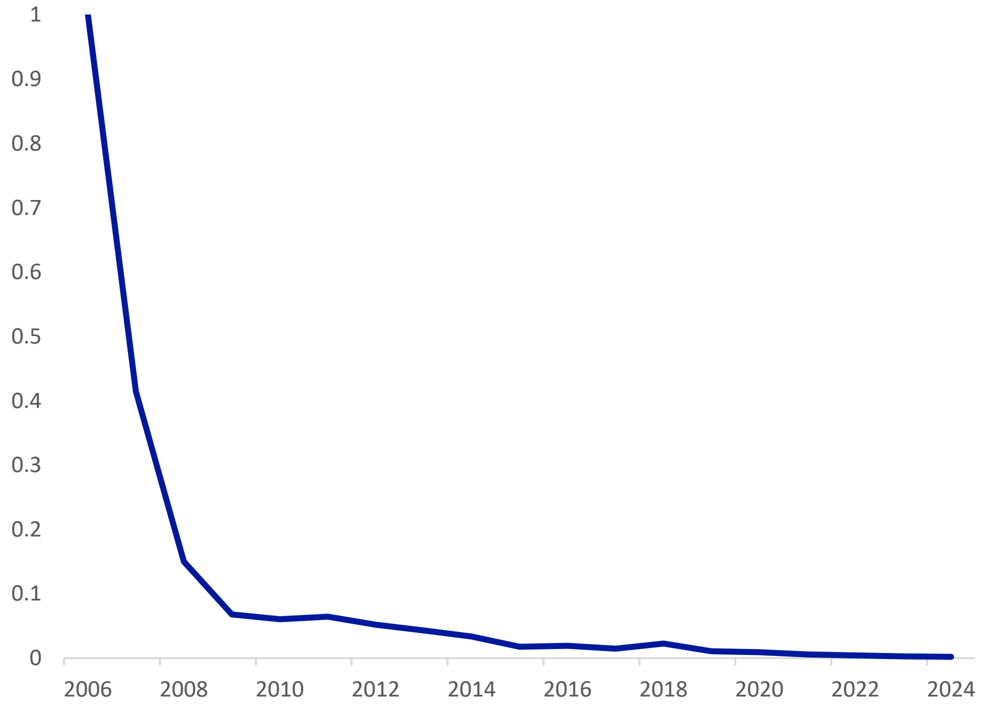


Source: OECD Data Explorer, Deutsche Bank Research

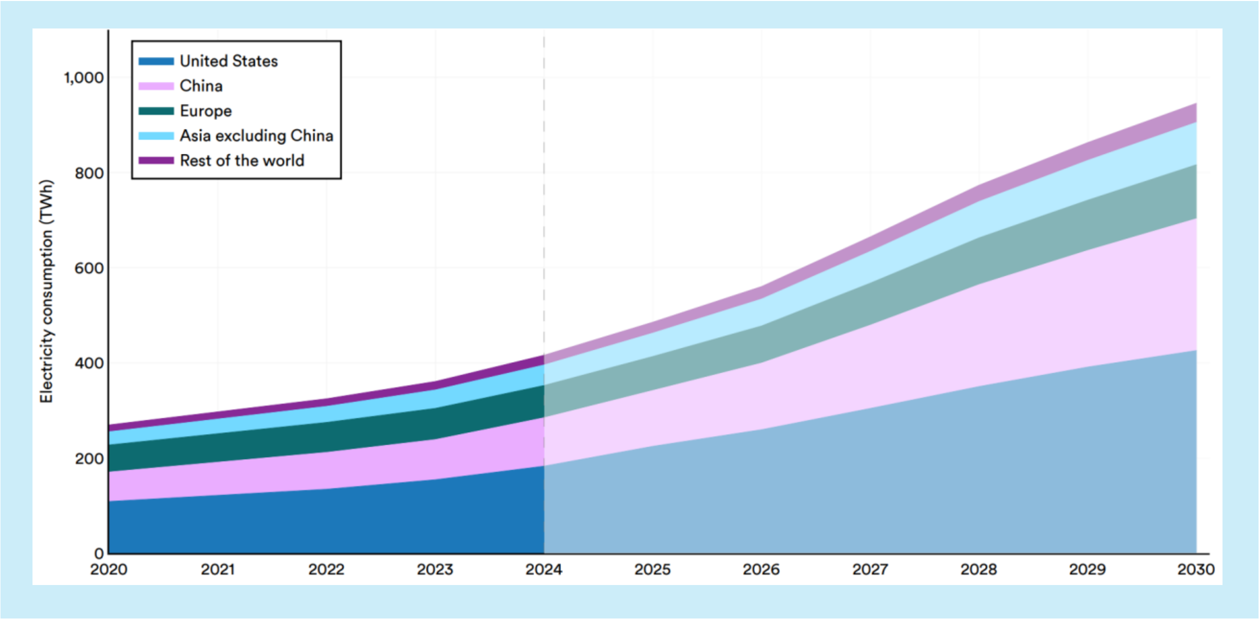
5. Making AI dramatically cheaper will likely increase, not decrease, spending

Jevons Paradox is alive: GPU computation cost has fallen by more than 99% since 2006, yet electricity consumption by data centres is set to double from 2024 to 2030 as demand surges

GPU chip computation cost index (2006=1)



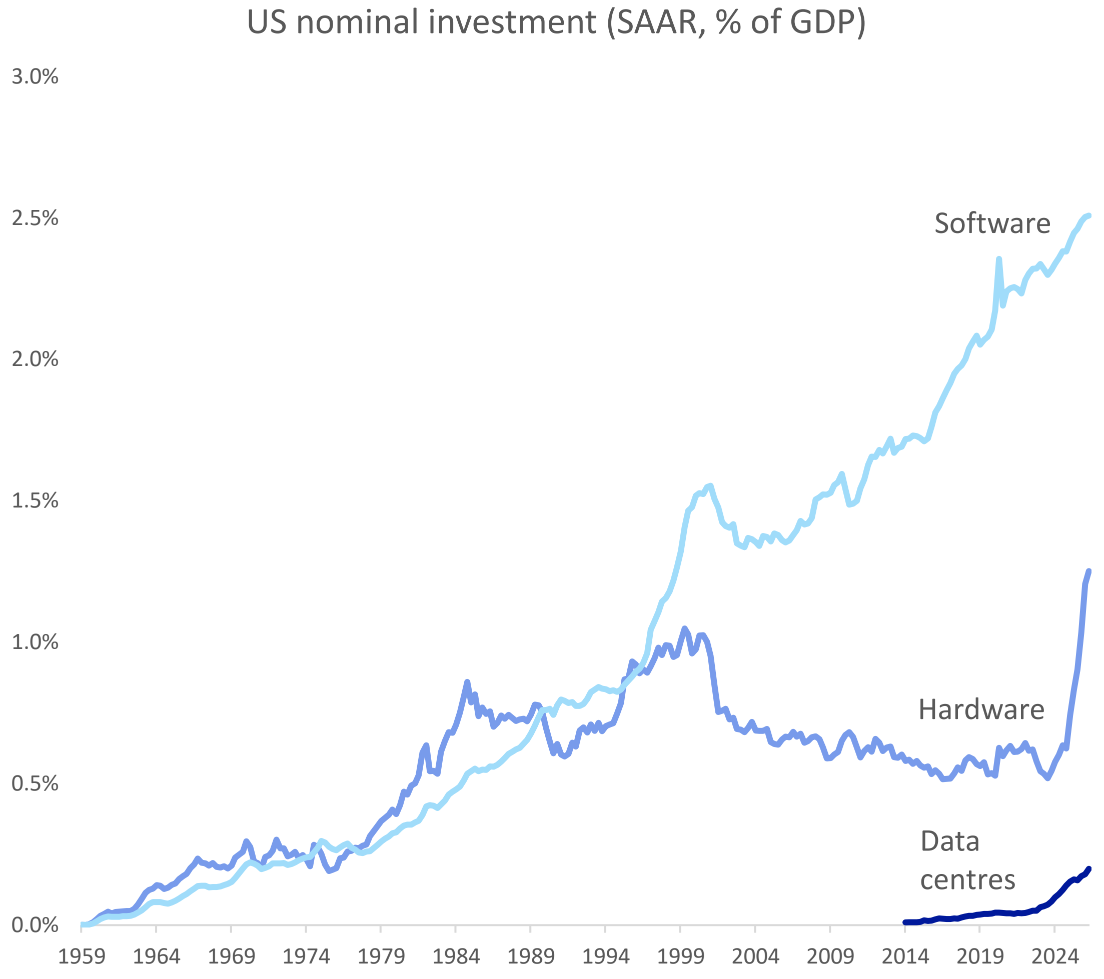
Data centre electricity consumption by region 2020-2030 (IEA projections)

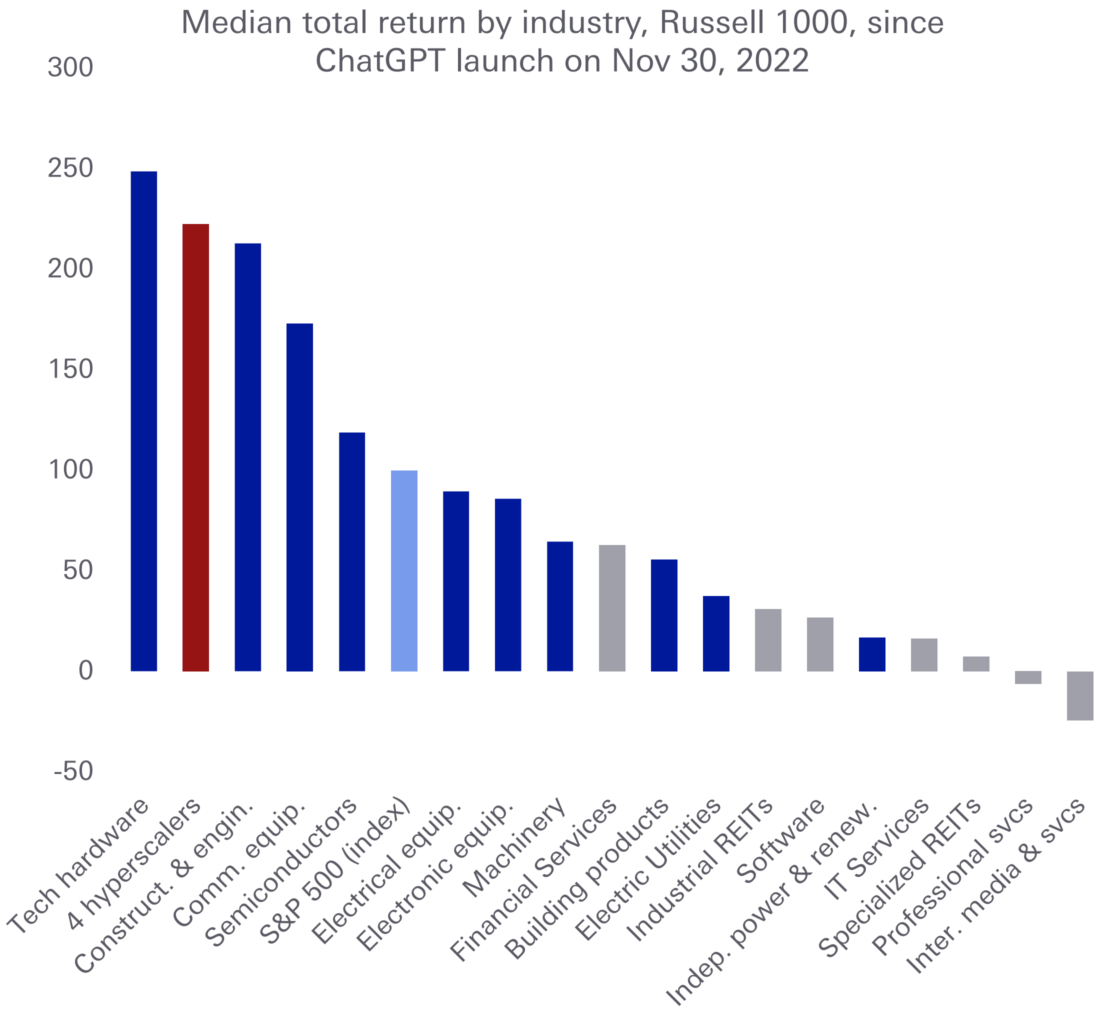


Source: International Energy Agency (2025), Stanford HAI AI Index 2026, Deutsche Bank Research

6. Software revolutions depend on heavy hardware investment

Hardware, construction, chip and equipment makers have led gains in the AI boom so far



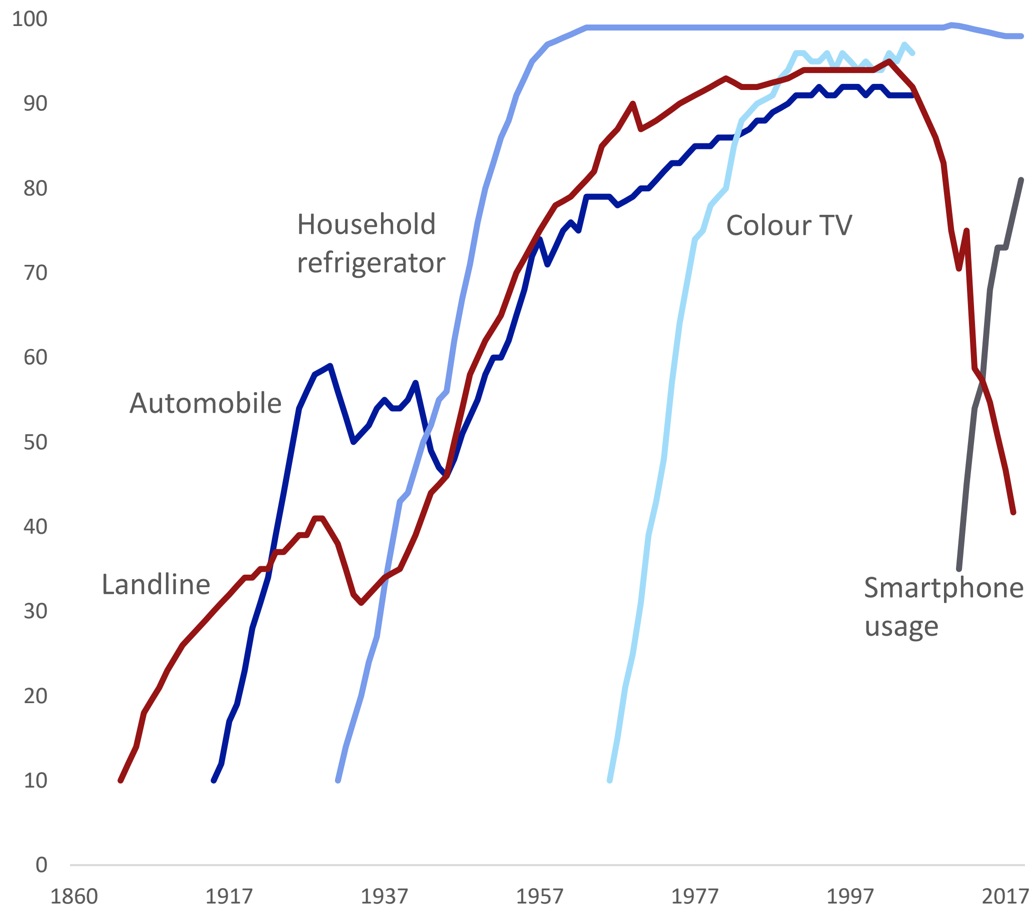


Source: Federal Reserve Bank of St Louis, Haver Analytics, Bloomberg Finance LP, Deutsche Bank Research

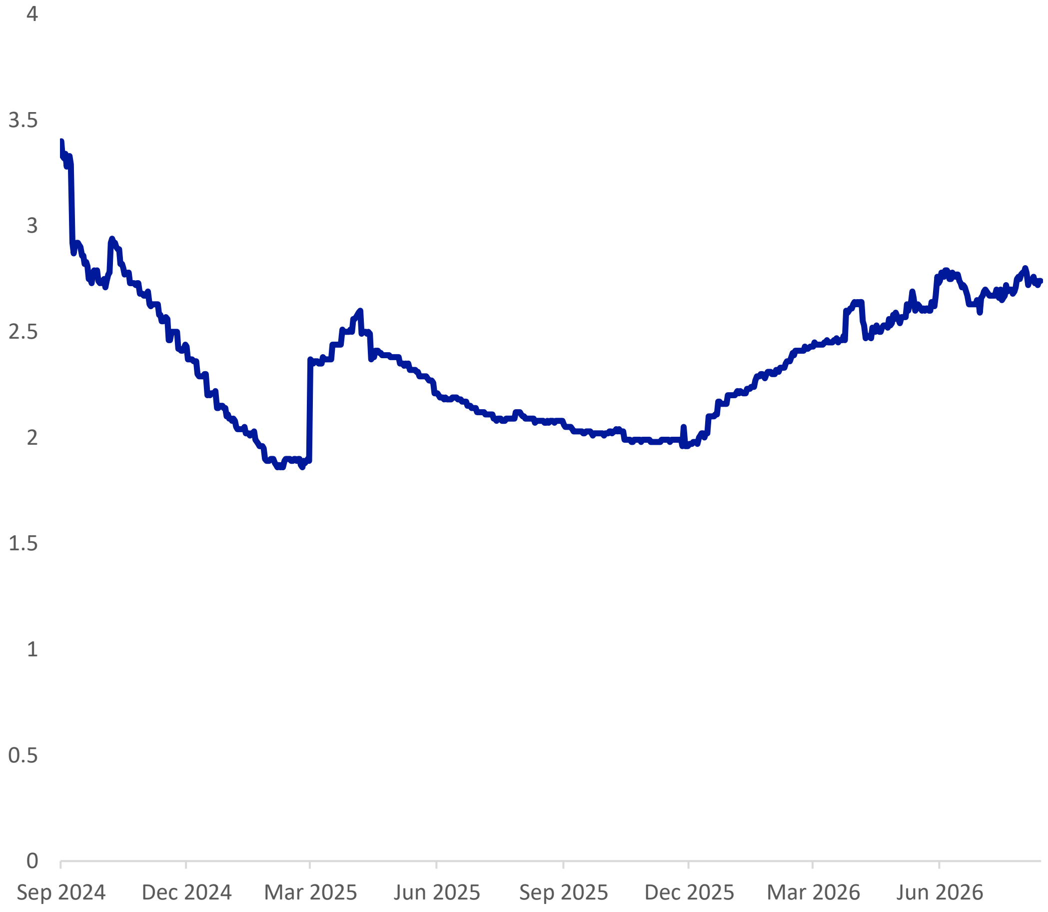
7. Hardware is often a bottleneck – but for once the delay is with producers not consumers

Past booms often relied on getting devices to consumers; now it's about getting scarce hardware online to serve them remotely

Share of US households using specific technologies (%)

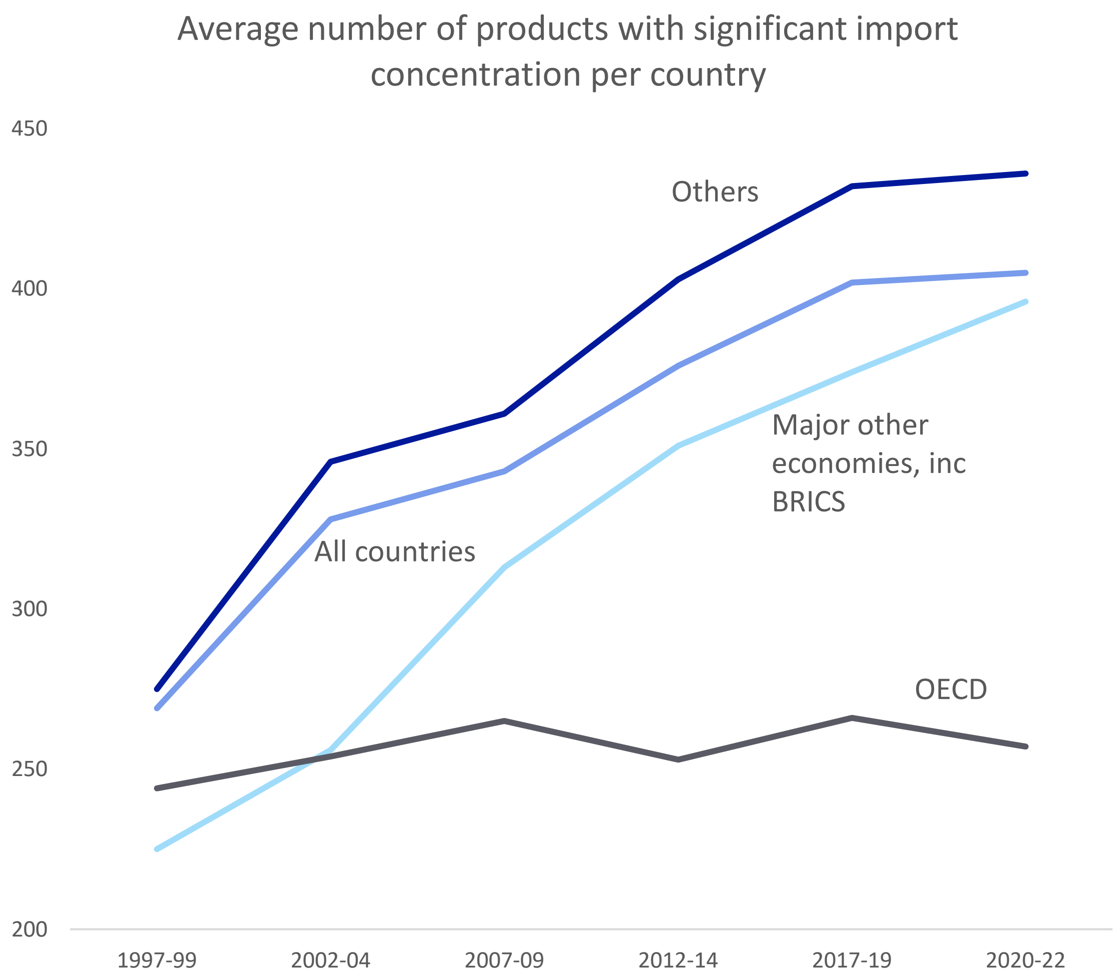


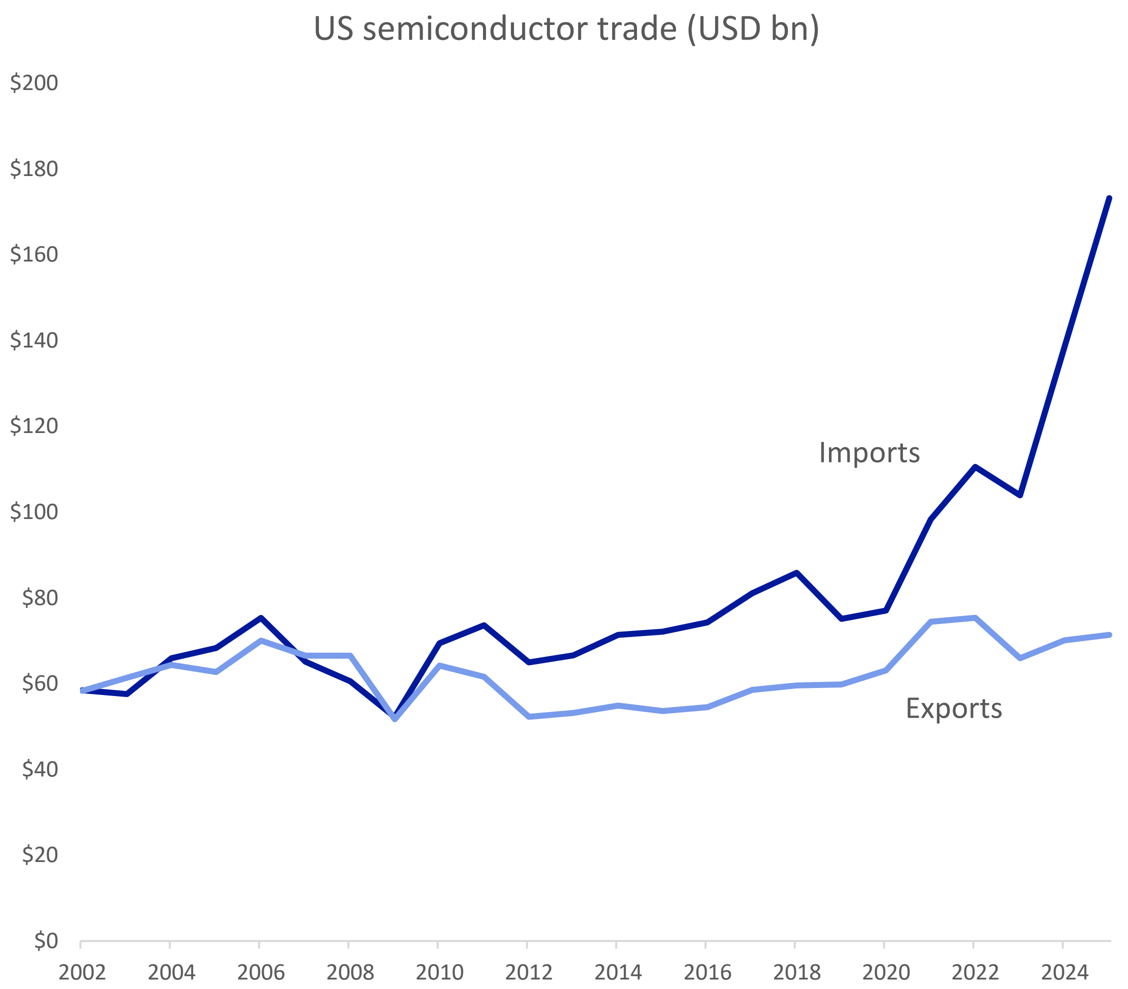
Hourly rental price of Nvidia H100 GPU chips (USD)



Source: Our World in Data, Federal Reserve Bank of St Louis, Haver Analytics, Bloomberg Finance LP, Deutsche Bank Research

8. Globalisation made tech cheaper – but also increased supply chain risks
There has been a surge in US imports of semiconductors as the AI boom has taken off





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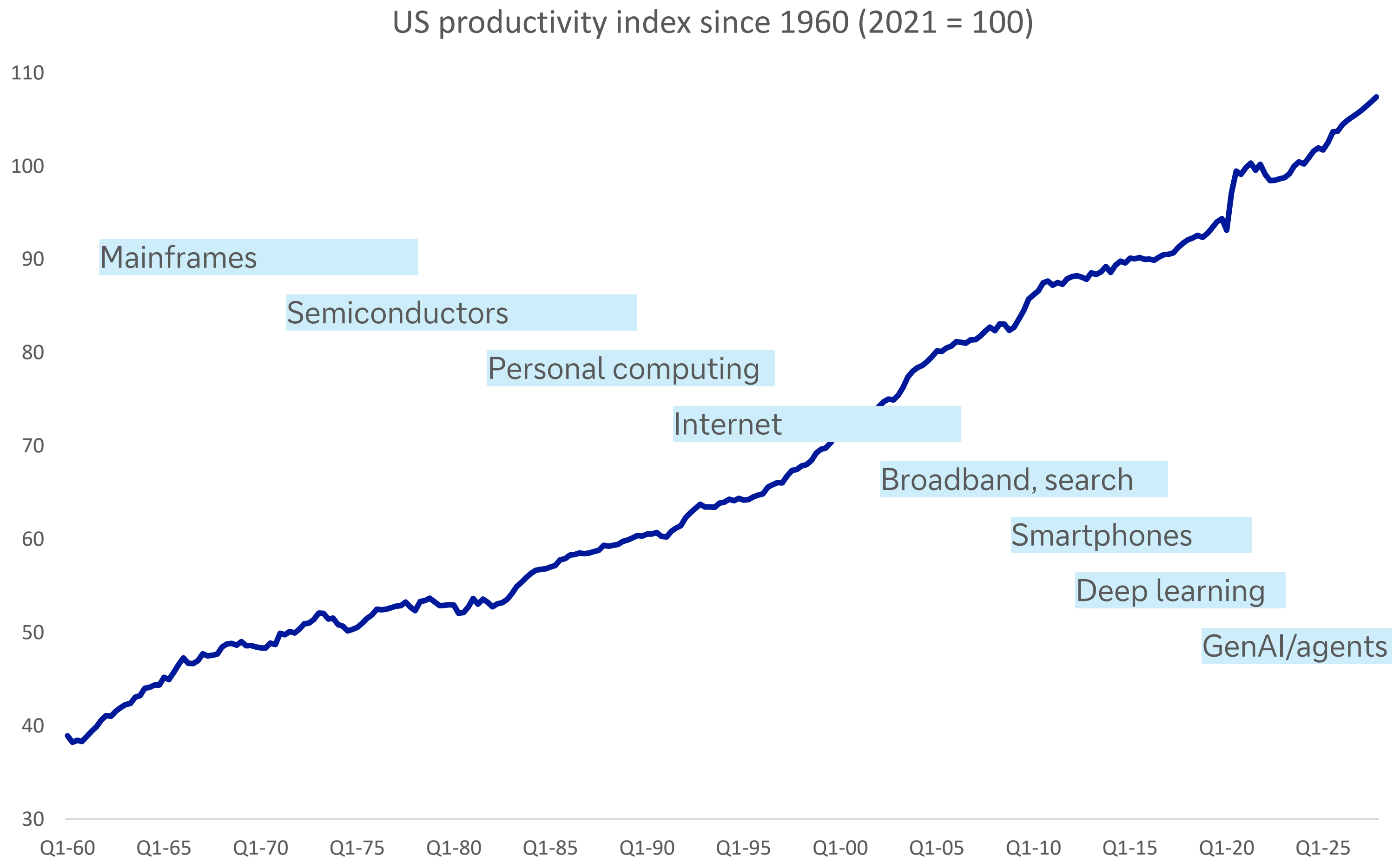
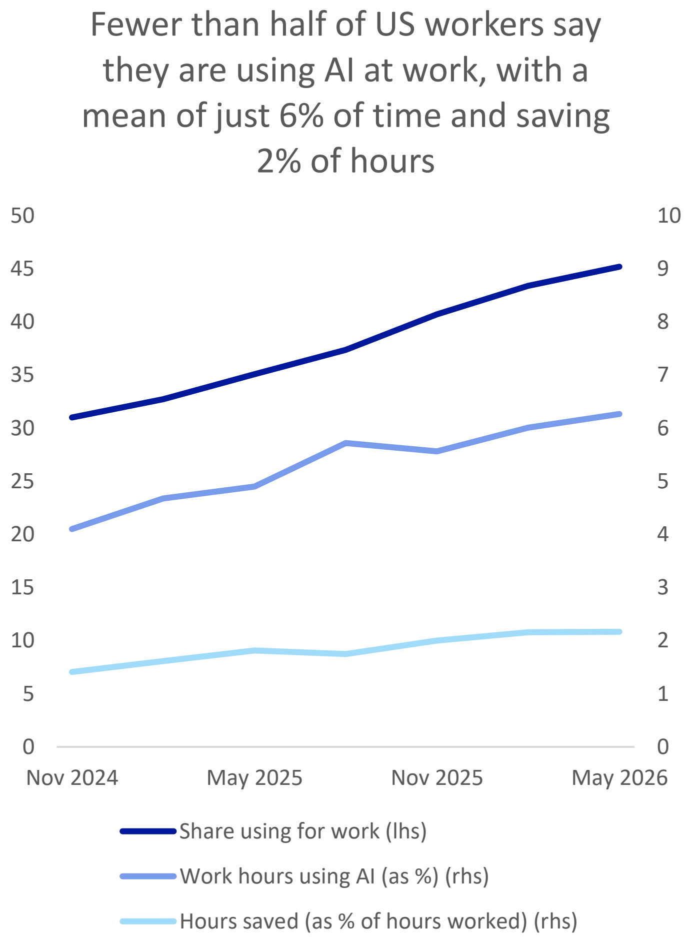
9. Tech revolutions take time to pay off, making them hard to see in long-term data

General purpose technologies have a productivity J-curve, generating costs before results



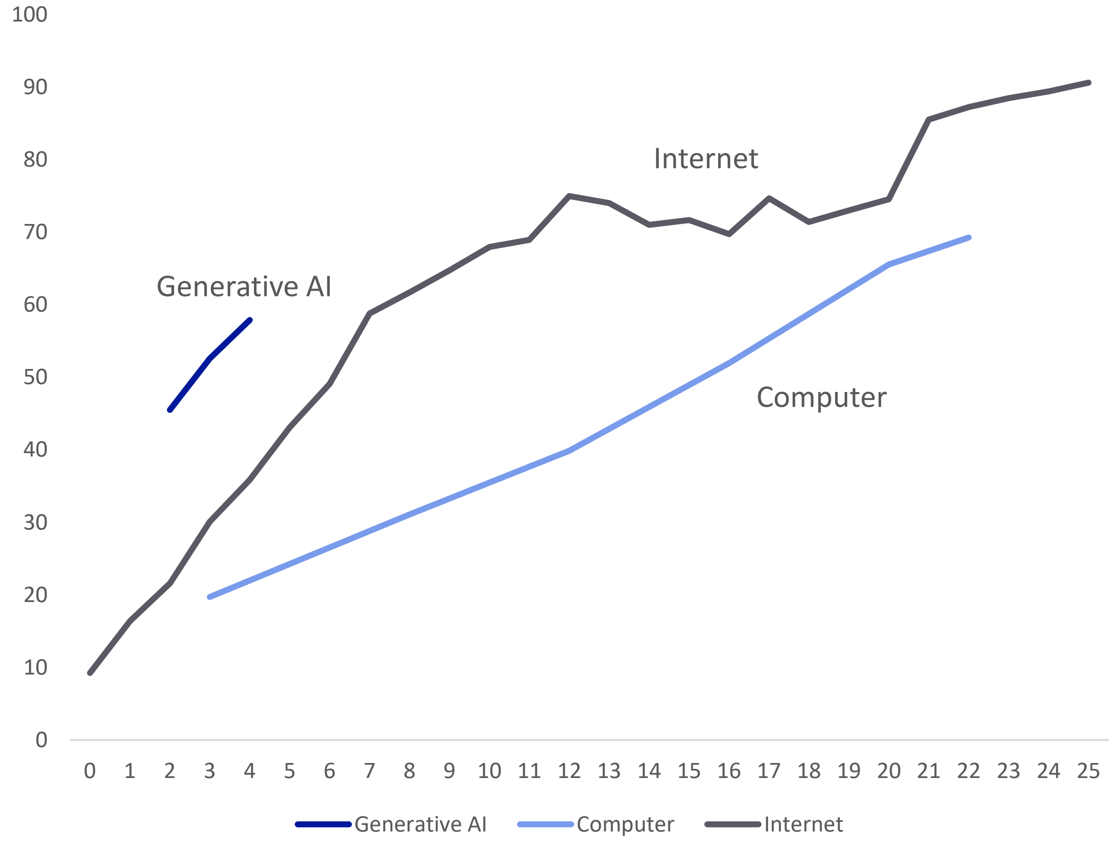


Source: OECD, GenAI Adoption Tracker, Deutsche Bank Research; blue bars in productivity chart give rough indication of notable tech eras

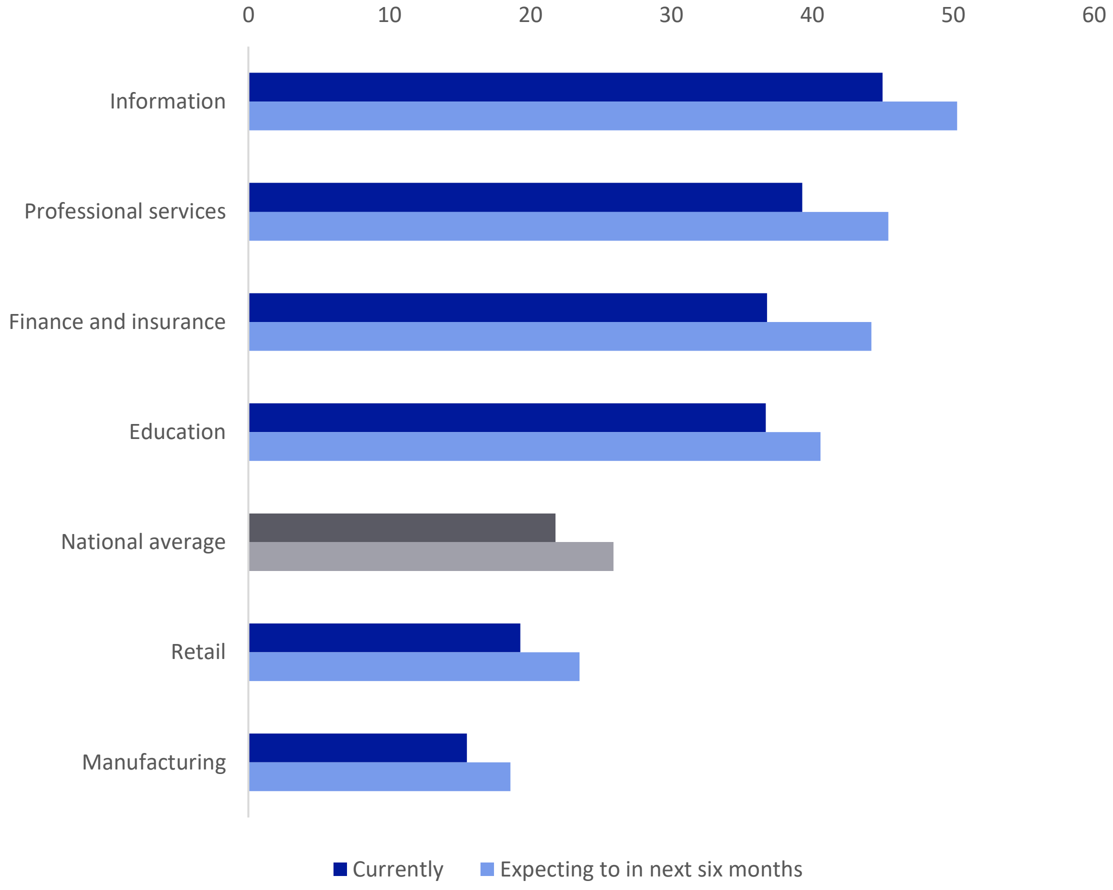
10. Consumers have adopted AI at a record pace but monetisable business use is slower

Downloading an app takes minutes; reorganising enterprises round new tech takes years

Speed of AI adoption by technology (% of population in years since introduction of first mass-market product)



US companies using/expecting to use AI (%, Aug 2026)



Source: GenAI Adoption Tracker, US Census Bureau, Deutsche Bank Research

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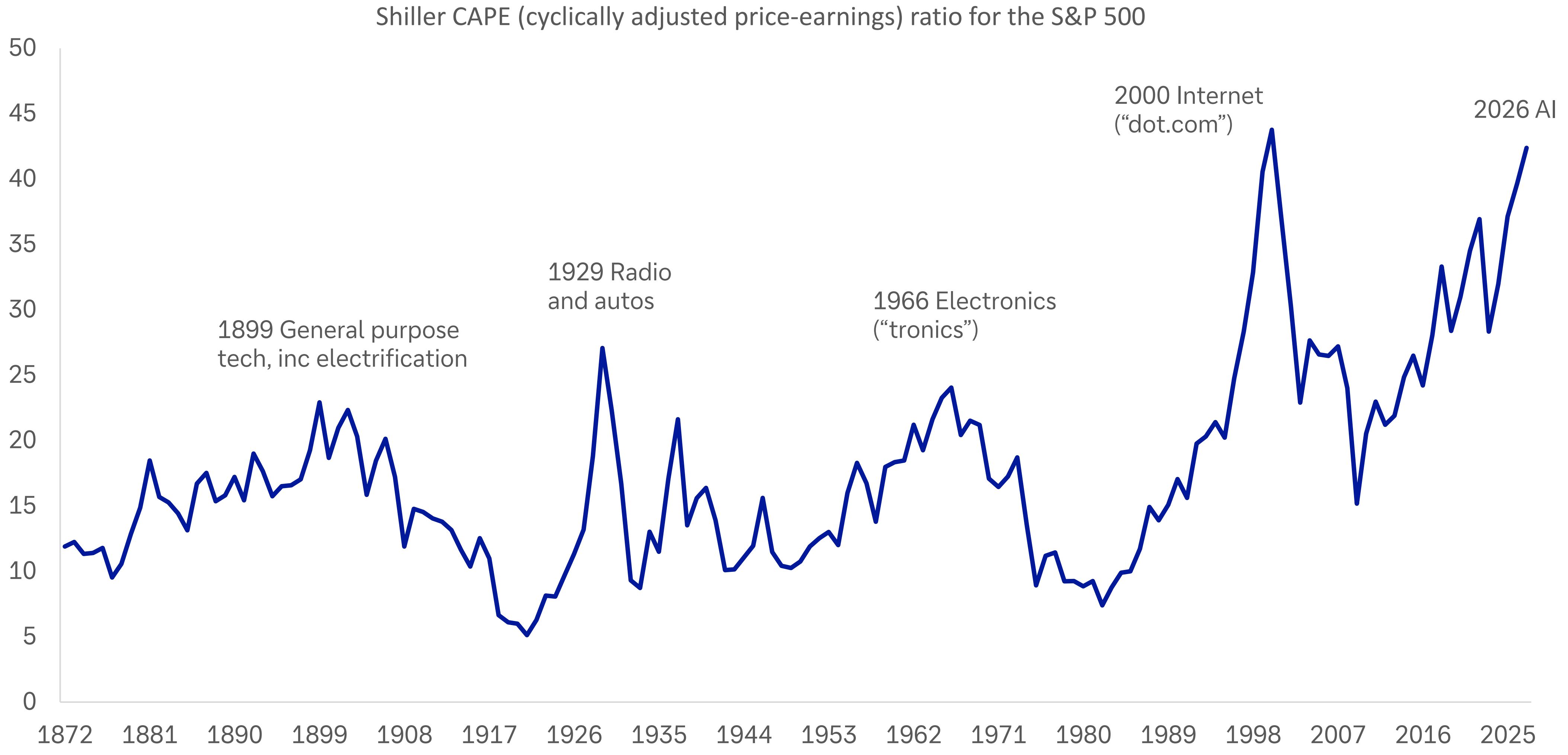
11. The biggest valuation booms are often tech booms – like today

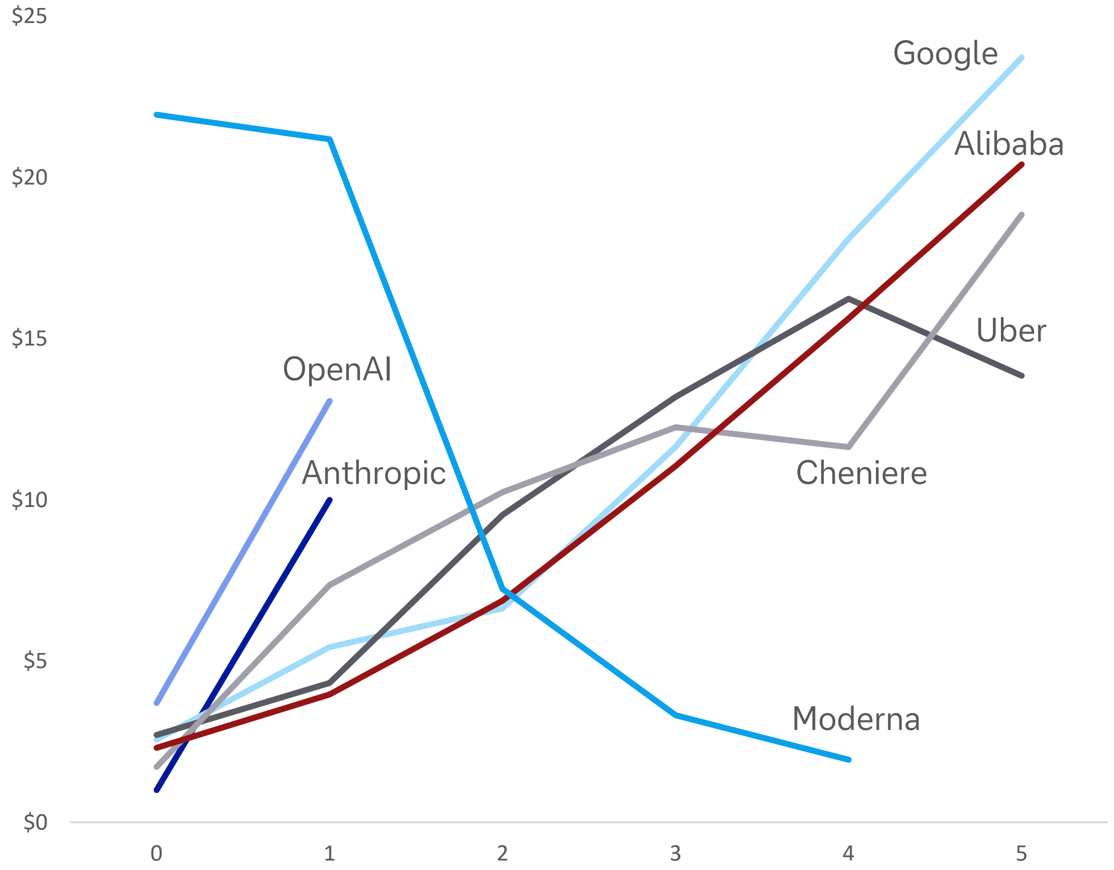
Peaks have mostly coincided with transformative technologies



Source: Robert Shiller, Deutsche Bank Research

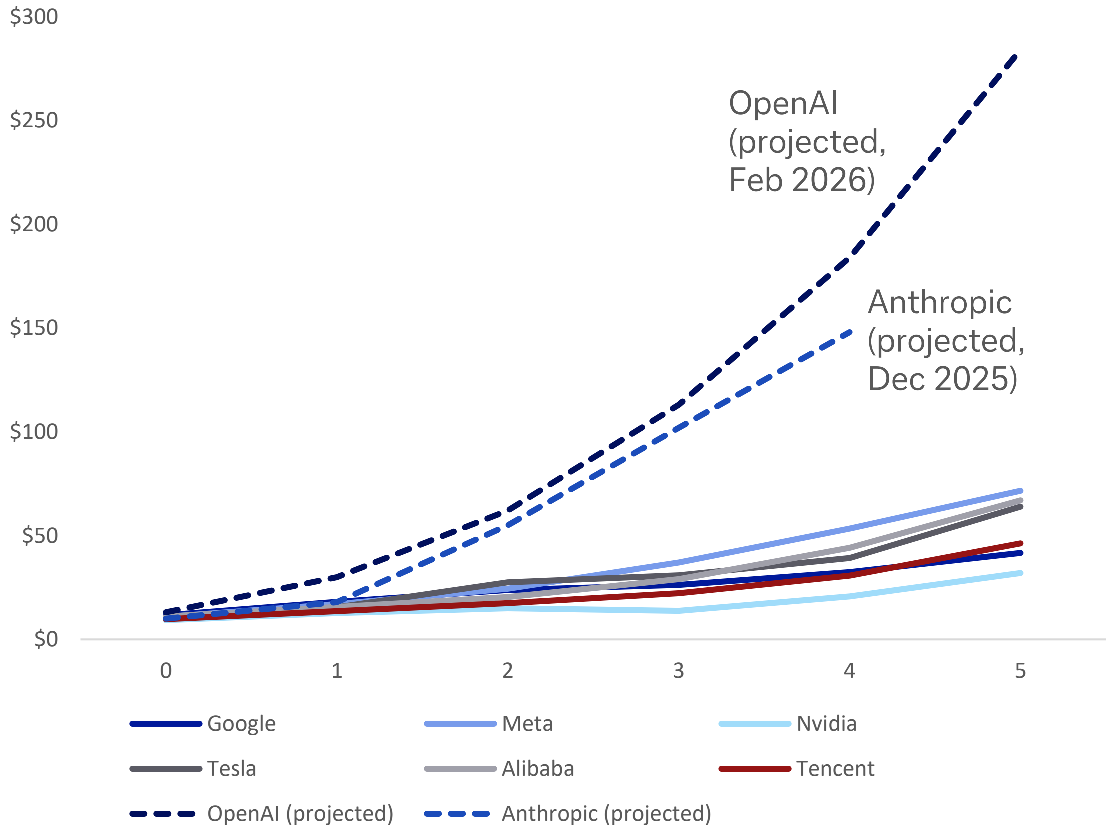
12. This time is different – but it will need to be even more different to meet projections
OpenAI and Anthropic are exceeding record revenue growth set by peers, at least for now

Annual revenue since exceeding $1bn milestone (USD bn)



Source: Bloomberg Finance LP, media reports, Deutsche Bank Research. Note: Year 0 shown as first year revenue exceeded $1bn (2025 USD). For OpenAI and Anthropic that was reportedly 2025. Moderna nominal revenue spiked from $800m in 2020 to $18.5bn in 2021, due to Covid vaccine. Cheniere revenue spiked with 2022 energy crisis.

Annual revenue since exceeding approx $10bn revenue, plus projected revenue for OpenAI/Anthropic (USD 2025)

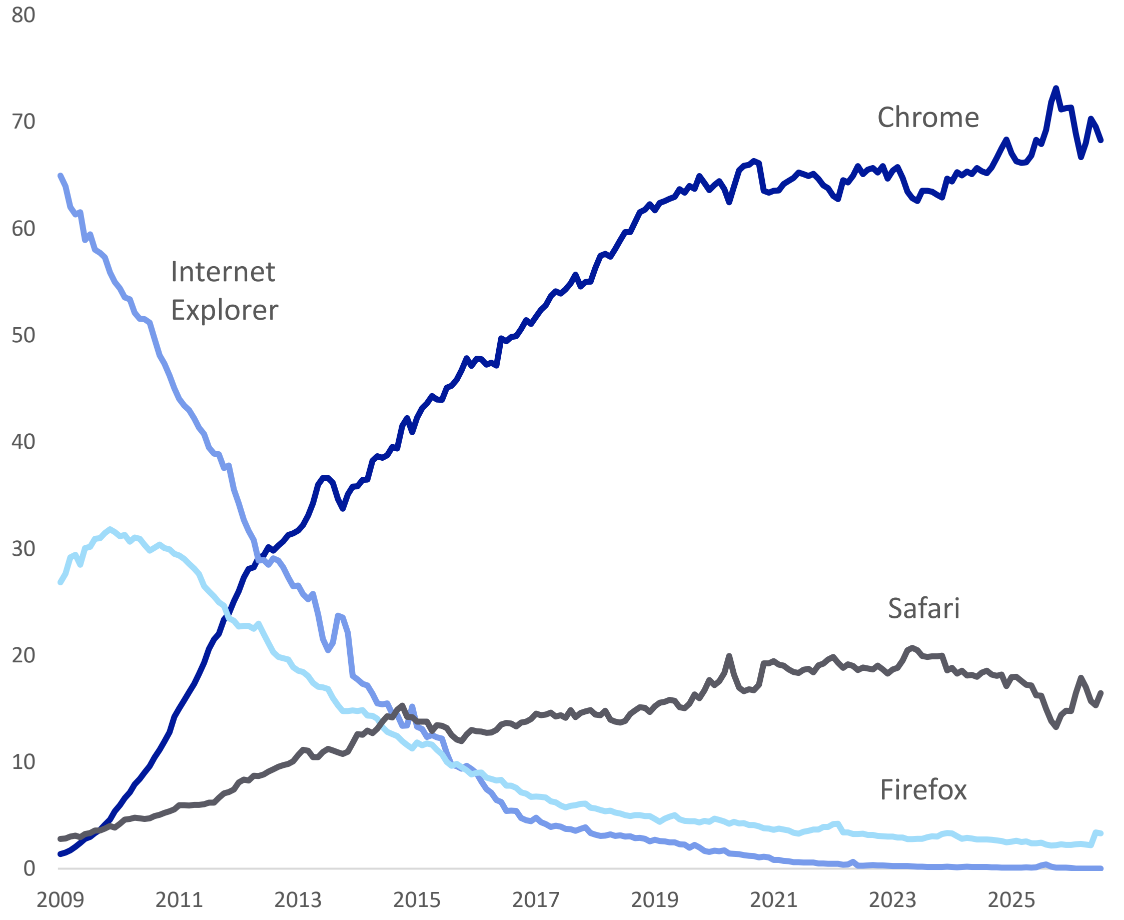


Source: Epoch, Bloomberg Finance LP, The Information (Feb 2026, citing recent updates to investors from OpenAI and Anthropic), Deutsche Bank Research. Note: Year 0 is first year revenue was approx $10bn (2025 USD)

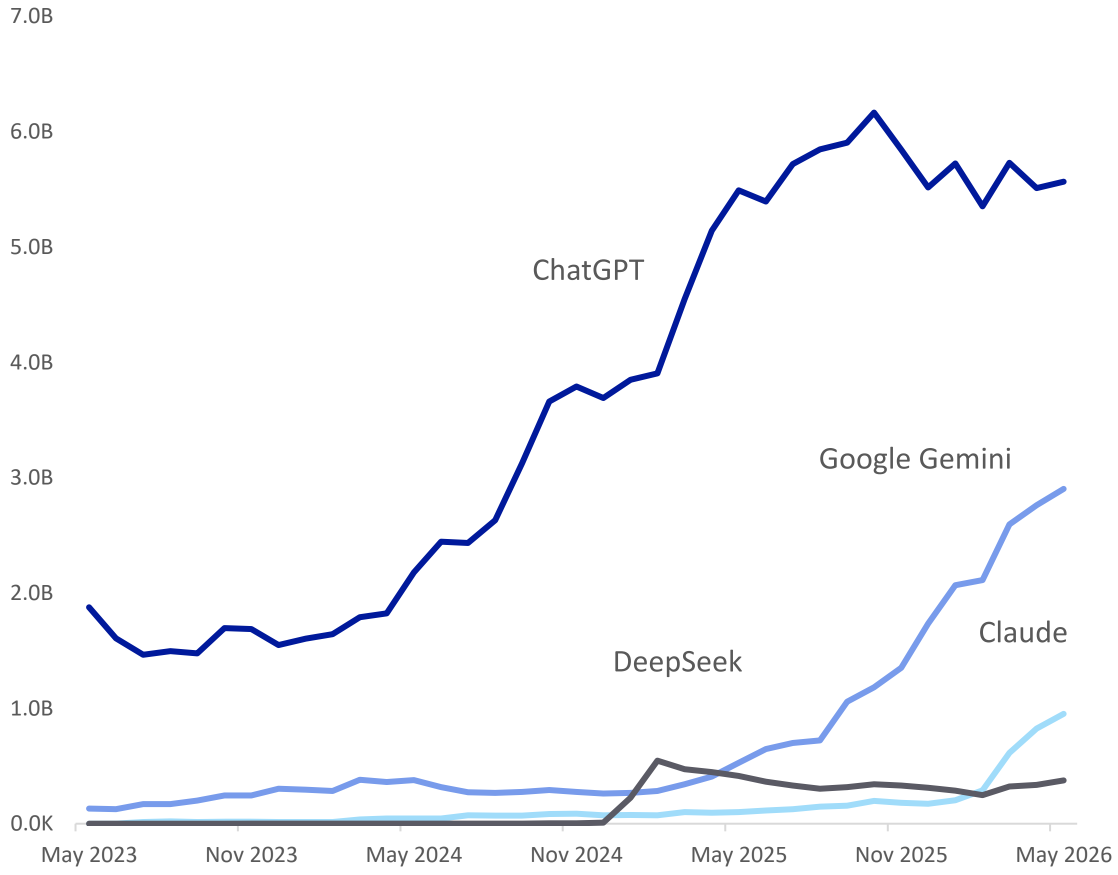
13. Early leads do not guarantee long-term winners

For example, top browser Internet Explorer overtook Netscape, then it too was overtaken

Browser market shares (%) (2009 to today)



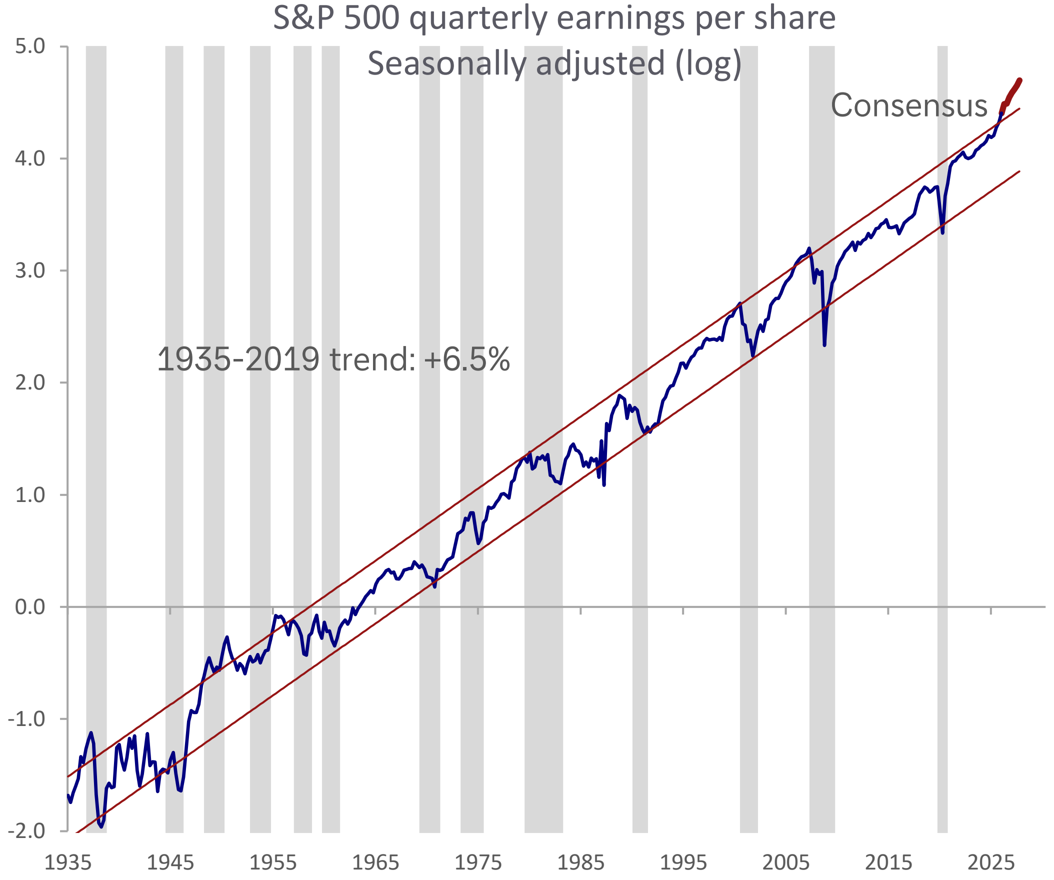
Top four generative AI platforms: monthly visits worldwide (desktop and mobile)

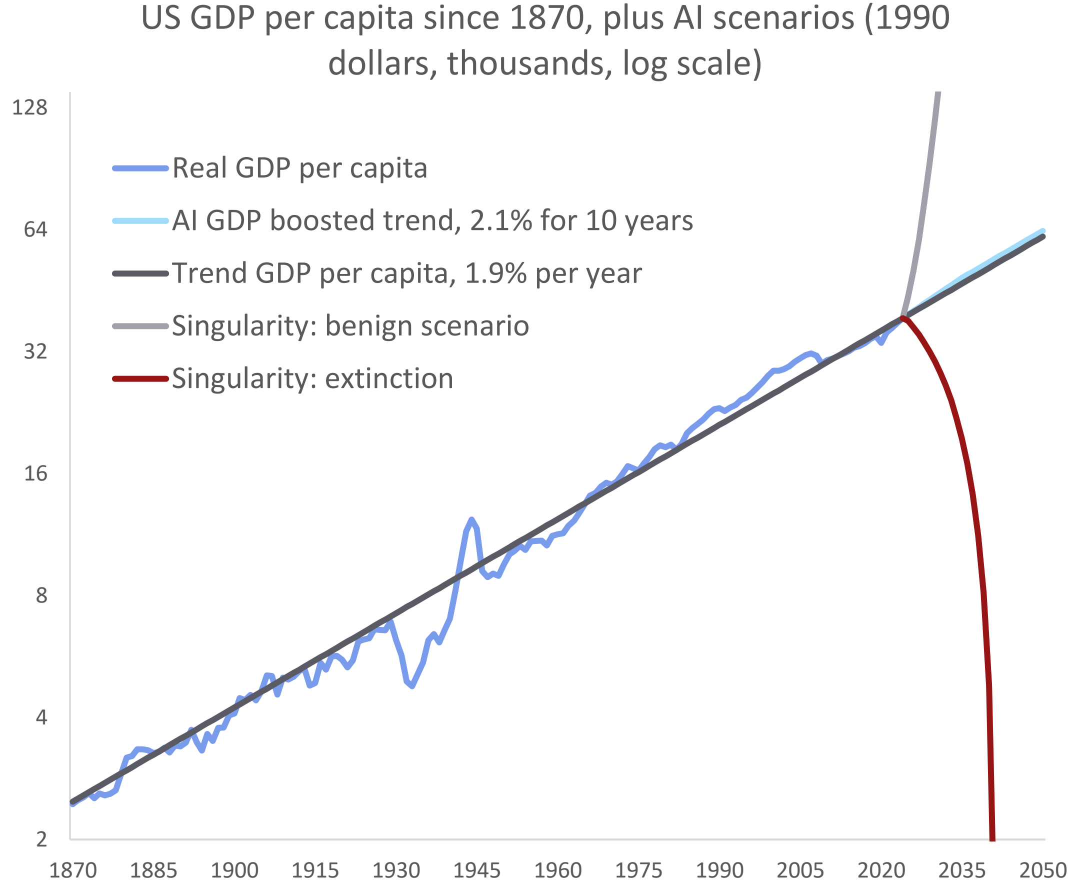


Source: Statcounter, Similarweb, Deutsche Bank Research

14. Long-term trends tend to persist through wars, recessions – and tech revolutions

Even so, some are considering more extreme scenarios this time… from utopia to extinction





Source: Bloomberg Finance LP, Federal Reserve Bank of Dallas, Deutsche Bank Research. Grey shaded bars in S&P quarterly earnings chart represent recessions

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Deal or no deal, investors are both concerned and excited

August 11, 2026

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Executive summary

Five months into the US-Iran conflict and the strength of corporate earnings makes it seem like the optimists have won. But things are more complicated. This report analyses proprietary data from our dbDataInsights team and shows that the issue of investing against the backdrop of a volatile energy market is highly polarising for people. Indeed, of all the investment themes we track, energy has one of the highest rates of both optimism and pessimism.

So, we have an odd situation: On one hand, global corporate earnings are incredibly strong despite the energy shock this year. On the other hand, the Iran conflict is still a massive problem for the world economy. An amicable agreement thus remains elusive.

Three findings from our analysis stand out.

— Energy – whether measured through prices or transition – is now one of the most-cited investment themes for US investors behind AI, and is prompting portfolio changes on par with technology and geopolitics.

— Energy prices carry one of the highest perceived risks of any theme we tested among UK investors.

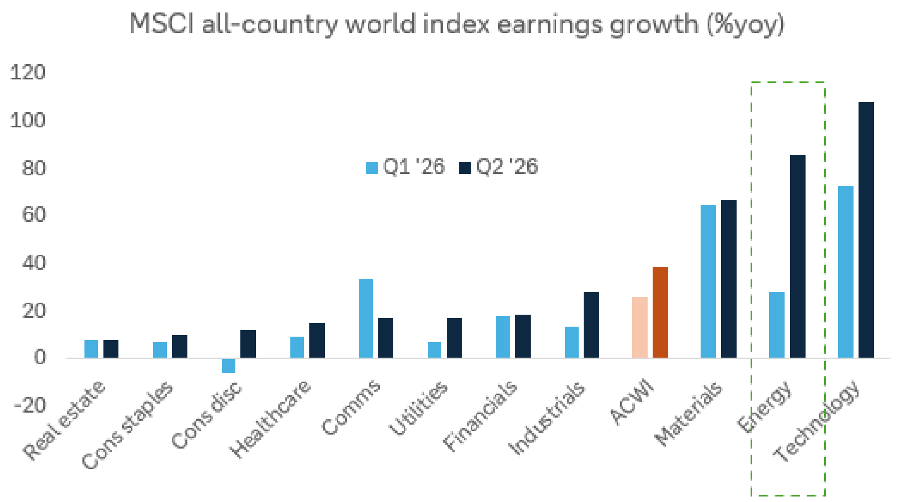
— Our megatrend model shows that energy disruption (as a medium-term trend) has remained a positive force for the world economy this year. That pattern closely echoes the 1970s oil shocks.

Energy shocks are unambiguously bad for consumers and growth in the short term. But over longer time frames, our model shows that energy disruption has tended to be a net positive for markets and economies because it accelerates diversification of supply and pushes economies to generate more GDP per unit of energy consumed. We see early evidence of a similar dynamic beginning today.

A serious issue ... but surprising resilience

The closure of the Strait of Hormuz has not been the economic catastrophe that many expected five months ago. In fact, corporates earnings have just had one of their best quarters outside of recession recoveries. For the companies in the MSCI all-country world index, median earnings growth is running at 16.5% yoy in Q2, well above the 14.7% in Q1. Energy has been a big part of that, with earnings growth, at a sector level, surging 86% in Q2, about triple the growth recorded in the prior quarter.

Figure 1: Earnings growth has been very strong in Q2



Source: Bloomberg Finance L.P., Deutsche Bank Asset Allocation

Economies have fared less well than companies, but they remain remarkably resilient given the harsh predictions that came from some quarters in early March. Of course, stockpile releases and other buffers have helped smooth the disruption – and as these wear down further, larger problems may arise.

So, now we have a curious juxtaposition of outlooks. On one hand, corporate earnings are proving resilient and the long-term focus on energy security has helped make many of the associated equities good investments. On the other hand, the Iran conflict is still a massive problem for the world economy.

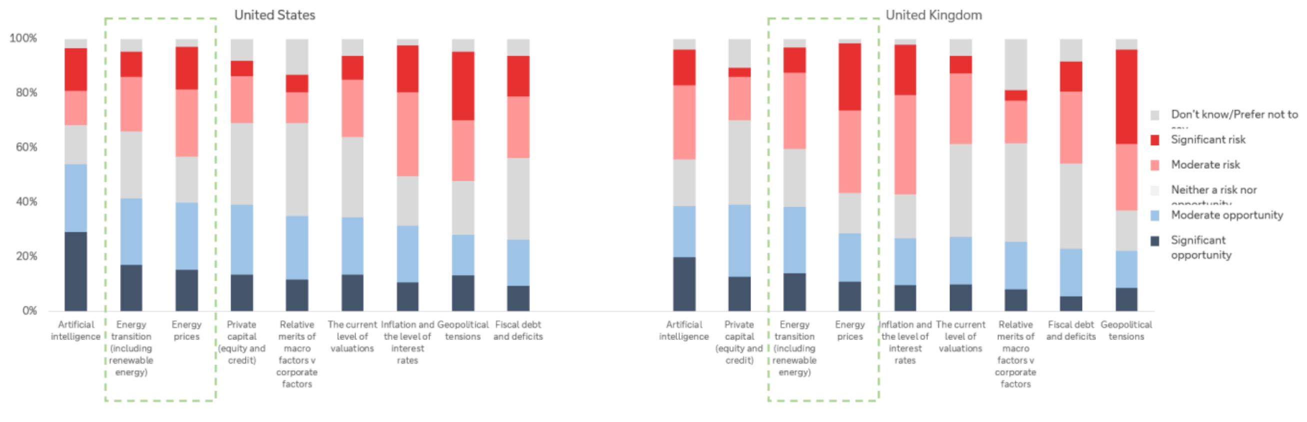
What investors think about the effect of energy on their portfolio

Our dbDataInsights survey on the topic was carried out in April and May of this year and polled a representative sample of people in the US and the UK. The responses indicate that energy has become one of the two dominant investment themes of 2026 – alongside, and closely intertwined with, artificial intelligence.

Energy is a high-conviction theme – in both directions

When we asked investors whether they see various investment themes as a risk or an opportunity for 2026, energy transition and energy prices ranked just behind artificial intelligence as the most-cited opportunities in both the US and the UK. The below chart shows a selection of the themes.

Figure 2: Survey question: "For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026"



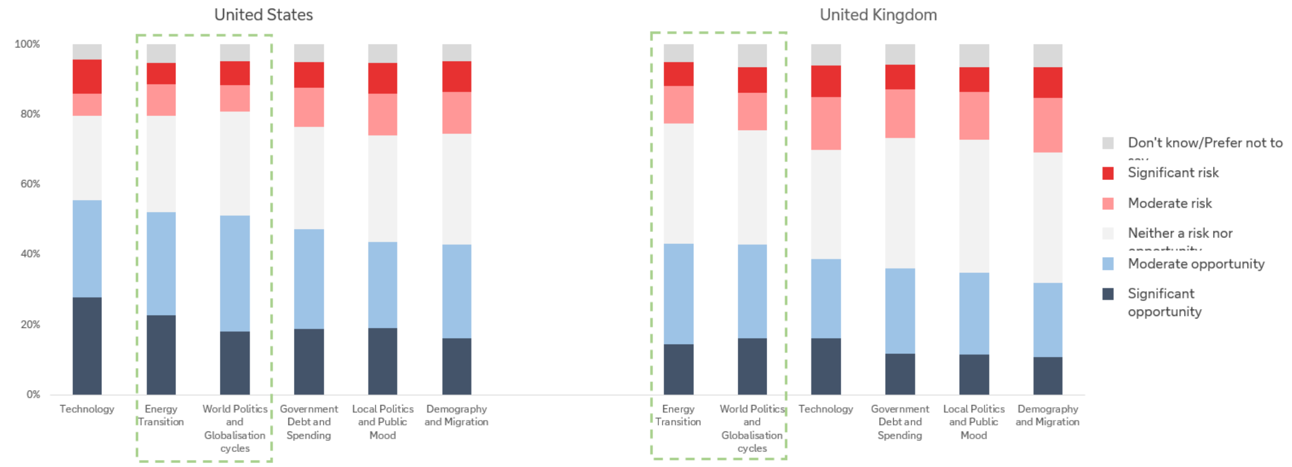
Source: dbDataInsights, Deutsche Bank

Energy is already changing portfolios

Conviction about energy appears to be translating into action – particularly when we think about it from a medium- or long-term point of view. To assess this, we fell back on our megatrend model that assesses the development of the six key megatrends that are driving the world's economies and markets.

When we asked investors how likely they are to change their portfolio over the next 12 months because of each megatrend, a dominant theme was energy and geopolitics as the below chart shows.

Figure 3: Survey question: "Within the next 12 months, how likely are you to make changes to your investment portfolio due to the following thematic megatrends?"

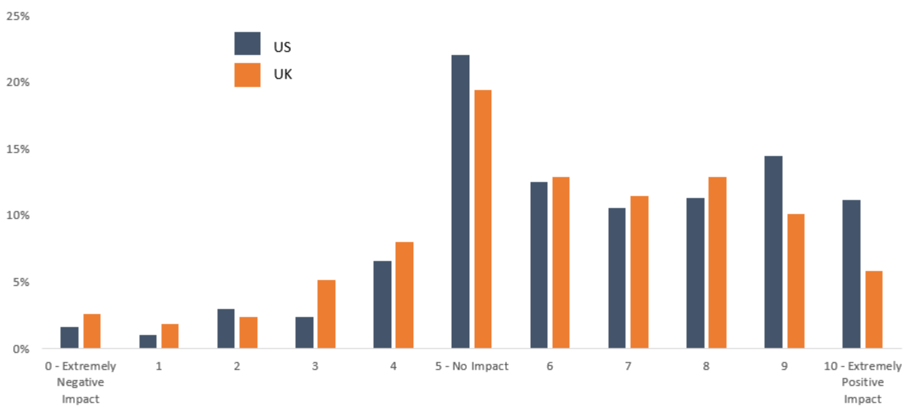


Source: dbDataInsights, Deutsche Bank

Taken together, the two survey questions tell a consistent story: energy has moved from a background, structural megatrend to one that investors are actively trading around, much as technology has been for the AI boom. The difference is that, unlike technology, energy's near-term path is being set as much by missile strikes and shipping insurance premia as by fundamentals.

Investors appear to sense this: when we asked how the energy transition will affect their main portfolio over the next three years, responses skewed firmly positive.

Figure 4: Investors' anticipated impact of the energy transition on their main investment portfolio over the next 3 years

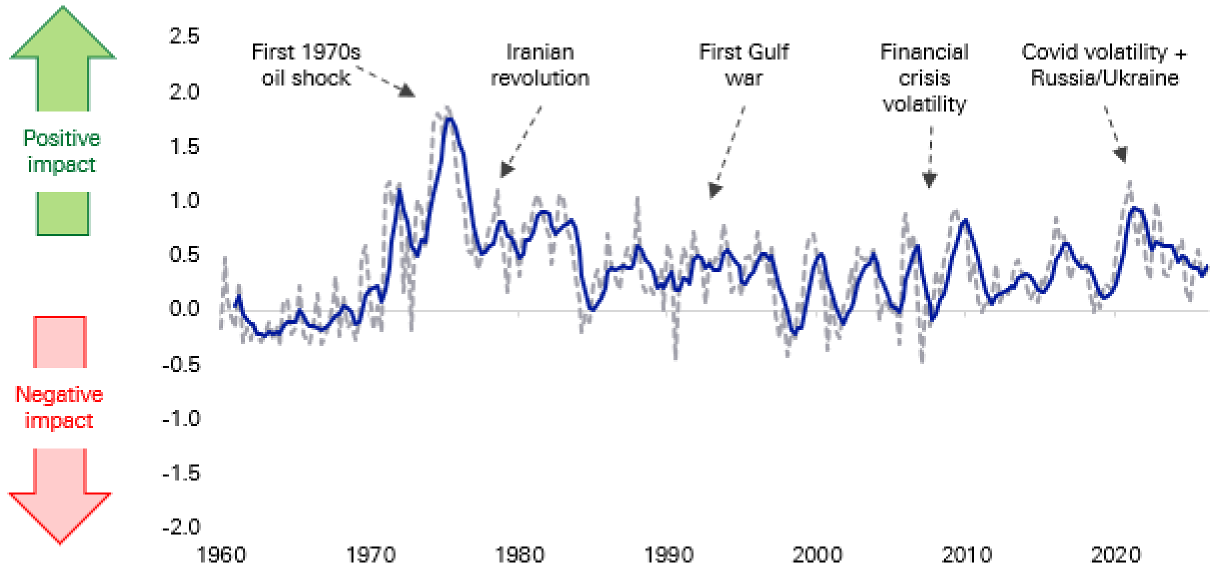


Source: dbDataInsights

Energy as a megatrend: why the medium-term signal is so positive

Energy is one of the six long-term trends that feeds into our megatrend model that uses almost 100 data series since 1957 to track their impact on economies and markets. Energy is a nuanced megatrend as, unlike several others, there is a big difference between the effect of short-term disruption and long-term gain. Indeed, prior periods of energy disruption have been incredibly good for the long-term energy trend, even if the short-term disruption was incredibly painful for consumers and economies.

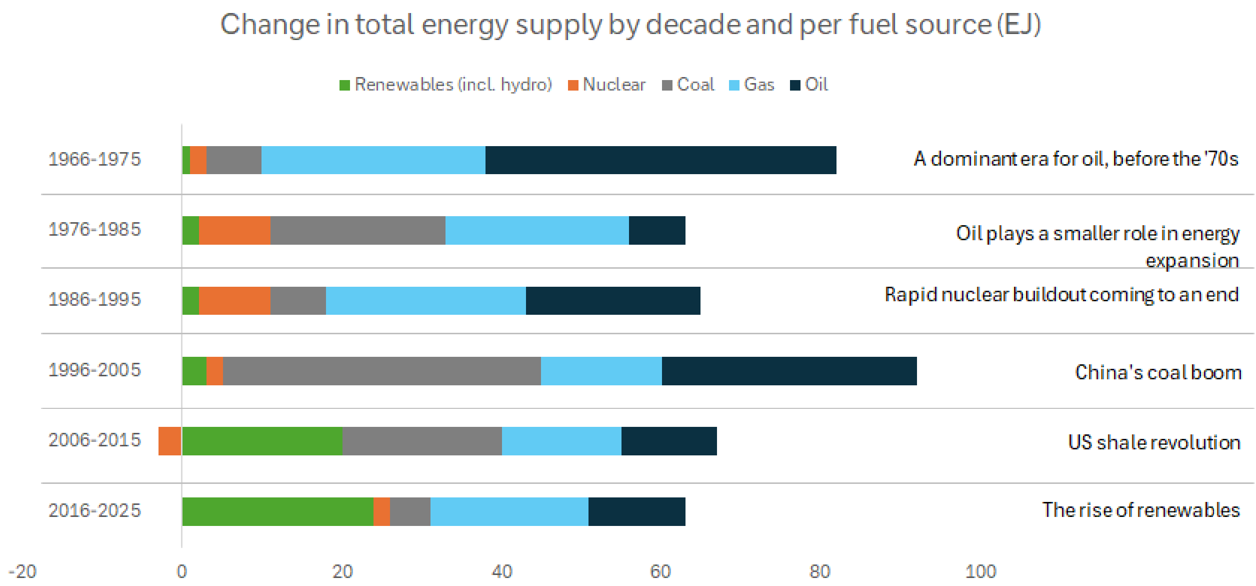
Figure 5: Our energy megatrend indicator has generally been positive as the world's economies diversify their supplies at the same time that innovation makes the economy more energy efficient



Source: Factset, Bloomberg Finance L.P., Fineon, EIA, Deutsche Bank Research

Viewed through a medium-term lens, the last five months of conflict in the Middle East can be viewed as likely being a long-term positive for economies and markets. That is because the Iran war is accelerating exactly the kind of diversification our model rewards: European and Asian buyers have moved further into US, Guyanese and West African crude and LNG since the Strait's closure; several G20 governments have fast-tracked strategic reserve releases and non-Gulf supply agreements; and renewed urgency has attached to the 'friend-sufficient' push for critical minerals and energy inputs that began during the US-China strategic rivalry.

Figure 6: Shocks over the decades have defined several eras for the evolution of the energy mix



Source: Energy Institute

The 1970s is something of a parallel

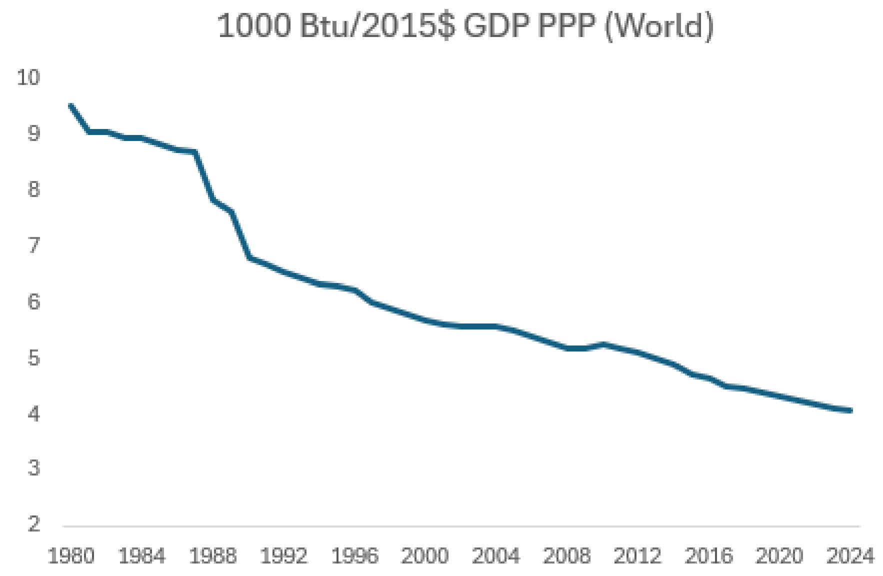
The closure of the Strait of Hormuz in 2026 and the disruption to Russian gas flows into Europe in 2022 both echo the twin oil shocks of 1973 and 1979, when OPEC's move to set its own prices followed by the Iranian revolution, sent crude to ten times its early-1970s level and permanently altered the trajectory of global oil demand.

Global oil consumption growth slowed from 7.8% before 1973 to 2.2% pa for most of the rest of the decade (on calculations from the Energy Institute). It outright declined after the second and took a decade for global oil consumption to recover to its 1979 level.

Whilst the 1970s shocks created havoc for both individuals and economies, they accelerated the pace at which the world changed how it generated economic output. The energy intensity of GDP fell 3.3% in 1980 (the largest single-year improvement in the Energy Institute's data series). That adjustment reflected both behavioural change and, over the following decade, structural investment in fuel efficiency and the phase-out of oil-fired power generation.

Since 1980, the energy intensity of GDP has roughly halved as shown in the following chart.

Figure 7: The energy intensity of GDP has been steadily falling – partly because of efficiency and innovation



Source: EIA

This trend is captured in our megatrend model which shows the medium-term payoff from decades of energy disruption. Those forces continue today. Every episode of chokepoint risk, from Ukraine in 2022 to Hormuz in 2026, adds another increment to the diversification of global energy supply and another incentive for energy innovation across economies.

Conclusion

Five months into the US-Iran conflict war, investors are highly polarised about the investment merits of investing against the backdrop of energy volatility. However, those that are positive are very positive. Our megatrend model suggests that reaction is rational: like the 1970s oil shocks, the medium-term legacy of this conflict is likely to be faster energy diversification and a further, durable improvement in the energy intensity of GDP, even though the near-term path for prices is likely to stay volatile.

As a result, energy security and supply diversification should keep attracting capital regardless of where the oil price settles. The beneficiaries are likely to cascade up and down the value chain including, LNG infrastructure, non-Gulf production, renewables, strategic reserves and grid resilience as countries and companies all try to diversify their chokepoint risk.

Appendix 1

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